

Physiology of Cardiovascular System

PC-II, May 2026

Introduction

- **The CVS:** A system of **pumping organ** and **Vessels** that transport **blood** to and from tissues throughout the body.
- The heart is the **life-beating, always-thumping muscle** in your chest.
- From **inside the womb** until **death**, the thump goes on.
- The heart for the average human will contract about **Three billion times**;
 - **never resting, never stopping** to take a break *except* for a fraction of a second b/n beats.

Introduction, cont'd

- Many believe that the heart is **the first organ to become functional**.
- **Within weeks of conception**, the heart starts its mission of supplying the body with nutrients.
- **The primary function of the heart** is to **pump blood** through the arteries, capillaries, and veins.
- There is an estimated **60,000 miles of vessels** throughout an adult body.
- **Blood then transports** O₂, nutrients, disease-causing viruses, bacteria, hormones, and has other important functions.
- The heart is the **pump** that keeps blood circulating properly.

General Consideration

- Physiology started as an experimental science in 1628 when William Harvey demonstrated that the CVS forms a circle in which blood was pumped out of the heart (the central pump) continuously through one set of vessels and returned to the heart through another set.
- Two circuits (pulmonary & systemic) both of which originate and terminate in the heart serve as routes of blood movement
- The end of the circulation is the microcirculation.
 - It lies within all tissues where exchanges between blood and cells take place.

Types of Circulation

- Blood circulation is an **evolutionary** consequence of body size.
- Isolated small cells & small organisms with **no circulatory system** meet metabolic needs by simple processes of **diffusion** and **convection** of solutes.
- In lower animals & vertebrates the circulation may be:
 - **Open or closed circulation**
 - **Single or double circulation**

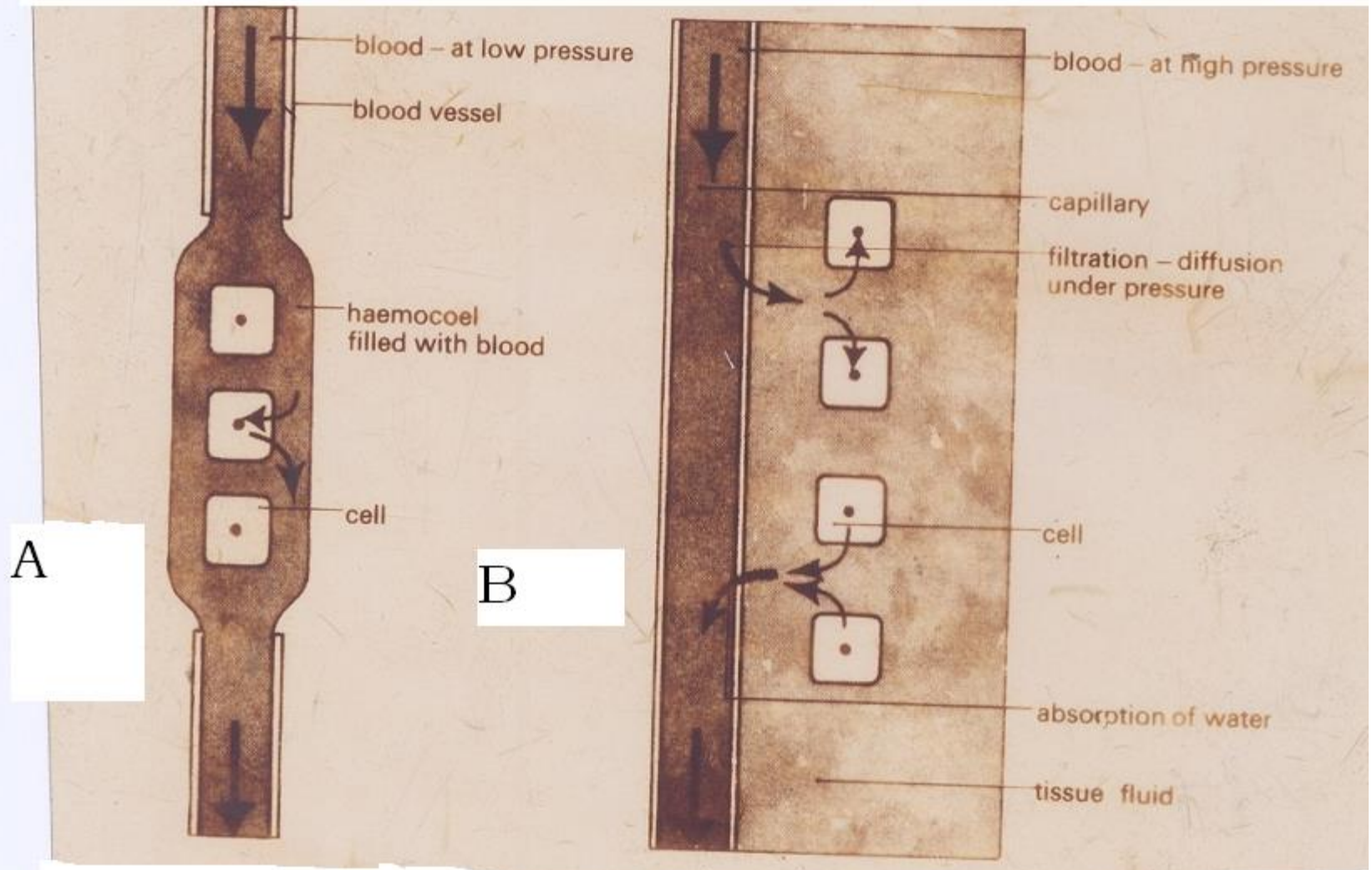
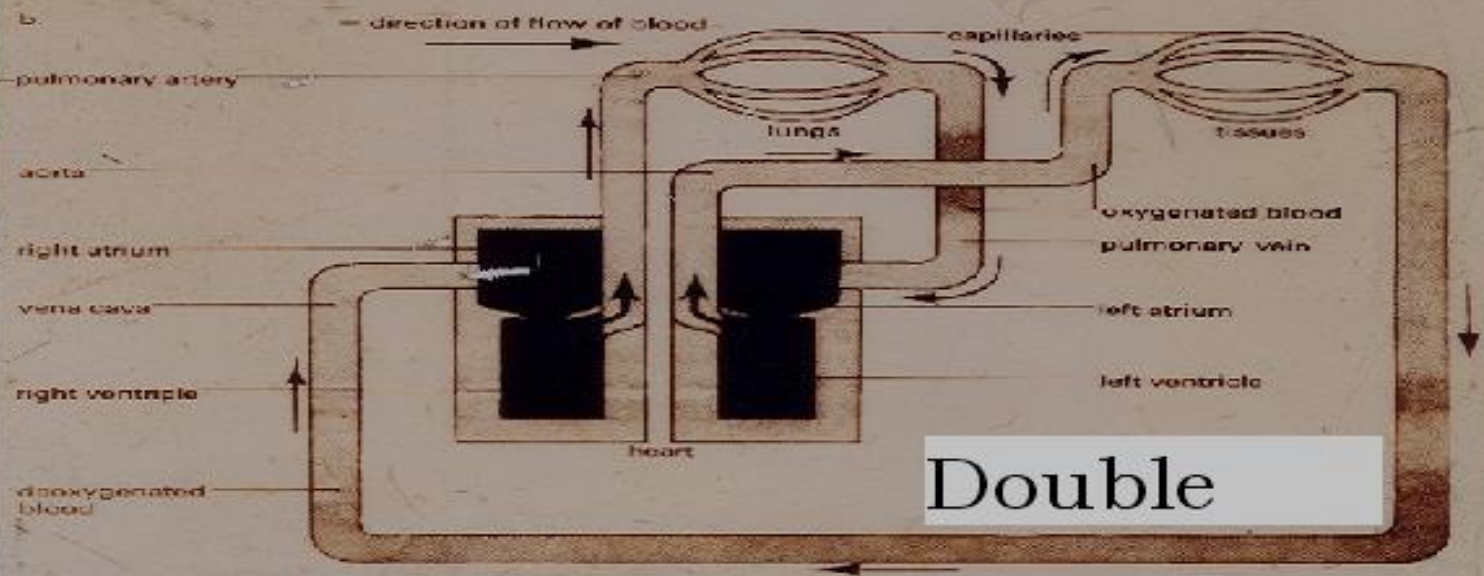
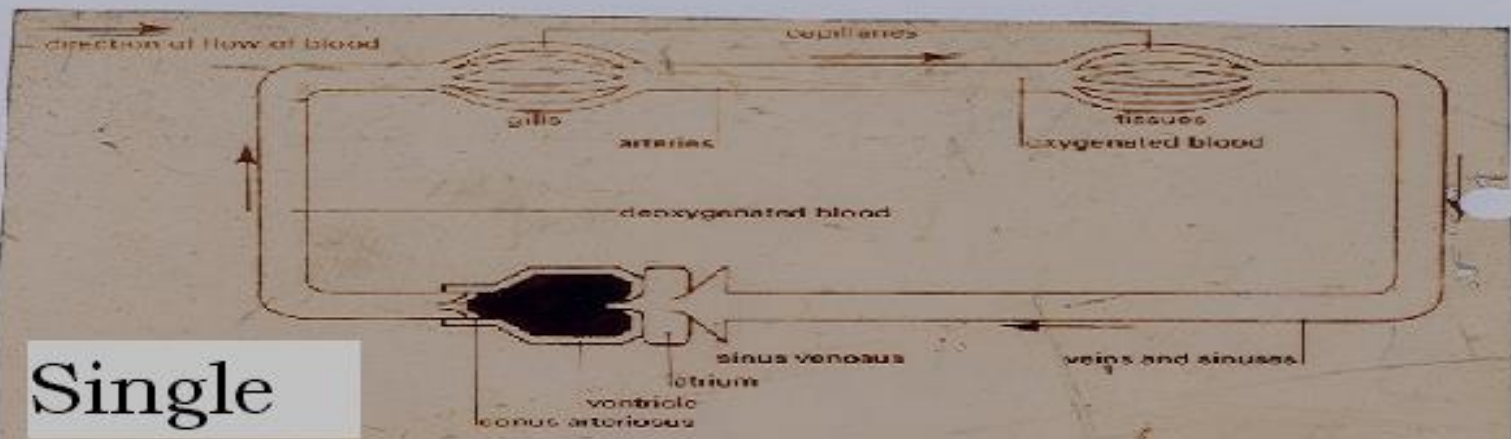


Fig..Open (A) and closed(B) circulations

Single



Double

Fig..Single and double circulations

Movement of Blood

- LV → Arteries of systemic (greater) circulation → Capillaries → Venules and veins → RA → RV → Pulmonary (lesser) circulation → LA
- General Setup of the CVS (fig next..)

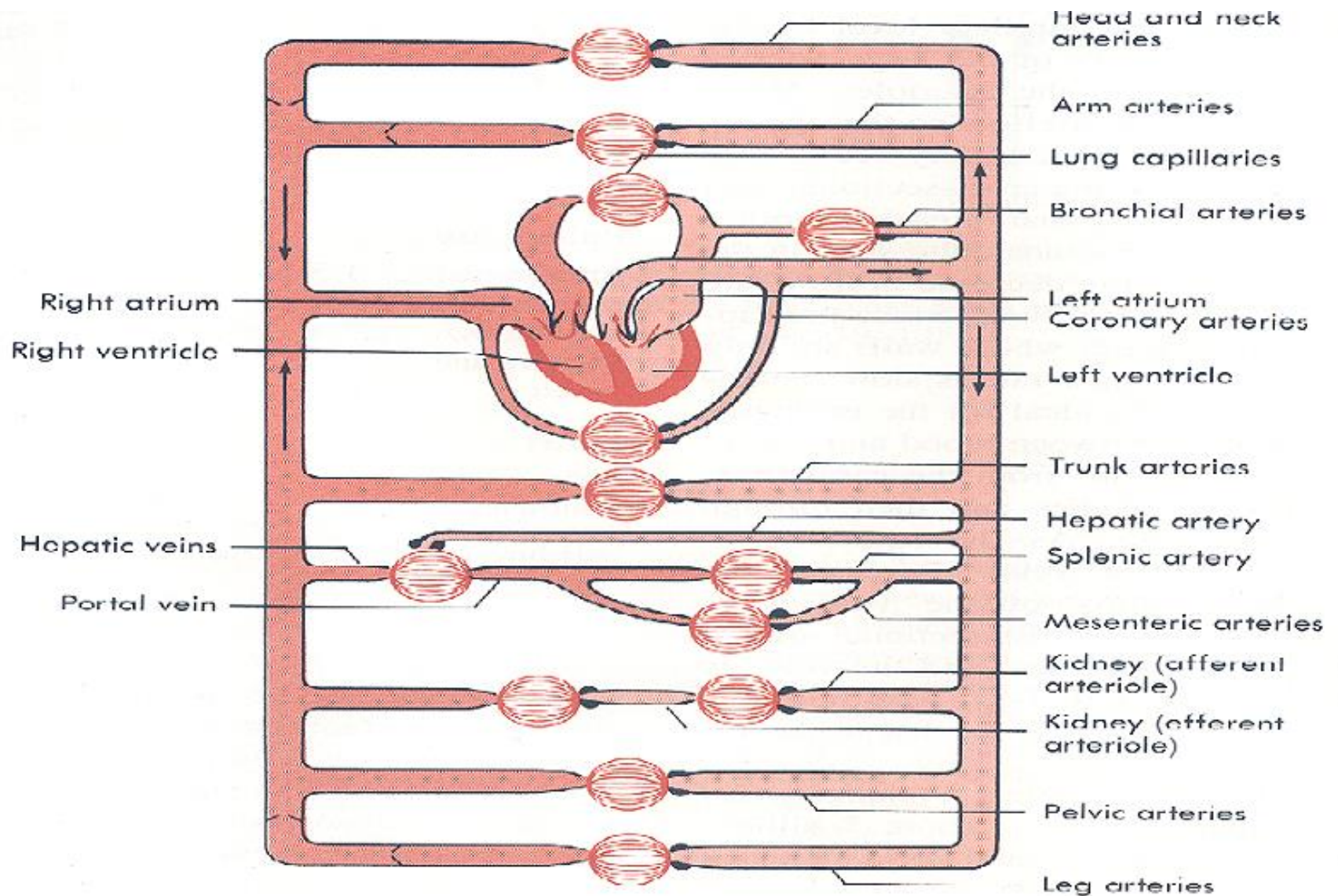


Fig.. Parallel and series arrangement of vessels of CVS

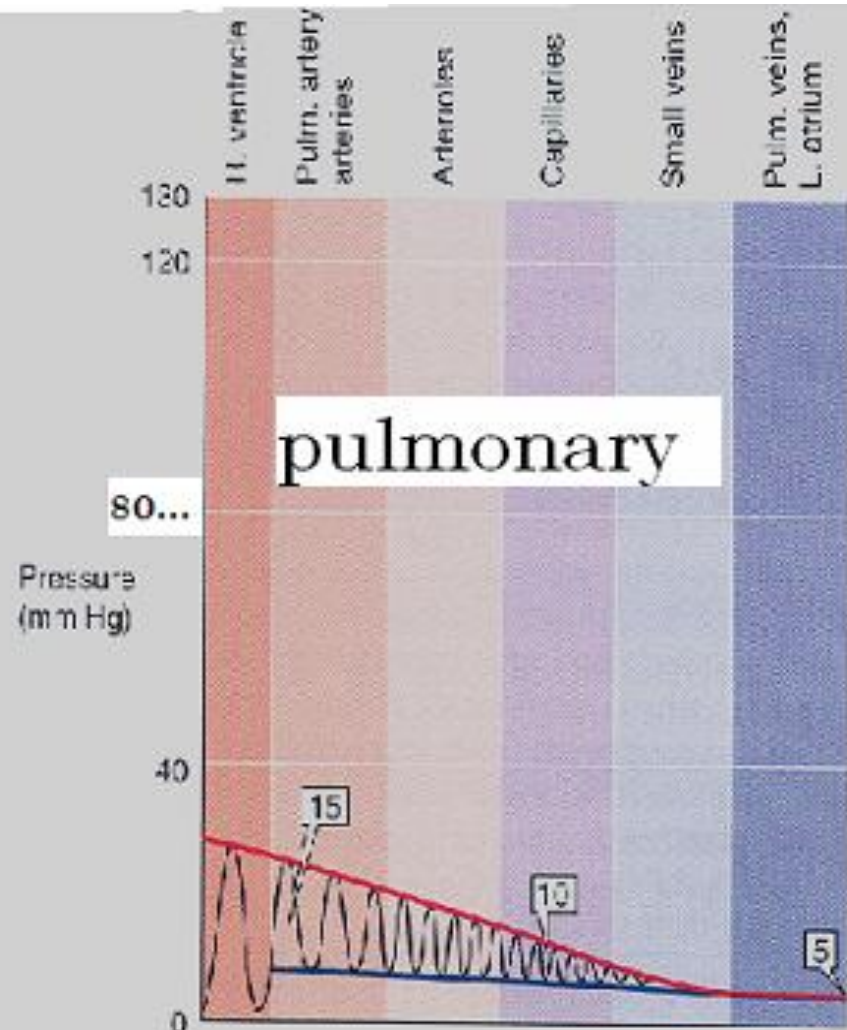
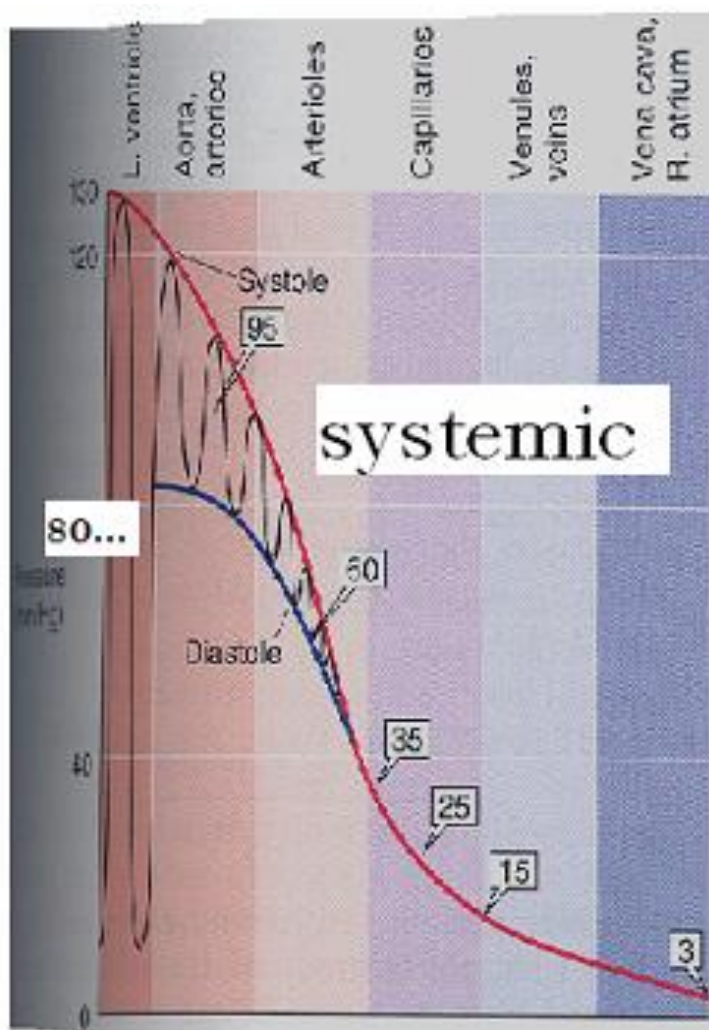


Fig..High and Low Pressure Systems of the CVS

The Functions of CVS

1. Transport & distribute essential materials (subs) to all cells of the body.
 - Bringing and supply O₂ & nutrients.
2. Removal of CO₂, metabolic by-products.
3. Maintenance of an optimum pH.
4. Regulation of the body temperature and blood flow.
5. Vital role in reproduction-hydraulic mechanism for penile erection.
6. Contribution to defense mechanisms by delivering antibodies, platelets & leucocytes to affected areas of the body.
7. Hormonal communication & function [[secret ANP](#)].

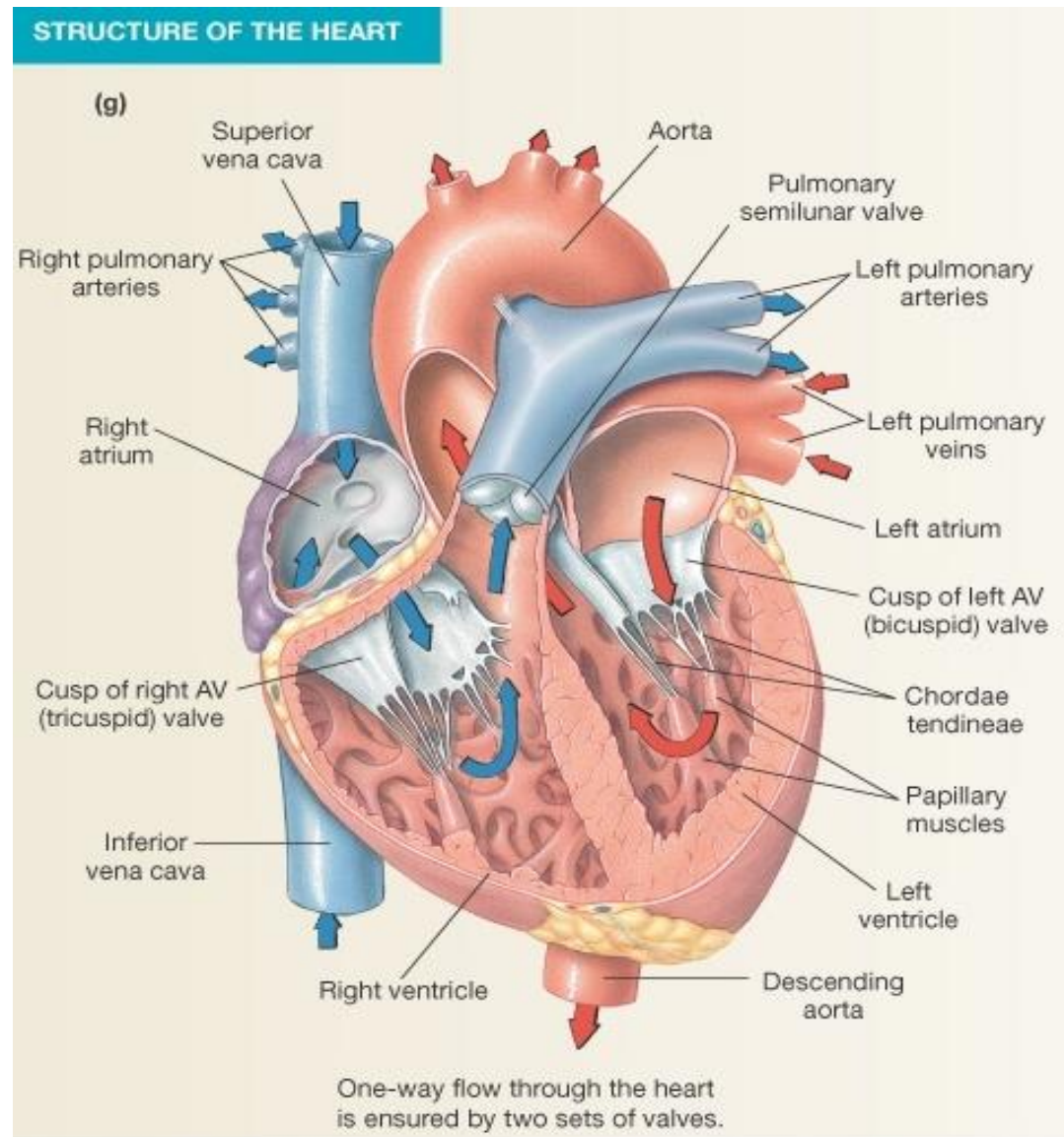
Components of the CVS

□ The CVS composed of:

- **Heart:** - pumping center.
- **Blood vessels:**
 - ♦ **Arteries** - distributing system
 - ♦ **Capillaries (MICROCIRCULATION)** - exchange system
 - ♦ **Veins** - Blood reservoirs and avenues for return of blood to the heart
- **Blood:** - circulating connective tissue.

The Heart

- @ Heart is the hollow, muscular organ that plays a **central pumping role**.
- @ Has a **dual pump** that drives blood in **two serial circuits**, the **systemic** & **pulmonary** circulations,
 - @ Receives blood from the rest of the body through the vena cavae (fig..).
- @ **Located** in the center of the thoracic cavity (**mediastenum**).



Gross Functional Structure of The Heart

- ✚ **Size:** The heart has the size of a clenched fist, weighs about 320gm in males (0.5% of body wt) and about 250 gm in the female (0.45% of body weight).
- ✚ **Parts:** The heart contains **four chambers** (atria and ventricles) with structures including valves, chordae tendineae, papillary muscle, trabeculae carneae, and blood vessels (fig..below).

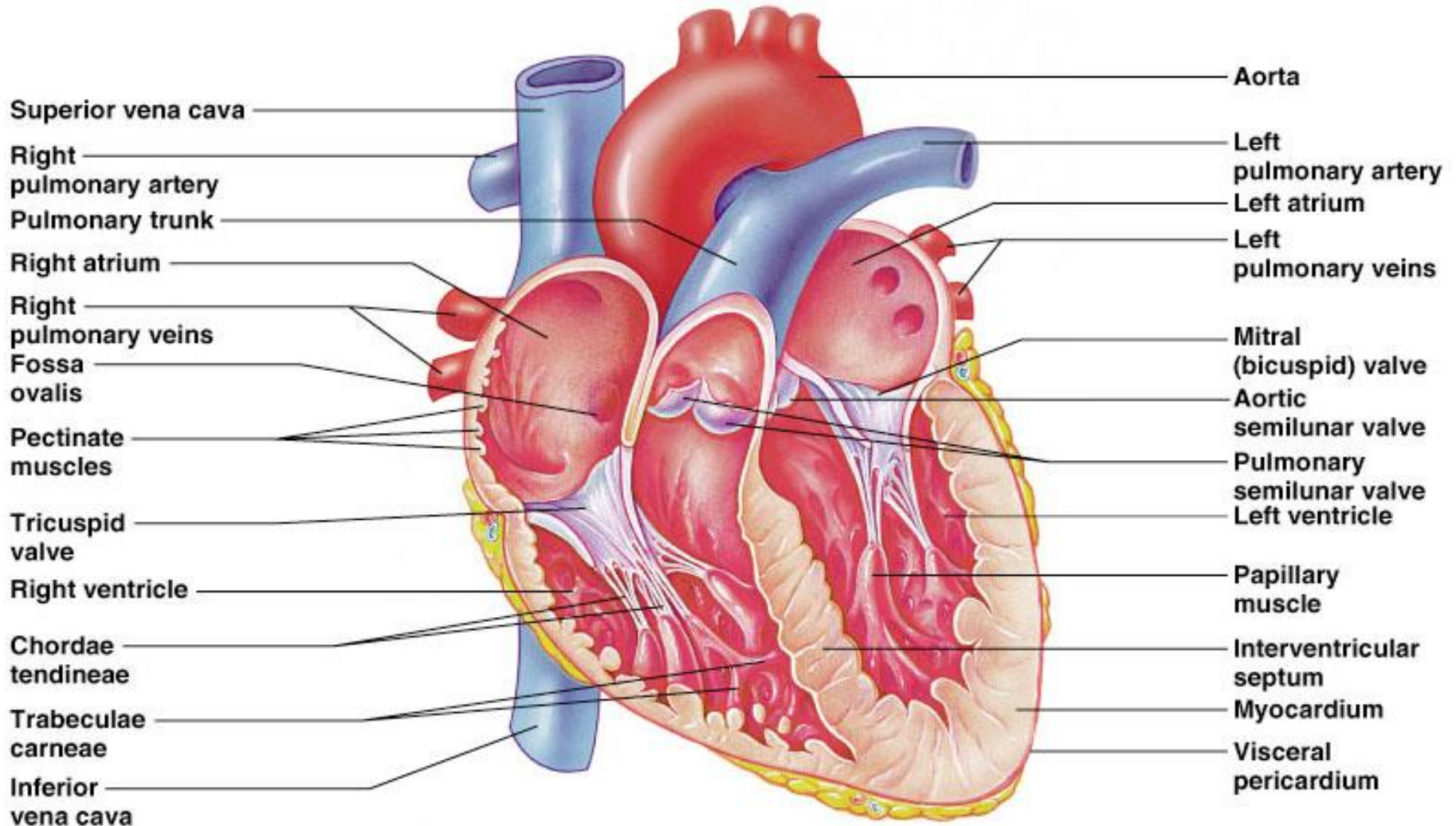


Figure. The diagram of the Heart

Surfaces/Layers

- **Outer surface: Pericardium** (fig..below)
 - Is a fibrous closed sac investing the entire heart and cardiac portion of great vessels.
 - It contains:
 - i. **visceral layer (epicardium)** immediately lining the outer surface of the heart and
 - ii. **parietal layer** forming outer lining of the pericardial sac containing a small amount of **fluid** for lubrication - facilitating the continuous movement of the enclosed heart.
- **Inner surface** of atria & ventricles (myocardium) is covered with **endocardium**, an endothelial connective tissue extending over valves and lining cavities of heart.

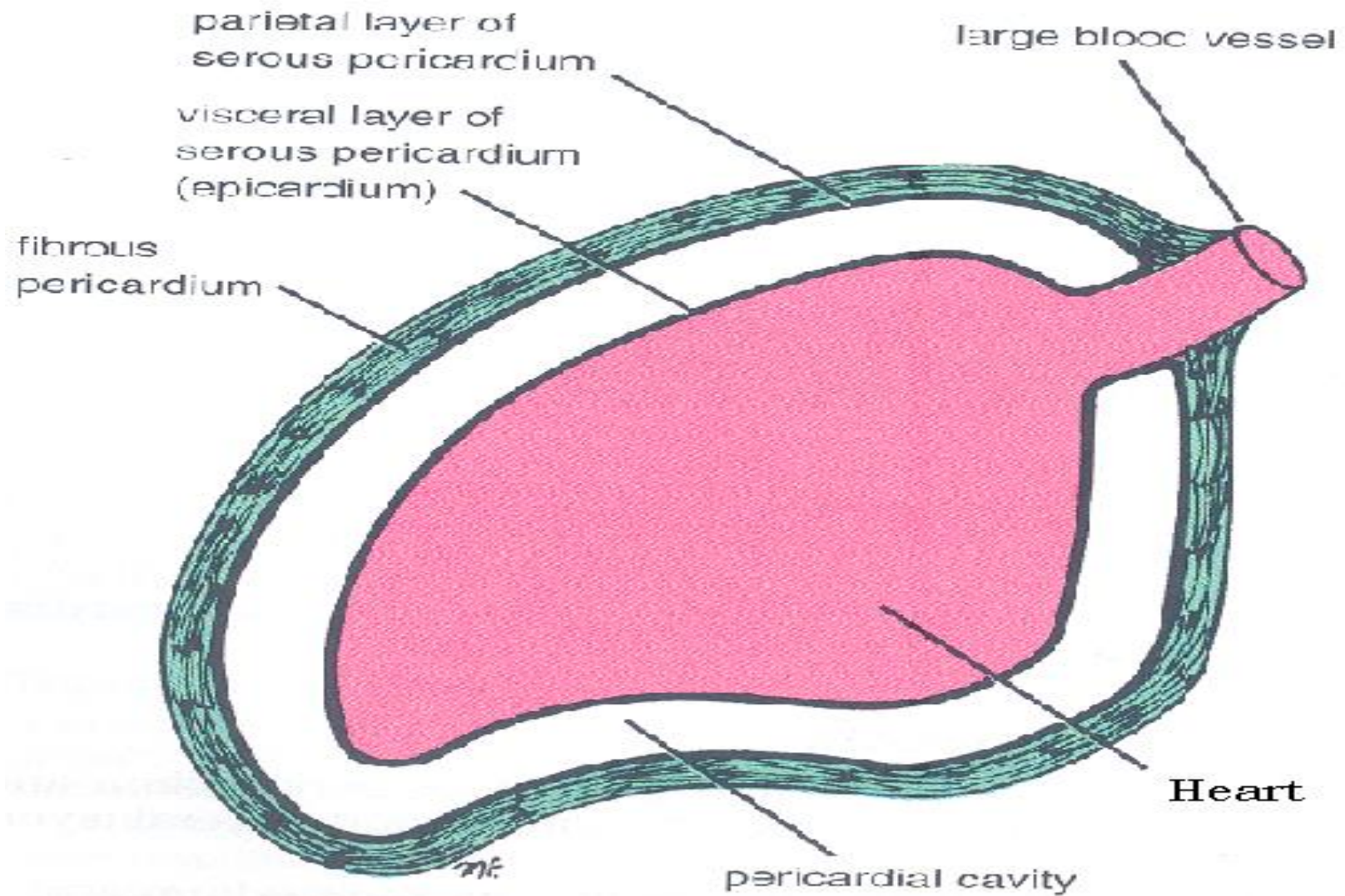


Fig..Different layers of the pericardium

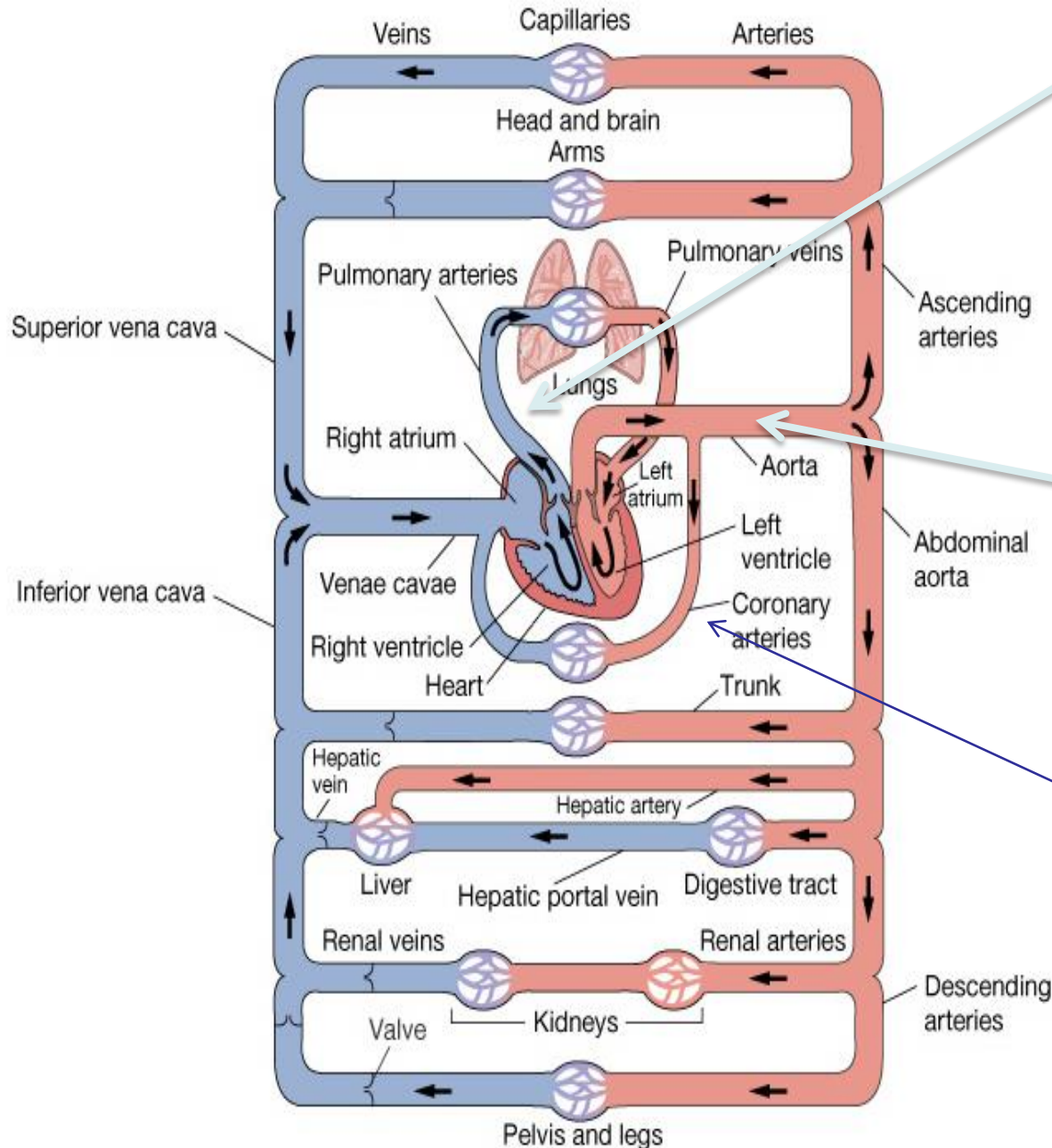
Divisions of the Circulatory System

- In the CVS, blood passes through two (double) circulations:
 - ✓ **Systemic circulation**
 - ✓ **Pulmonary circulation**
- The two circuits are arranged in **series**: the output of one becomes the input of the other.
- **Left side of the heart** pumps **oxygenated blood** into the systemic circulation to all tissues of the body **EXCEPT** the alveoli of lungs.
- **Right side of the heart** pumps **deoxygenated blood** into the pulmonary circulation to the alveoli of the lungs.

The left heart, systemic arteries, peripheral/systemic capillary, and systemic veins make up the
systemic circulation

The right heart, pulmonary artery, lungs, pulmonary capillaries, and pulmonary veins are collectively referred to as the
pulmonary circulation

Pathway of Blood via the Heart & Lungs



Pulmonary circulation:

1. Low Resistance

2. Low Pressure area (25/10 mmHg)

Systemic circulation:

1. High Resistance

2. High Pressure zone (120/80 mmHg)

5% of CO

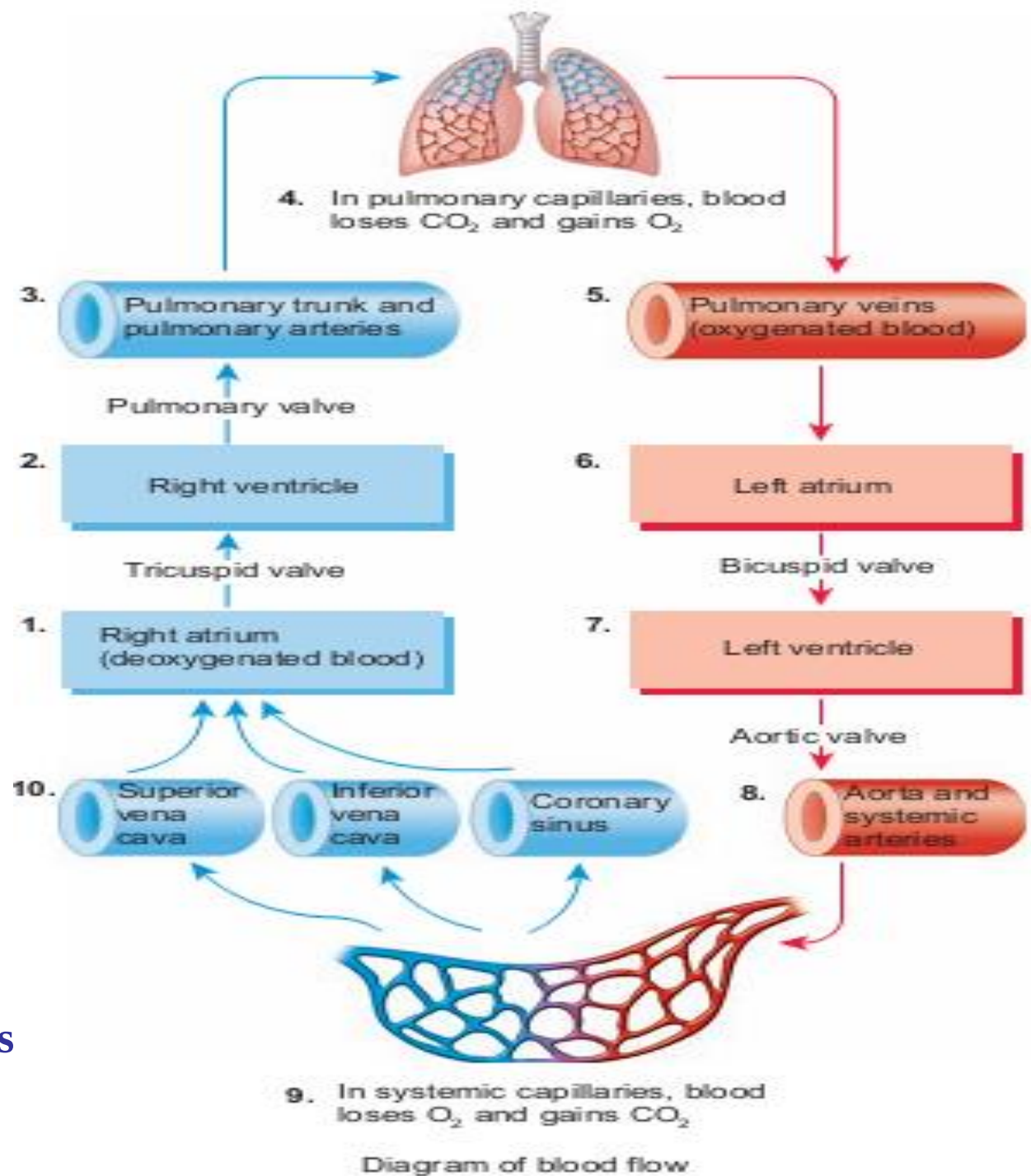
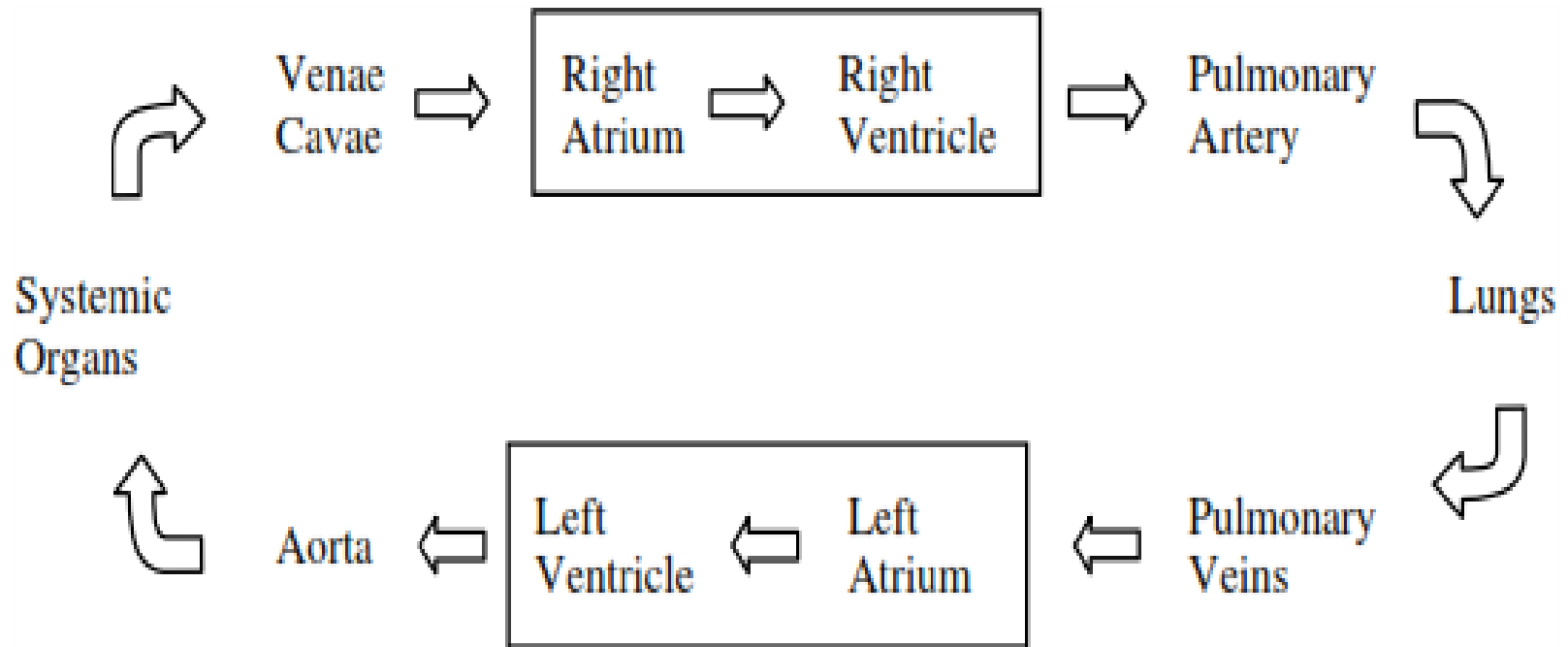


Figure. Systemic and Pulmonary Circulations

Pathways of Blood Through the Heart & Lungs & Valves

- Ⓢ Right atrium → tricuspid valve → right ventricle → pulmonary semilunar valve → pulmonary arteries → Lungs = Pulmonary circulation.
- Ⓢ Left atrium → bicuspid valve → left ventricle → aortic semilunar valve → aorta = Systemic circulation.
- Because the *ventricles pump blood under higher pressure* over *greater distances*, their walls are **thicker**.
- The **left ventricle** pumps blood into *great distances* to all other parts of the body at *higher pressure*, and *resistance to blood flow is larger*.
- Therefore, *the left ventricle works much harder* than the right ventricle to maintain the same rate of blood flow.

Table: Route of Blood Flow through the Heart



In summary, the heart is a single organ consisting of two pumps; the right heart delivers blood to the lungs & the left heart delivers blood to the rest of the body. Both pumps work simultaneously.

Blood Flow After The Heart

Aorta-Arteries-Arterioles-Capillaries-Venules-Veins-Vena Cava.

Pathways of blood flow

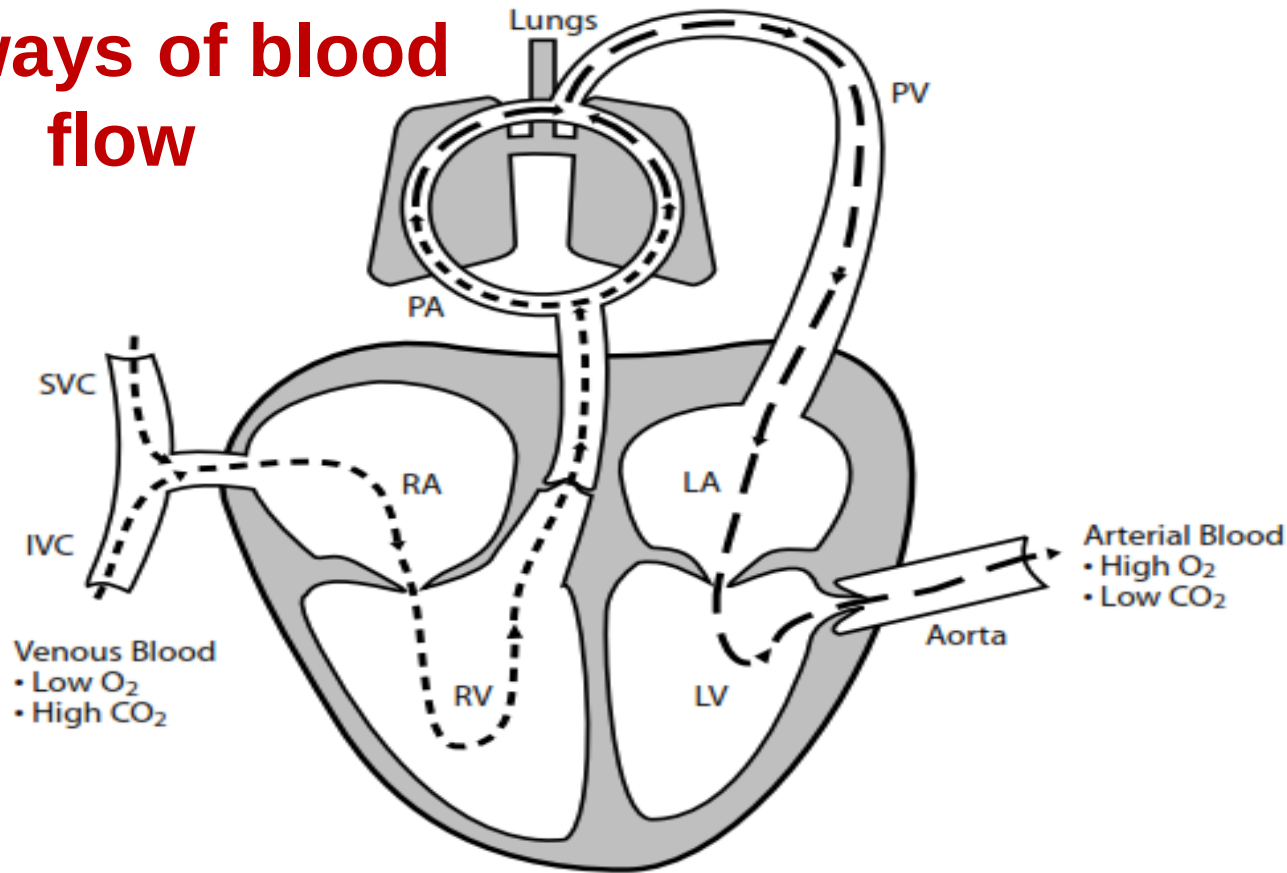
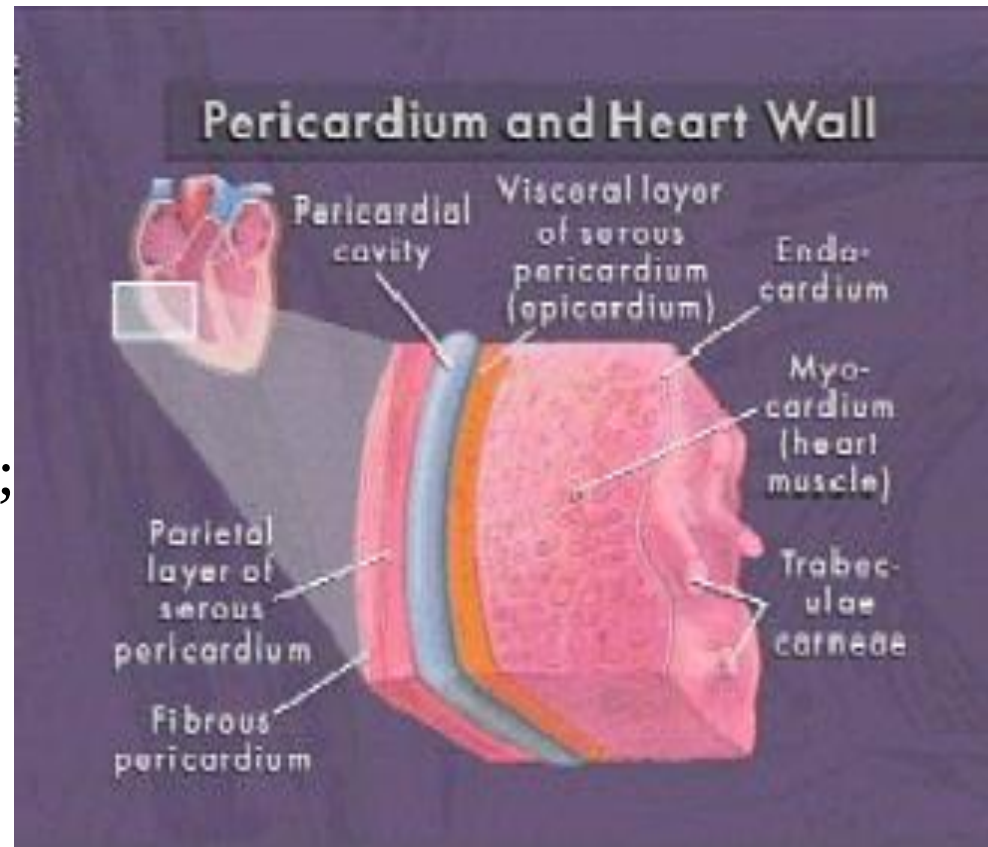


Figure. Route of blood flow through the heart. Systemic blood returns to the heart by way of the superior (SVC) & inferior (IVC) venae cavae. This blood, which is **low in O₂ & high in CO₂**, enters the right atrium (RA) & then the right ventricle (RV). The RV pumps the blood through the pulmonary artery (PA) to the **pulmonary circulation**. It is **within the lungs that gas exchange takes place**. Next, this blood, which is **high in O₂ and low in CO₂**, returns to the heart by way of the pulmonary veins (PV). It enters the left atrium (LA) & then the left ventricle (LV). The LV pumps the blood through the aorta to the **systemic circulation** & the **peripheral tissues**.

Histology of the Heart

- A *double layered closed sac* that surrounds the heart is known as **pericardium**.
- **Pericardial fluid** in pericardial cavity reduces friction in the beating heart.
- **Wall of the heart** has *3 layers*:
 1. **Endocardium**: *innermost* layer; epithelial tissue that lines the entire circulatory system.
 2. **Myocardium**: **thickest** layer; consists of the cardiac muscles.
 3. **Epicardium**: **thin**, external membrane around the heart.



Cardiac Muscle (Myocardium)

- The heart is a **strong muscular pump** that contracts and relaxes in all life.
- **Myocardium:-** *composed of 2 types of cardiac muscles:*
 - ❖ **Atrial muscle:-** low pressure reservoirs.
 - ❖ **Ventricular muscle:-** pumping chambers.
 - ❖ **Specialized excitatory & conductive muscle fibres.**
- *Atrial & ventricular muscles* are the **contractile components**.
- *Specialized excitatory muscles* are **weakly contractile** but qualified for **autorhythmicity**.

Cardiac Muscle (Myocardium)

❑ In cardiac muscle there are 2 types of cells: (based on functional criteria)

1. Autorhythmic (Automatic) cells

- Are self-stimulating cells without any external stimulation.
 - Have specialized excitatory & conductive systems & having 2 functions:
 - **Generate** excitatory impulse & **send** to working myocardial cells.
1. **Specialized excitatory cells** → → They generate rhythmic cardiac impulse.
 - ✓ Are pacemakers
 2. **Conductive muscle fibres** → → They conduct cardiac impulse
 - ✓ Are conductive systems

2. Contractile cells (Atrial & Ventricular muscles)

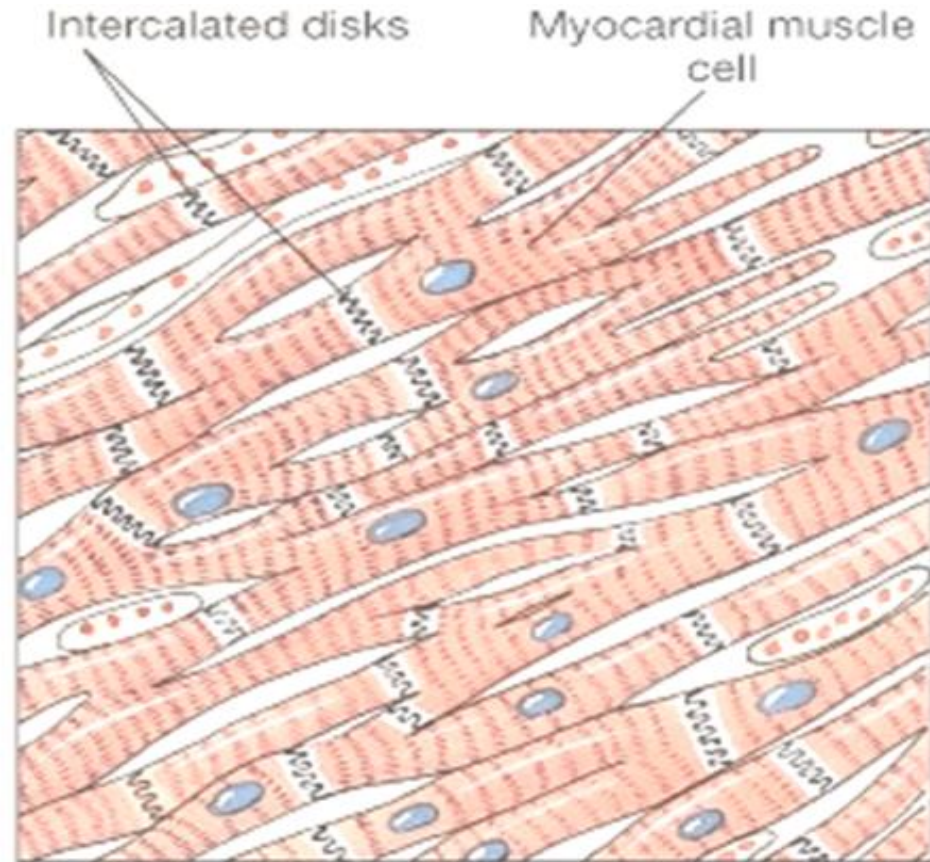
- Are working myocardium
 - Do mechanical work of pumping
 - ✓ They contract when stimulated.

Structural properties of cardiac muscle cells

- Cardiac muscle tissue forms two **functional syncytia** or units that contract in a **synchronous or wavelike** fashion important for the pumping action of the heart.
 - The **atrial syncytium** constitutes walls of atria.
 - The **ventricular syncytium** constitutes walls of ventricles.
- APs are conducted from atrial syncytium to ventricular syncytium thru the AV bundle.

Cardiac Muscle

- ✦ Adjacent cardiac muscle cells join **end-to-end** at structures called the *intercalated discs*.
- ✦ **Intercalated discs**: contain *two types* of specialized junctions:
 - desmosomes** - hold the cells tightly together.
 - gap junctions** - permit spread of AP (electrical impulse) from one muscle cell to another.



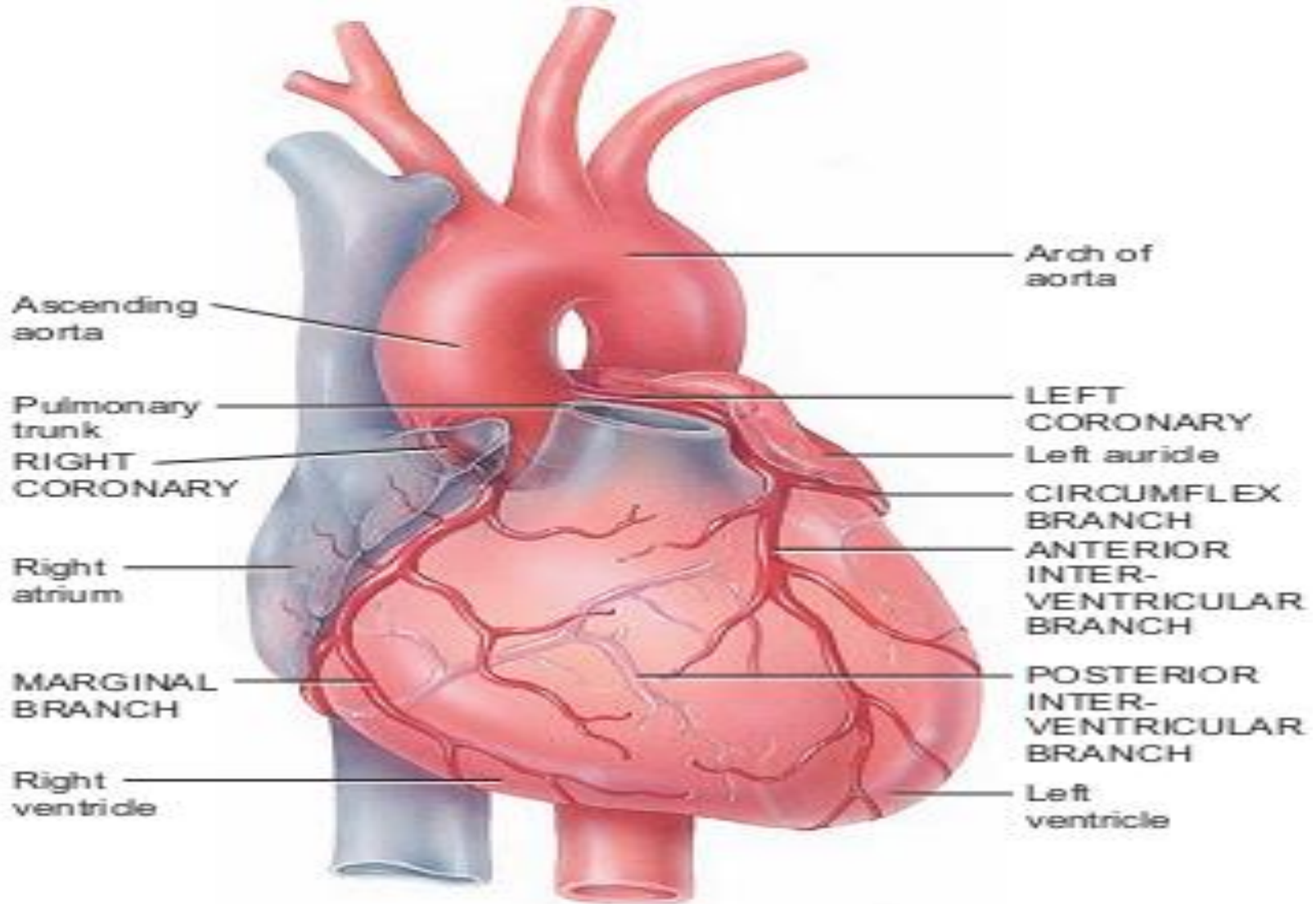
Myocardial muscle cells are branched, have a single nucleus, and are attached to each other by specialized junctions known as intercalated disks.

❖ *A single adequate stimulus for AP in one myocyte results in the rapid spread of excitation to all myocytes via gap junctions.*

Coronary Circulation: Blood supply to the heart

- The heart receives arterial blood from *coronary artery*, which is the branch of ascending aorta.
- **While the heart is contracting**, little blood flows in coronary arteries b/c they are closed.
- **When the heart relaxes**, high blood pressure in aorta propels blood through coronary arteries, into the capillaries, and then into the coronary veins.
- **Resting coronary blood flow** is **250 ml/min, 5% of CO**.
- *Blood flow decreased in the coronary arterial diseases, leading to Angina pectoris, myocardial infarction (MI).*

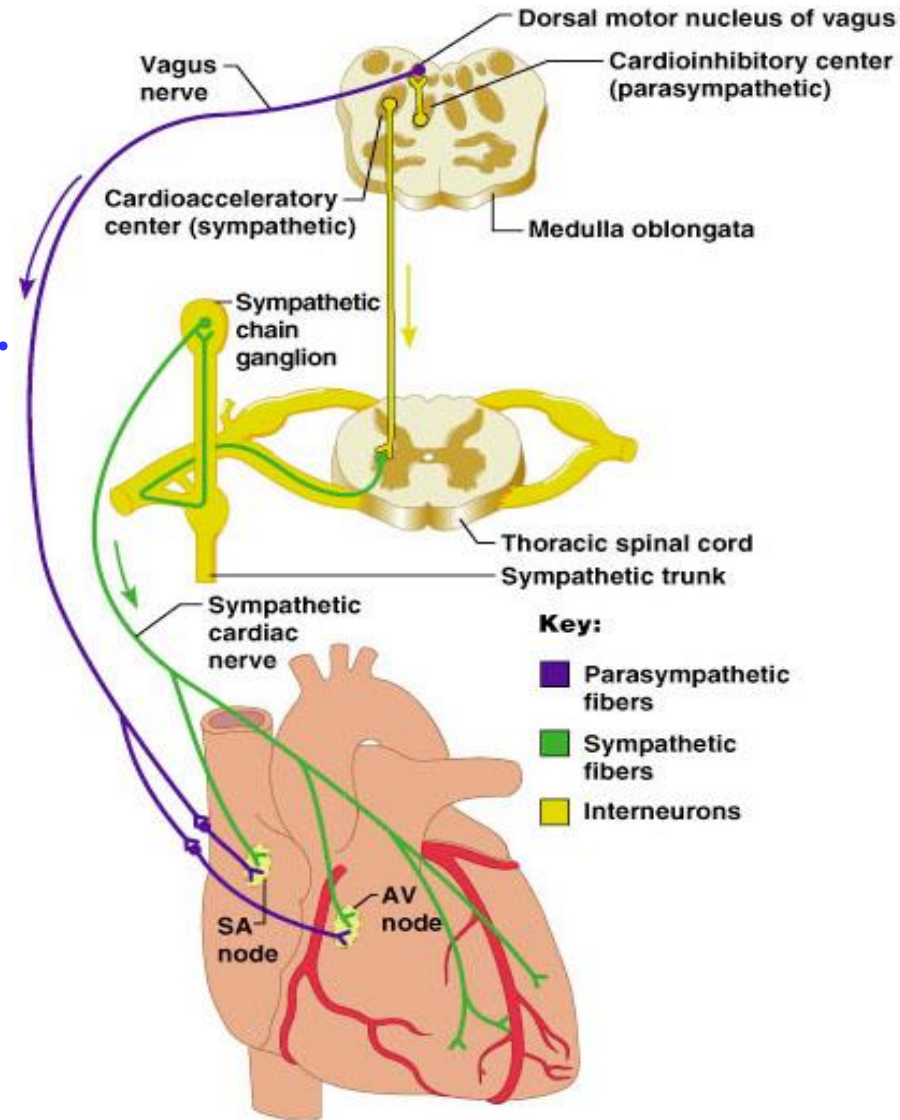
Coronary circulation, cont'd



Anterior view of coronary arteries

Innervation of the Heart

- Heart has **dual autonomic nerve** supply from both SNS & PNS.
- **Sympathetic nerve** to the heart is controlled by the **medullary vasoconstrictor/cardioaccelerator center**.
- **Sympathetic fibers** supply the **atria**, the **ventricles** & the **nodal areas**.
- **Parasympathetic fibers** supply to the heart is controlled by the **vasodilator/ cardioinhibitory center**.
- **Parasympathetic fibers** supply the **atria**, **SA-node** & **AV-node**.
- *The ventricles are not supplied by the vagus nerve.*



Innervation of the Heart, cont'd

1. Parasympathetic supply:

- Preganglionic parasympathetic fibers from cardio-inhibitory center in medulla → vagal fibers to relay in terminal ganglia in wall of atria → short postganglionic fibers → SA-node and AV-node.
- The right vagus has a strong influence on SA node, while the left vagus has dominant effect on AV node.
- Ventricles are not supplied by vagus nerve.

Innervation of the Heart, cont'd

2. Sympathetic supply:

- Innervation of the heart by the sympathetic NS receives signals from the medulla through the bulbo-spinal pathway synapsing with its pre-ganglionic fibre in the intermediolateral gray column of the thoracic cord.
- Receptors signaling to sensory components of the cranial nerves IX and X are located in the aortic arch and the carotid sinus.

Effects of sympathetic and parasympathetic stimulation on the heart and the circulation

	<i>Sympathetic stimulation</i>	<i>Parasympathetic stimulation</i>
Heart		
Heart rate	Increased	Decreased
Contractility	Increased	Decreased
Conduction	Increased	Decreased
Splanchnic vessels	Constricted	
Coronary arteries		
Direct effect	Constricted	Dilated
Demand-mediated effect	Dilated	Constricted

Properties of Cardiac Muscles

- *The cardiac muscles have 4 physiological properties, which are essential for the function of the heart as the central pumping organ.*
- These are
 - ✓ **Autorhythmicity:** ability to generate cardiac impulse.
 - ✓ **Excitability:** ability to receive & respond to a stimulus.
 - ✓ **Conductivity:** ability to transmit cardiac impulse.
 - ✓ **Contractility:** ability to contract forcefully when stimulated by an action potential.

Authorhythmicity

- It is the ability of the heart to generate cardiac impulse & to beat regularly independent of any external stimulus.
- **An inherent & rhythmical electrical activity** is the reason for the heart's lifelong beat.
- **Source** of this electrical activity is a network of **specialized cardiac muscle fibers**, called **autorhythmic fibers** (auto- = self), because they are self-excitabile.
- These **specialized excitatory & conductive muscle system** of the heart have two functions:
 1. **Generate rhythmic cardiac impulse (act as pacemaker)**
 2. **Conduct the cardiac impulse**

Specialized Excitatory & Conduction System of the Heart

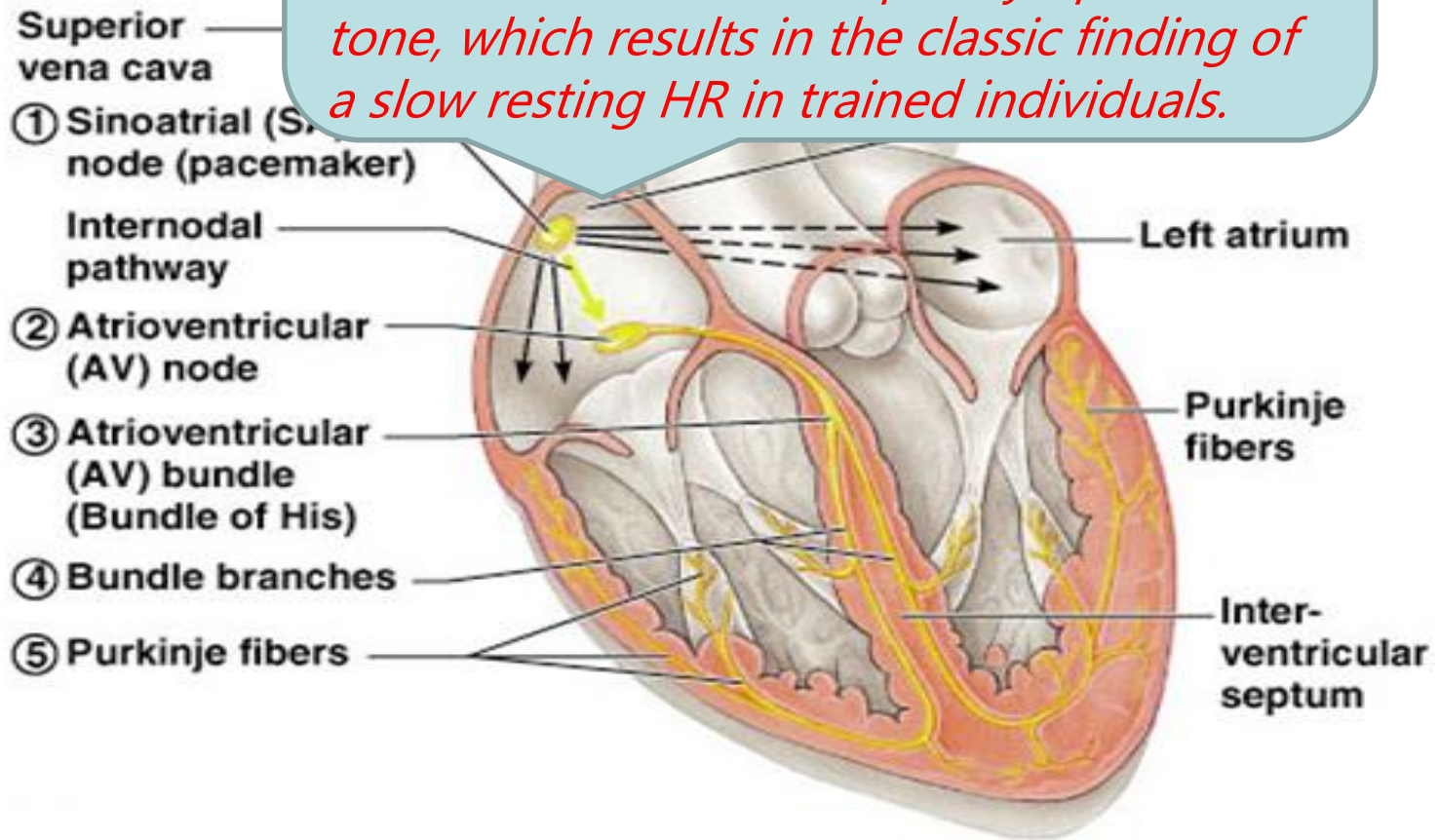
- 1. Sino-atrial node (SA-node):** a normal *rhythmical self-excitatory impulse* is generated in the SA-node.
- 2. Internodal pathways:** conduct impulse from SA-node to AV-node
- 3. Atrioventricular node (AV-node):** in which impulse from the atria *delayed* to be conducted into the ventricles.
 - So, it is the *site of nodal delay*.
 - The only *electrical communication* b/n atria & ventricles.
- 4. Atrioventricular bundle (bundle of His):** conducts impulse from the atria to the ventricles.
- 5. Purkinje fibers:** conduct cardiac impulse into the ventricles.

Excitatory & conductive system, cont'd

- Various autonomic influences affect the heart's electrical activity:

- ▶ SA node =
- ▶ AV node,
- ▶ Bundle br

The intrinsic rate of SA node firing is about 100 beats/min. Normal parasympathetic tone reduces the SA node firing rate to about 70 beats/min at rest. Endurance athletes have enhanced parasympathetic tone, which results in the classic finding of a slow resting HR in trained individuals.



Sinoatrial node (Pacemaker)

- Located on the posterior wall of the R-atrium inferior to the opening of the SVC.
- The **sinoatrial (SA) node** is the **normal pacemaker of the heart** and the origin of each normal heartbeat.
- SA-nodal fibres have a *special capability to discharge repetitive & rhythmic cardiac impulses.*
- It generates impulse at a frequency of **80-120/min.**
- Determines the **frequency of heart beat.**
- The **RMP** of SA-nodal fibers is **-55 to -60 mV**, but there is *no stable period of RMP.*
- SA-nodal discharge **decreases with age.**

Conduction Velocity of AP in the Cardiac Muscles

Structures	Conduction velocity (m/s)
SA-node	0.05
Internodal fibers	1.0
Atrial muscle	0.3
AV-node	0.05
AV-bundle	1.0
Purkinje fibres	4.0
Ventricular muscles	1.0

Factors Affecting Autorhythmicity

1. ANS:

- **SNS**:- Innervates SA-node, AV-node, conductive and contractile muscles. Increase rhythmicity, **+ve chronotropic effect**.
- **PNS**: - Innervates SA-node, AV-node & atrial muscles.
 - Inhibits autorhythmicity, **-ve chronotropic effect**.
 - **Strong vagal stimulation** leads to **cardiac arrest**.

2. Temperature: - $\uparrow$ BT by 1°C , $\uparrow$ HR by 20 beats/min. - $\uparrow$ BT, increases SA-nodal discharge.

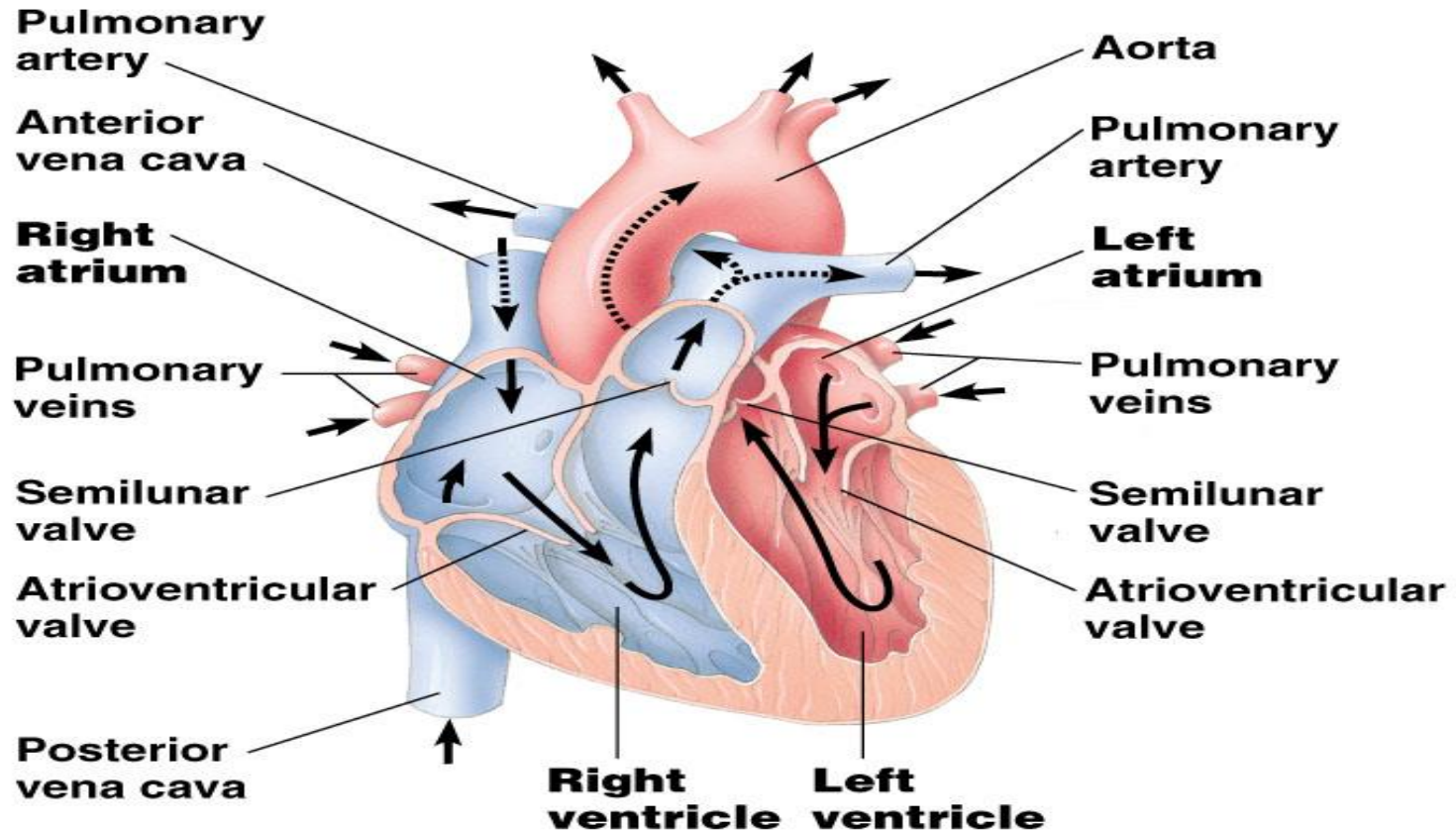
3. Drugs:- AD, ND & T3/T4, increase the cardiac rhythmicity. - They have **+ve chronotropic effect**.

- **Cholinergic drugs**: - Like Ach & pilocarpine, decrease HR.
 - They have **-ve chronotropic effect**.

Heart Valves & Circulation of the Blood

- ⌚ As each chamber of the heart contracts, it pushes a volume of blood into a ventricle or out of the heart into an artery.
- ⌚ *Valves open & close* in the response to **pressure changes** as the **heart contracts & relaxes**.
- ⌚ *Each of the four valves* ensure the **one-way, or forward, flow of blood** by opening to let blood through & then closing to prevent its backflow.

- **Two sets of valves** in the heart maintain the *one-way flow of blood* as it passes through the heart chambers:
 - **Atrioventricular (AV) valves**
 - **Semilunar valves**



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Figure. Heart Valves & Blood Flow

The Cardiac Valves

- Cardiac valves are made up of **fibrous connective tissues**.
- *There are 4 cardiac valves:*

1. **Tricuspid valve (T-valve)**
2. **Bicuspid valve (M-valve)**
3. **Pulmonary semilunar valve (P-valve)**
4. **Aortic semilunar valve (A-valve)**

Function: prevent back flow of blood.

- **Atrioventricular valves** - are located b/n atria & ventricles.
- **Aortic & pulmonary valves** - are known as **semilunar valves** because they are made up of 3 *crescent moon-shaped cusps*.
- If the **AV valves are damaged**, the blood may **regurgitate** (backflow) into atria when ventricles contract.

Cardiac Valves, cont'd

- ⊕ All valves consist of **connective tissue** (not a cardiac muscle tissue) and, therefore, **open & close passively**.
- ⊕ Valves open & close in response to **changes in pressure**:
 - **AV valves**: open when pressure in atria is greater than pressure in ventricles (**during ventricular diastole**) & closed when pressure in ventricles is greater than pressure in atria (**during ventricular systole**).
 - **Semilunar valves**: open when the pressure in the ventricles is greater than pressure in the arteries (**during ventricular systole**) & closed when the pressure in the pulmonary trunk & aorta is greater than pressure in the ventricles (**during ventricular diastole**).
- **Chordae tendineae** (fine strong filaments **arising from the strong papillary muscles**) extend to valve cusps and are **attached to the free edges of atrio-ventricular valves**.
- **Papillary muscles** contract when ventricles contract.

The semilunar valves

- Constitute the **aortic and pulmonary valves** located at the exits of the right and left ventricles.
- **Open and close passively.**
- **Aortic valve is three-cusped** and allows blood to flow into the aortic tree and through their cusps to the left and right main **coronary arteries.**
- **Open** when pressure in the ventricles is greater than pressure in the arteries (i.e., **during ventricular systole**) and **close** when pressure in the pulmonary trunk and aorta is greater than pressure in the ventricles (i.e., **during ventricular diastole**).
- **Pulmonary valve** allows blood to flow into the pulmonary artery.

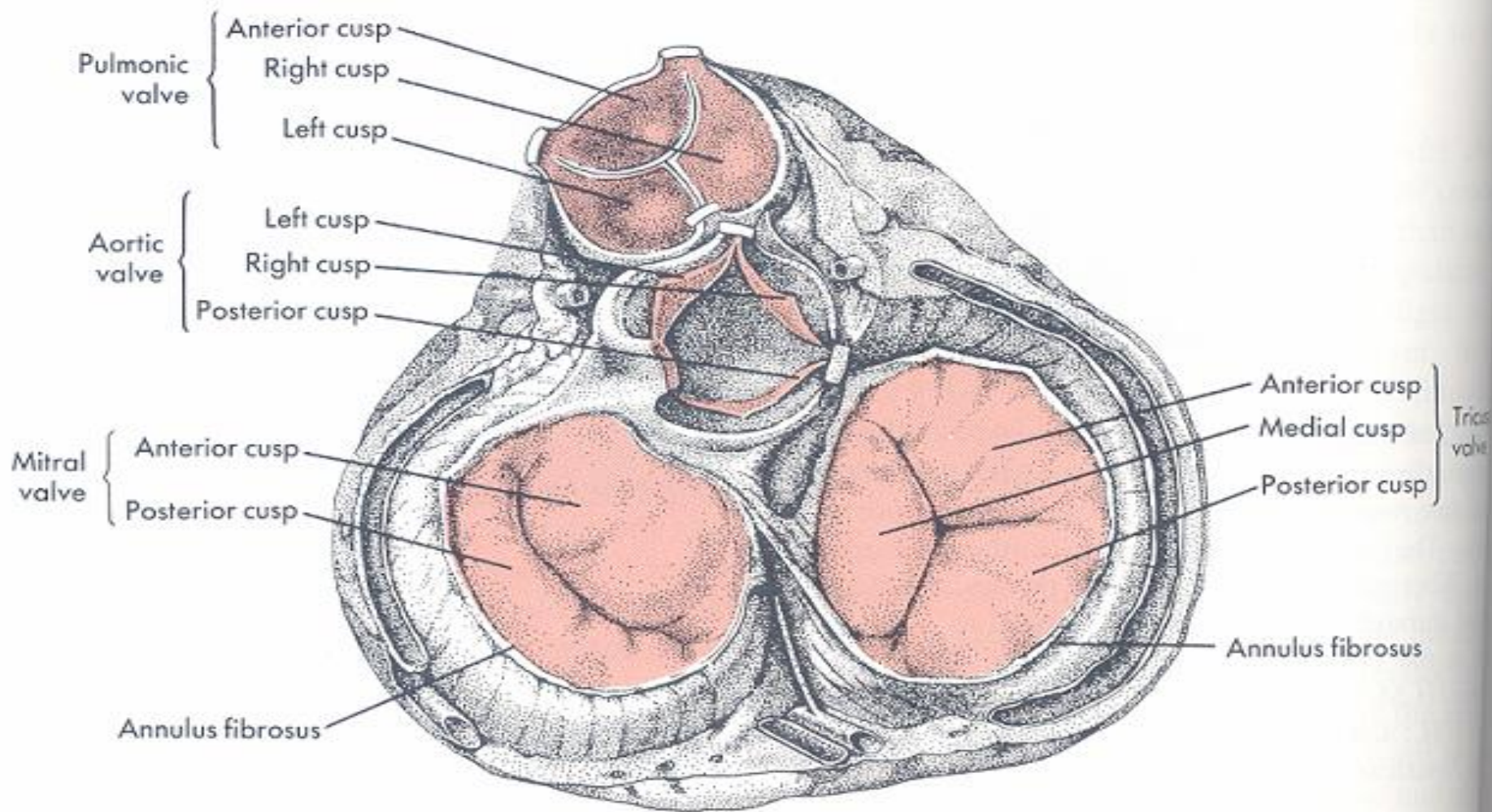


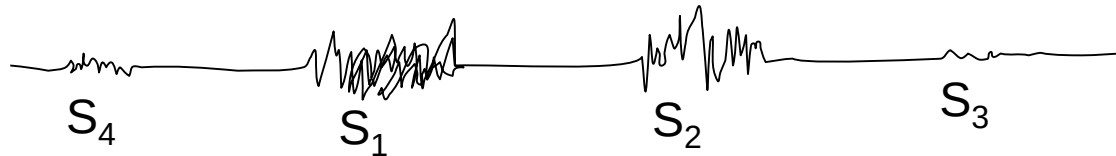
Fig. Four cardiac valves as viewed from the base of the heart. Note how the leaflets overlap in the closed valves.

Heart Sounds

- Heart sounds (lub-dub) are associated with closing of heart valves.
- 4-separate audible heart sounds (S_1 , S_2 , S_3 and S_4)
- Means of identification

1. **Auscultation:** - direct/immediate auscultation
- Stethoscope mediated auscultation

2. **Phonocardiography based recording**



S_1 : is always audible, has a LUB-sound

S_2 : is always audible, has DUB-sound

Continuous heart sound: Lub-Dub, Lub-Dub

Duration of Lub is shorter than that of Dub

S_3 : is audible in children & in adults during exercise

S_4 : is audible very rarely.

Heart Sounds, cont'd

S₁: First heart sound

- Cause: sudden closure of AV-valves.
- Timing: occurs at the beginning of ventricular systole.
- Best site of auscultation
 - ❖ 5th LIS (left intercostal space), at mid-clavicular line
 - ❖ 4th LIS, near the sternum

S₂: Second heart sound

- Cause: Sudden closure of semilunar valves
- Timing: occurs at the beginning of ventricular diastole
- Best site of auscultation
 - ❖ 2nd RIS, near the sternum
 - ❖ 2nd LIS, near the sternum

Heart Sounds (cont'd)

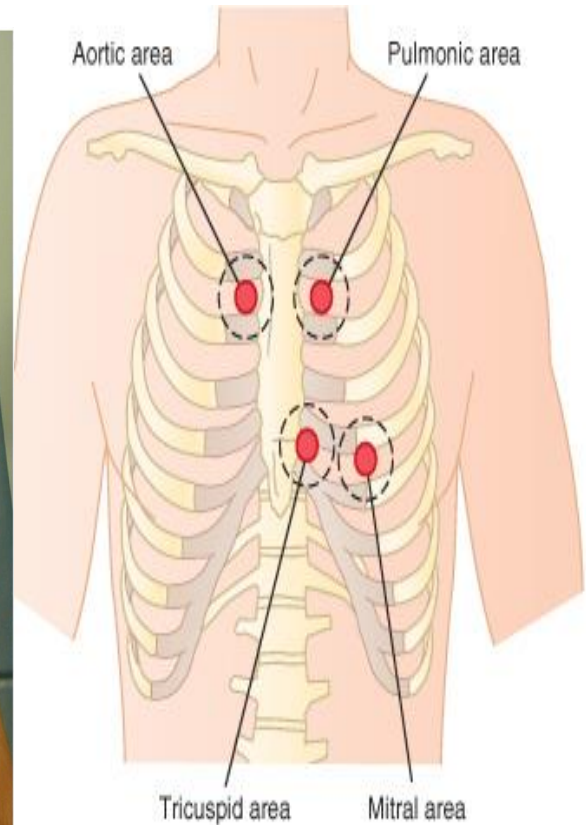
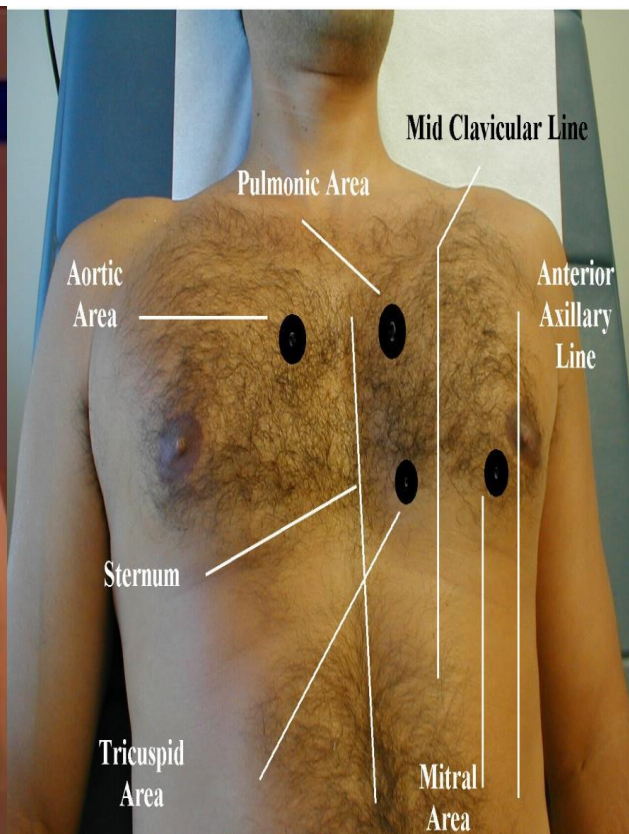
S₃: Third heart sound

- Cause: rapid filling of the ventricles with blood during ventricular diastole.
 - It is audible in children and in adults during exercise.
- Best site of auscultation: at the apex of the heart beneath the xiphoid process.

S₄: Fourth heart sound

- Cause: rapid ventricular filling during atrial systole
- Timing: during atrial systole
- Best site of auscultation: at the apex of the heart

Areas of Auscultation [areas of Listening]



Abnormal Heart Sounds

- ❑ **Abnormal heart sounds** are most commonly recorded when the heart valves are defective.
- ❑ Abnormal heart sounds are called **heart murmurs**.
- ❑ They are **two types**:
 - ⊕ **Systolic murmurs**: - AV-valve insufficiency
 - Semilunar valve stenosis
 - High CO (physiological)
 - ⊕ **Diastolic murmurs**: - Semilunar valve insufficiency
 - AV-valve stenosis
- Ⓢ **Valvular insufficiency** causes **regurgitation**.

Electrophysiology of the Heart

Phases & ionic bases of myocardial AP

- It has **5-phases**:

Phase-0: Rapid depolarization

Caused by rapid Na-influx

Phase-1: Early rapid repolarization

Caused by Cl^- influx

Phase-2: The **plateau**, a period of maintained depolarization.

Caused by slow Ca^{2+} influx

Phase-3: Repolarization

Caused by K^+ efflux

Phase-4: Complete repolarization

- RMP re-established.

Caused by Na-K ATPase

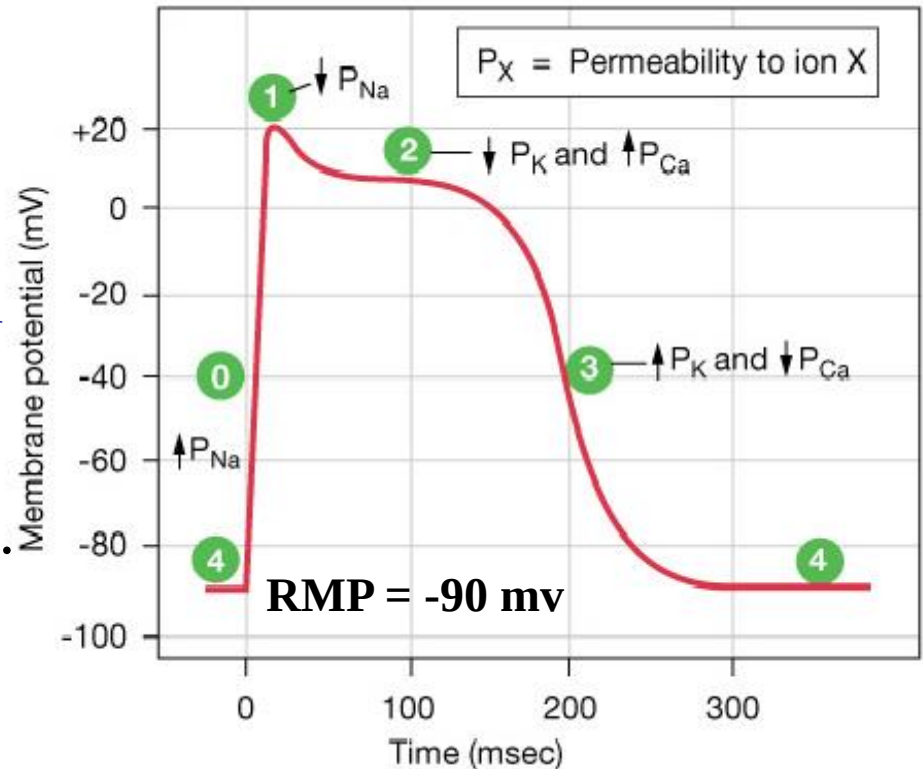
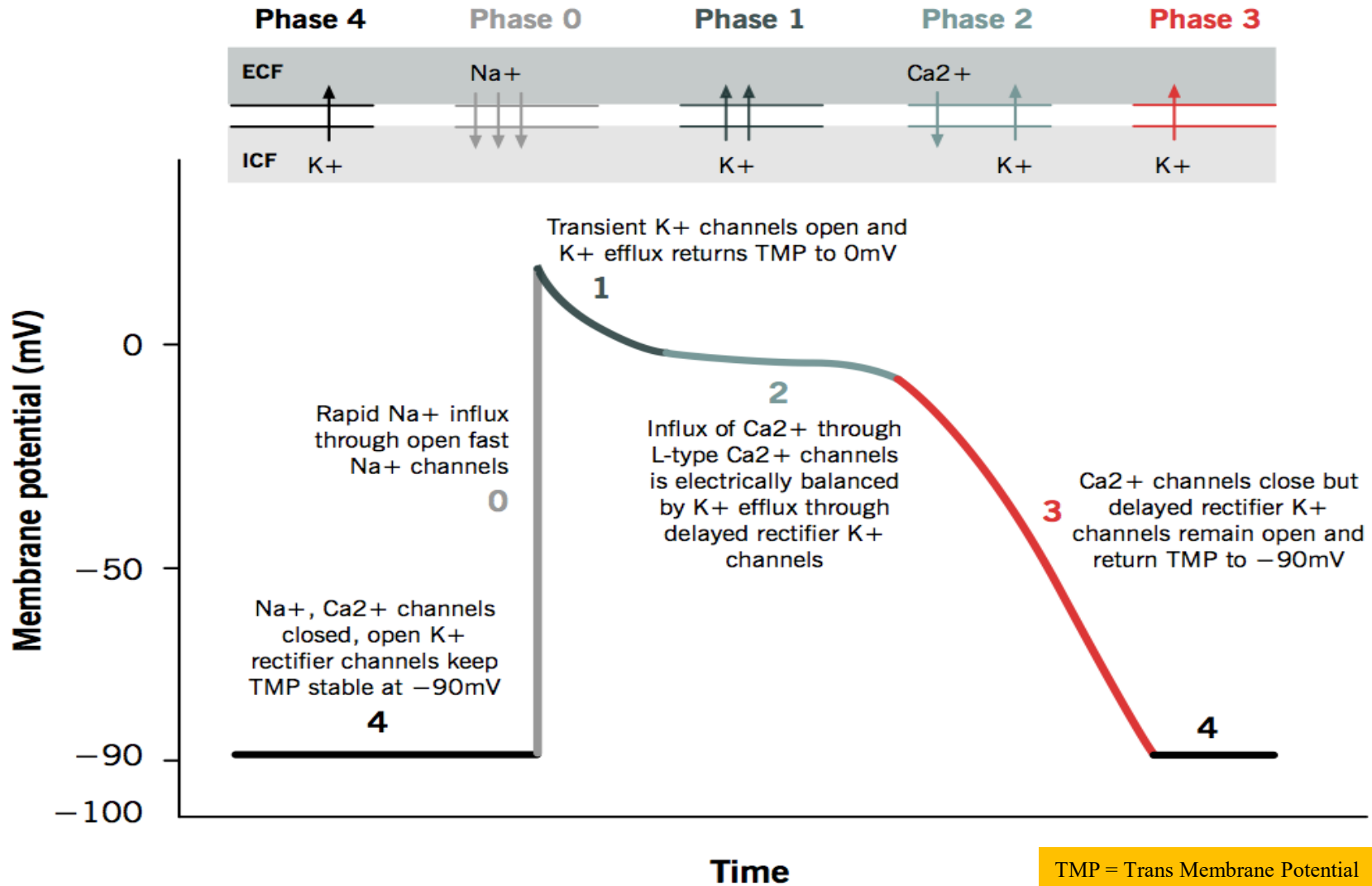


Fig. AP of ventricular muscles

**By comparison, depolarization in a neuron or skeletal muscle fiber is much briefer, about 1 msec (0.001 sec), b/c it lacks a plateau phase.*

Electrophysiology of the Heart...

Action potential of cardiac muscles



- ✓AP: one electrical event in a single cell & AP: recorded using intracellular electrode
- ✓ECG: is summed electrical activity of all cells & ECG: recorded from surface of body

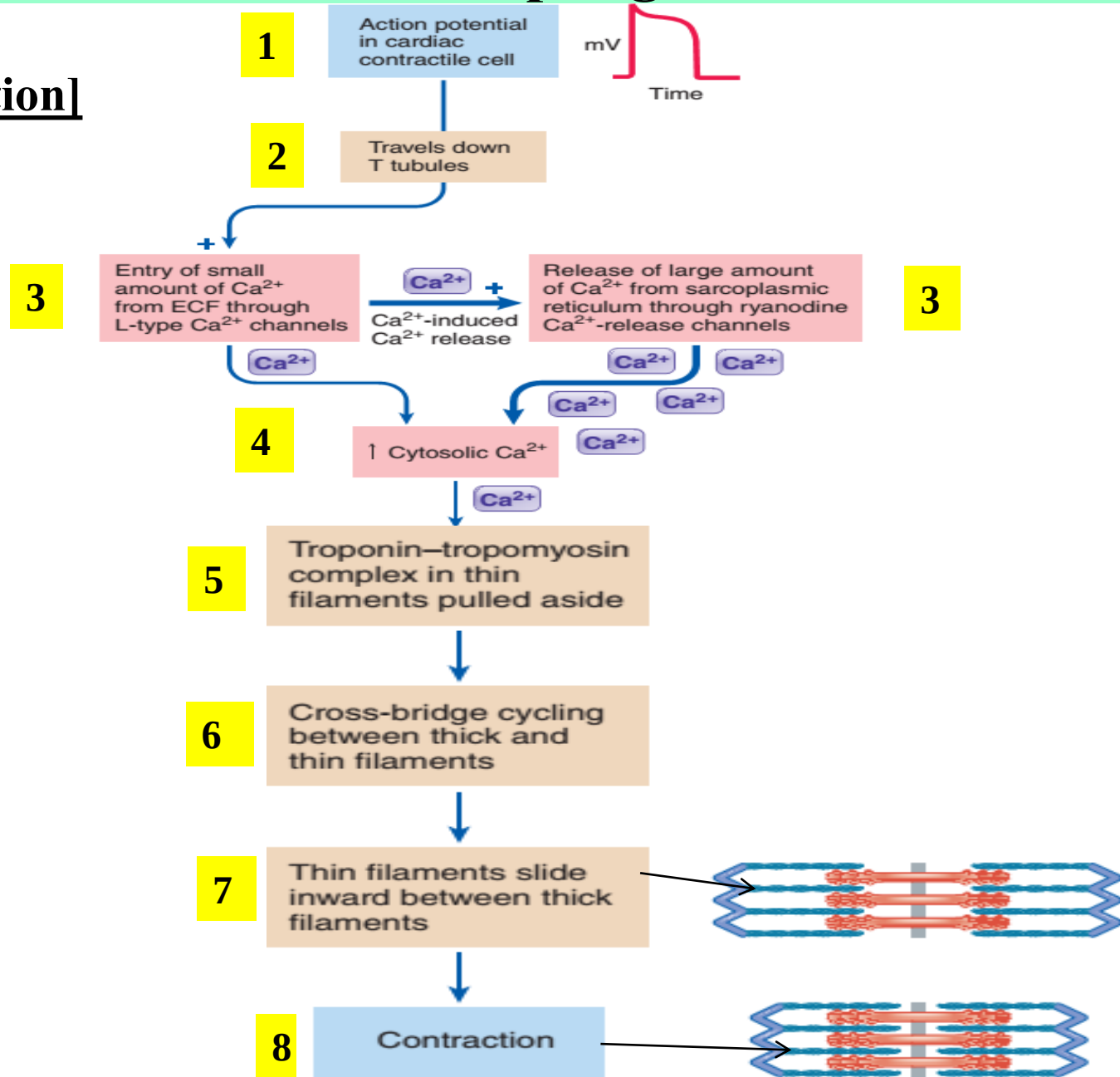
Excitation-contraction coupling in cardiac muscles

- It is a mechanism by which an action potential (AP) causes *contraction of the myocardium*.
- It has the following sequences:
- **Autorhythmicity** $\longrightarrow$ Membrane depolarization $\longrightarrow$ T-tubule depolarization $\longrightarrow$ Release of Ca^{2+} from SR + from T-tubule (ECF) $\longrightarrow$ Troponin-Tropomyosin complex in thin filaments pulled aside $\longrightarrow$ Ca^{2+} activates contractile molecules $\longrightarrow$ Sliding of the filaments $\longrightarrow$ **Contraction of the myocardium**.
- In muscle, **refractory period** is the time interval during which a second contraction cannot be triggered.
- **AP of a cardiac muscle** has a **prolonged refractory period**. As a result, another contraction cannot begin until relaxation is well underway.

Advantage: *prevents tetanic contraction of the heart. If the heart muscle could undergo tetanus, blood flow would cease.*

Excitation-contraction coupling in cardiac muscles...

AP [Excitation]



Contraction

Relaxation of cardiac muscle (diastole)

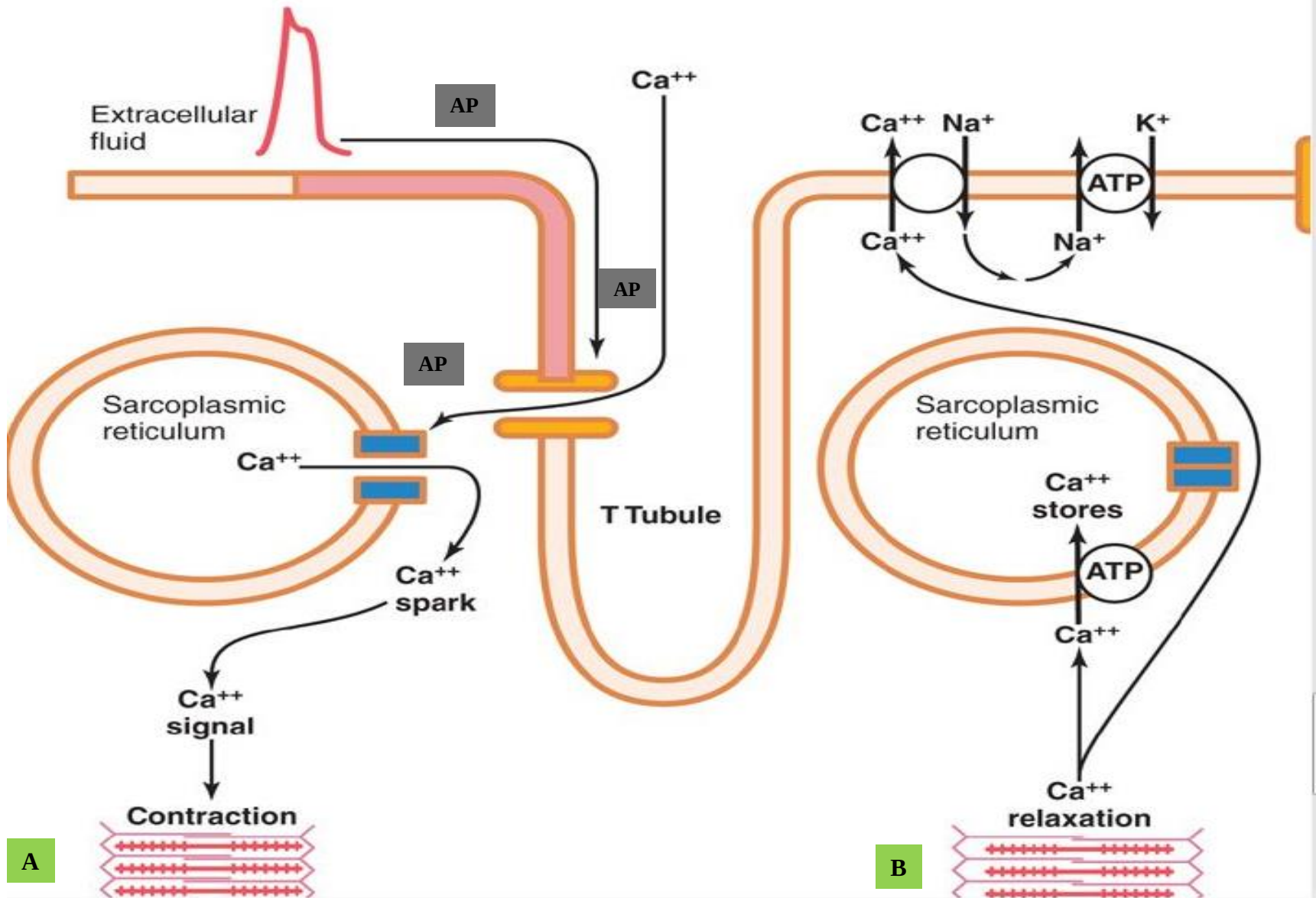
❑ As result of **Ca²⁺ removal** → Relaxation

❑ Ca²⁺ removed by:

1. **Uptake by SR** [by Ca²⁺ pump] → by the action of Ca²⁺-ATPase
2. **Discharged by Na⁺/Ca²⁺ exchange** [3Na⁺ in : 1Ca²⁺ out] → Via sarcolemma (from ICF to ECF).

- ❑ Excitation-contraction coupling by gaining **Ca²⁺** ions
- ❑ Relaxation by removing **Ca²⁺** from cytoplasm
- ❑ Strength of contraction can be modulated or related to **Ca²⁺** level
- ❑ Pace maker cells [SA-node] control **HR**

Excitation-contraction-coupling: Contraction & relaxation



The Cardiac Cycle (CC)

- All activities in the heart during a **single heart beat**.
 - *The integration of electrical activity, muscle contraction, and change in **volume**, **pressure**, and **flow** are integral components of a single CC.*
- **The contraction & relaxation** of the cardiac chambers.
- HR is a term used to describe the frequency of the CC. It is considered one of the four vital signs.
- **A single CC ('beat')** comprised of **3 major stages**: atrial systole, ventricular systole, and complete cardiac diastole.
- **75 cycles** completed per minute.
- **Duration** of each cycle = **0.8 second**
 - **Ventricular diastole** = 0.5 sec. - **Ventricular systole** = 0.3 sec.
- **Means** exploring activities accomplished during cardiac cycle:
 - Recording electrical activity of the heart: **ECG tracing**
 - Auscultation of heart sounds: **Phonography**
 - Measuring aortic pressure,
 - Measuring atrial pressure, and
 - Measuring ventricular pressure & volumes

Cardiac Cycle, cont'd

- The cardiac cycle is described as the **repetitive electrical and mechanical events** that occur with **each beat of the heart**.
- **Electrical events precede mechanical events**, w/c result from the entry of Ca^{2+} into the myocytes during cardiac APs.
- **ECG waves** can be correlated with **mechanical events**:
 - **P wave** precedes **atrial contraction** (atrial depolarization).
 - During PR segment, there is conduction through AV node & ventricular conducting system.
 - **QRS complex** precedes **ventricular contraction**.
 - **T wave** precedes **ventricular relaxation** (ventricular repolarization).

Cardiac cycle, cont'd

- Blood is pressurized by **ventricular contraction** & then ejected into the circulation during **systole**.
- **Ventricular muscle relaxation** is followed by **ventricular filling** during **diastole**.

The mechanical events of the cardiac cycle occur in 5 phases:

- 1. Ventricular diastole.** Throughout most of ventricular diastole, the atria & ventricles are relaxed. The **AV valves are open**, and **ventricles fill passively**.
- 2. Atrial systole.** During atrial systole, a small amount of additional blood is pumped into the ventricles.
- 3. Isovolumic ventricular contraction.** **Initial contraction increases ventricular pressure**, **closing the AV valves**. Blood is pressurized during isovolumic ventricular contraction.

Cardiac cycle, cont'd

4. **Ventricular ejection.** SL valves open when ventricular pressures exceed pressures in the aorta & pulmonary artery. Ventricular ejection (systole) of blood follows.
5. **Isovolumic ventricular relaxation.** SV valves close. Pressure in ventricles decreases. When pressure in ventricles decreases below atrial pressure, AV valves open. Atria fill with blood throughout ventricular systole, allowing rapid ventricular filling at the start of the next diastolic period.

❖ Throughout the cardiac cycle, the blood pressure increases and decreases. As ventricles contract the pressure rise, causing the AV valves to slam shut.

❖ Resting heart rates can be significantly lower in athletes, and significantly higher in the obese.

START

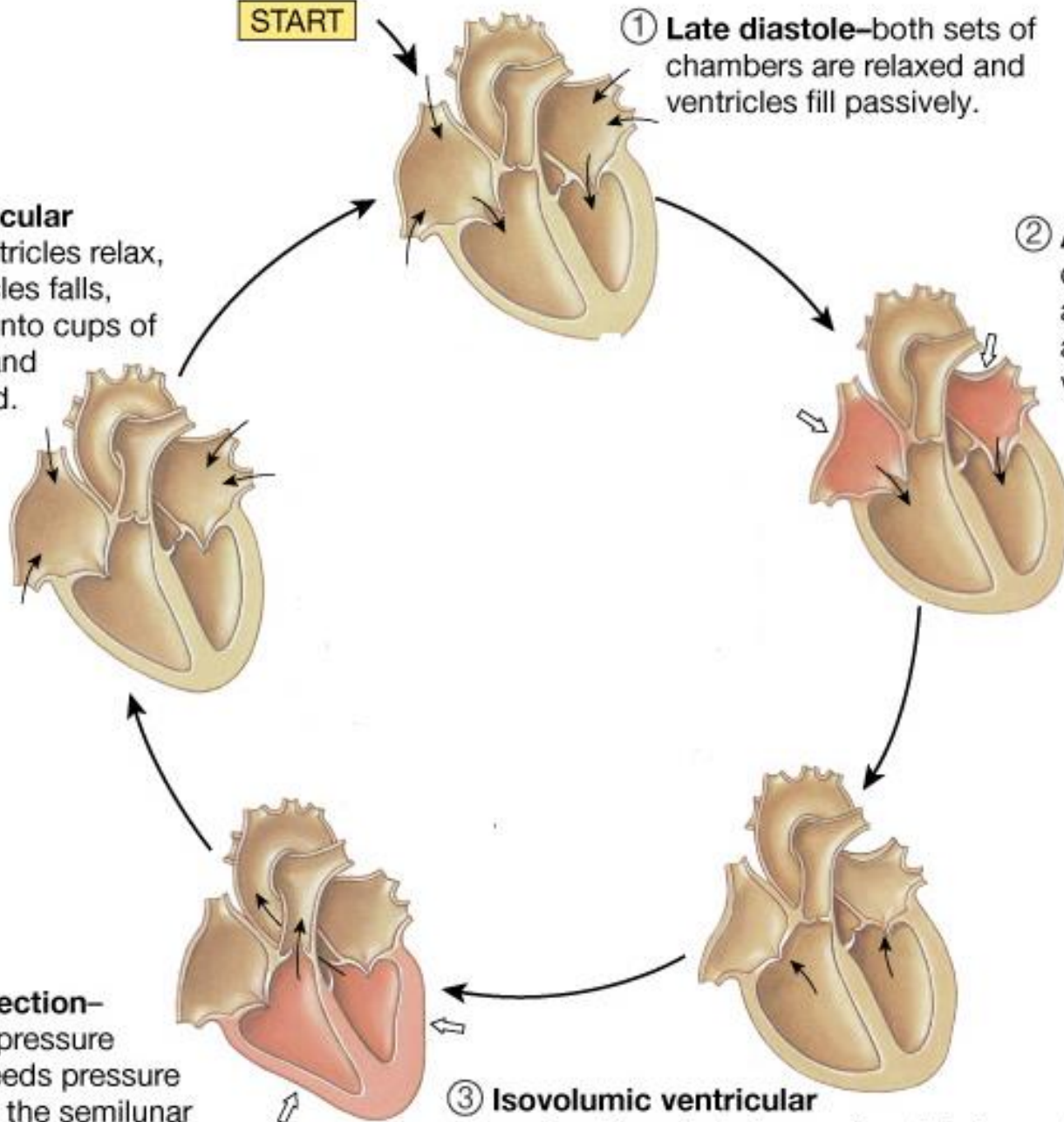
① **Late diastole**—both sets of chambers are relaxed and ventricles fill passively.

② **Atrial systole**—atrial contraction forces a small amount of additional blood into ventricles.

③ **Isovolumic ventricular contraction**—first phase of ventricular contraction pushes AV valves closed but does not create enough pressure to open semilunar valves.

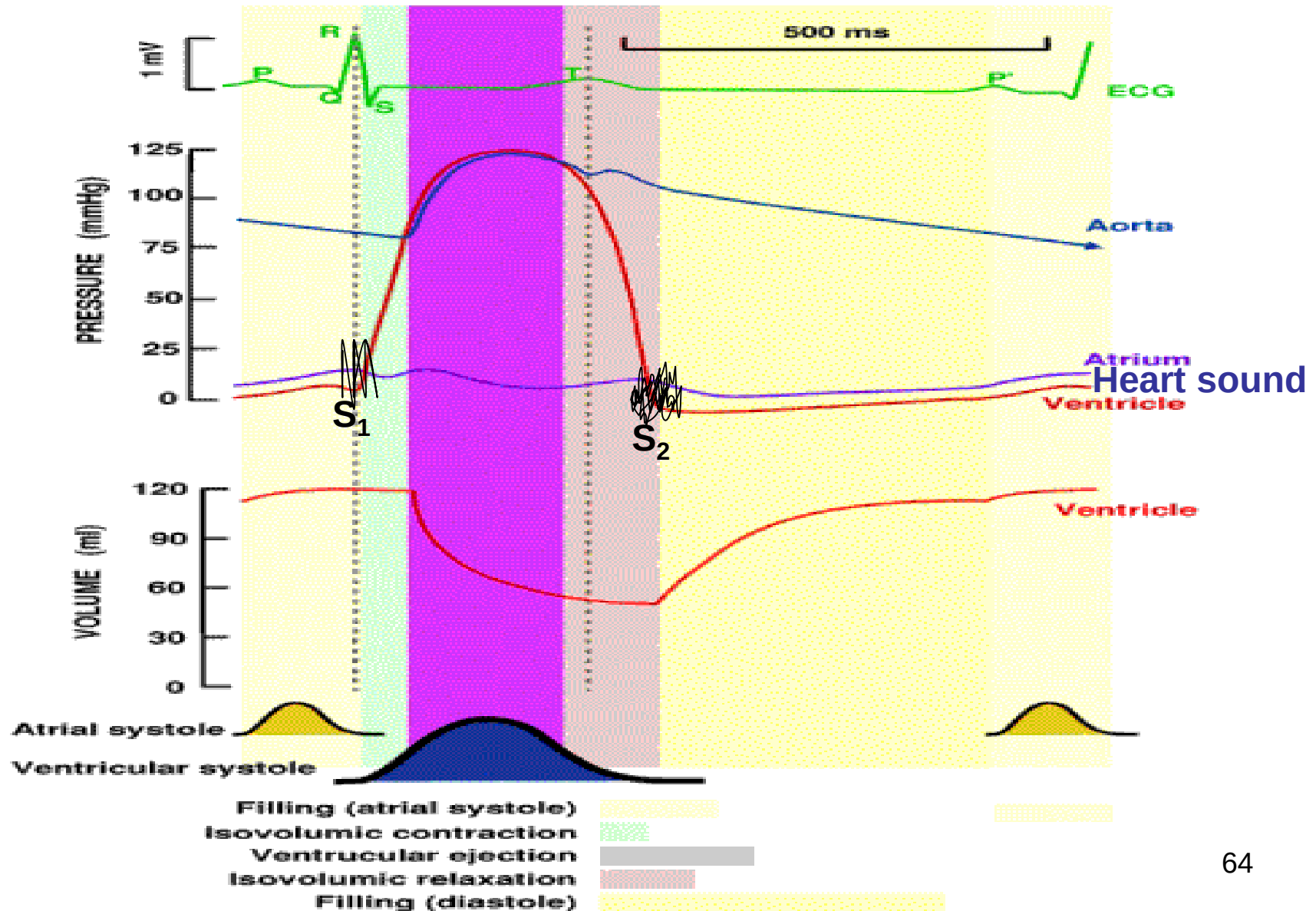
④ **Ventricular ejection**—as ventricular pressure rises and exceeds pressure in the arteries, the semilunar valves open and blood is ejected.

⑤ **Isovolumic ventricular relaxation**—as ventricles relax, pressure in ventricles falls, blood flows back into cups of semilunar valves and snaps them closed.



Summary of The Cardiac cycle

The cardiac cycle in the left side of the heart

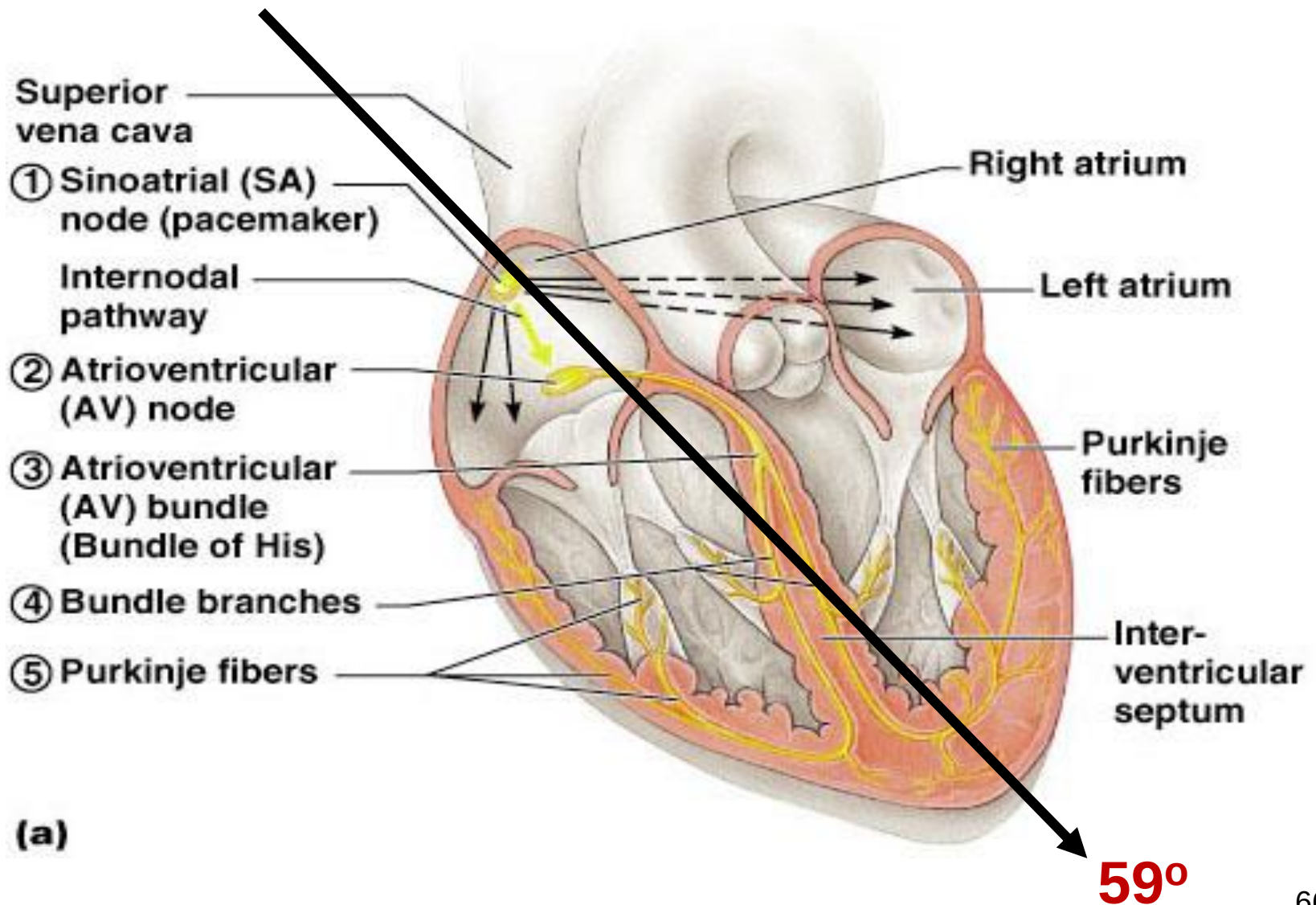


Cardiac Conduction System

◆ **Cardiac APs propagate through the conduction system in the following sequence:**

- 1) **Cardiac excitation normally begins in SA node**, located in the Rt atrial wall just inferior & lateral to the opening of the SVC.
 - SA node cells **do not have a stable resting potential**
 - Rather, they *repeatedly depolarize to threshold spontaneously*
 - The spontaneous depolarization is a **pacemaker potential**.
 - When the pacemaker potential reaches threshold, it triggers AP.
 - Each AP from the SA node propagates throughout both atria via **gap junctions** in the intercalated discs of atrial muscle fibers.
 - Following the AP, the atria contract.
- 2) **By conducting along atrial muscle fibers, the AP reaches the atrioventricular (AV) node.**

Sequences of Cardiac Excitation



The Conduction System, cont'd

3) From AV node, AP enters AV bundle (bundle of His)

- This bundle is the *only site where APs can conduct* from the atria to ventricles. (*Elsewhere, fibrous skeleton of the heart electrically insulates the atria from ventricles*).

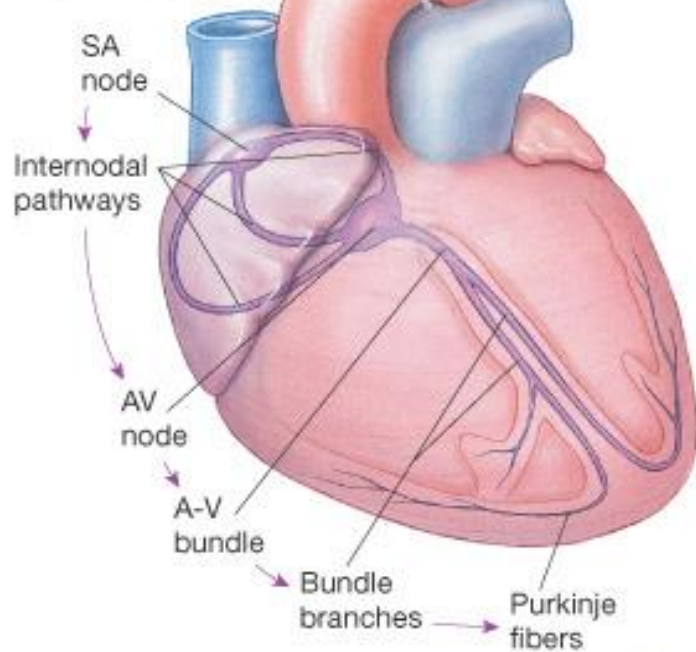
4) After propagating along the AV bundle, the AP enters both the Rt & Lt bundle branches.

- The bundle branches extend through the interventricular septum toward the apex of the heart.

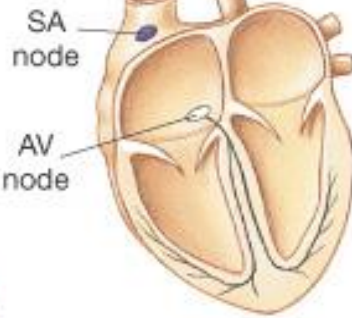
5) Finally, the *large-diameter Purkinje fibers rapidly conduct* the AP beginning at the apex of the heart up-ward to the remainder of the ventricular myocardium.

- Then the **ventricles contract**, pushing the blood upward toward the semilunar valves.

(a) The conducting system of the heart



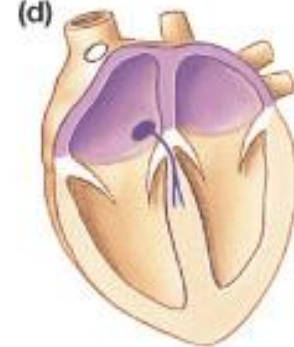
(b) SA node depolarizes.



(c) Electrical activity goes rapidly to AV node via internodal pathways.



(d) Depolarization spreads more slowly across atria. Conduction slows through AV node.



(e) Depolarization moves rapidly through ventricular conducting system to the apex of the heart.



(f) Depolarization wave spreads upward from the apex.



Electrocardiography

- ◆ **Electrocardiography** is the *electrical activity* of the heart that is conducted through the body fluid, and recorded from the surface of the body using highly sensitive *electrodes*.
- ◆ **Electrocardiogram (ECG)** of the heart is recorded from specific sites of the body in graphic form (Fig. below) *relating voltage (vertical axis) with time (horizontal axis)*.
- ◆ It is the *sum of all depolarization* occurring in the cardiac muscle fibers during each cardiac cycle.

Electrocardiography, cont'd

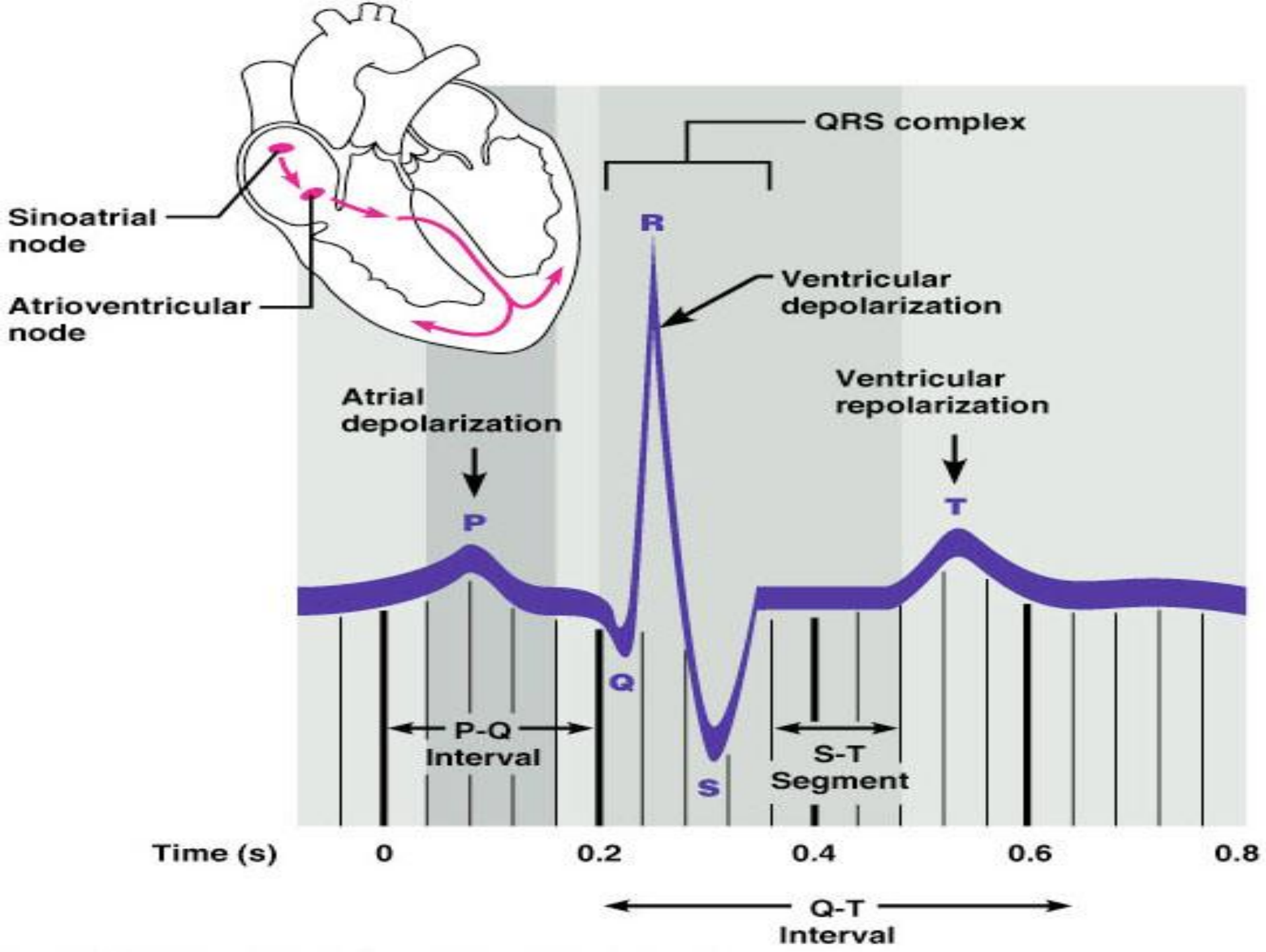
- ◆ Differences of electric potentials are recorded b/n pairs of points (electrodes) on the external surface of the body or b/n skin loci and indifferent or reference electrode.
- ◆ The technique was developed by the Dutch physiologist named **William Einthoven**.
- ◆ ECG provides events about *arrhythmia, hypertrophy, myocardial ischemia, necrosis & damage (MI)*.
- ◆ **Electrical axis** of the heart is **59°**

ECG Conventions

1. 1mV input→10mm deflection
 2. Paper speed =25mm/sec.
 3. Recording points = wrist, ankle, skin on chest
 4. Right leg = ground (earth)
- **Electrocardiogram (ECG):** Fluctuations of potential that represent the algebraic sum of the action potentials of myocardial fibres recorded extracellularly during the cardiac cycle.

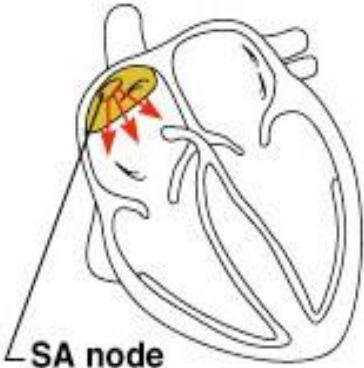
Relation b/n cardiac AP & ECG waves

- **Electrical activity** of the heart muscle is recorded by the **electrocardiogram (ECG)**.
- As shown below, there are **three main deflections**
- **Per cardiac cycle:**
 - **the P-wave:** corresponding to atrial depolarization;
 - **the QRS complex:** corresponding to ventricular depolarization and
 - **Q** = Septal part & **R** = Major part & **S** = basal part of Vt depolarization
 - **the T-wave:** corresponding to ventricular repolarization.
 - **Atrial repolarization** record is **masked** by the larger QRS complex.

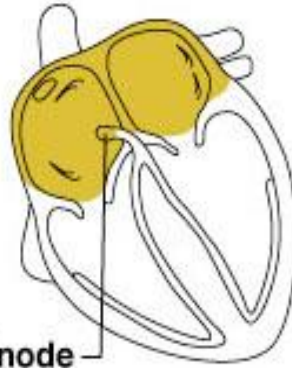


Heart Excitation Related to ECG

SA node generates impulse; atrial excitation begins



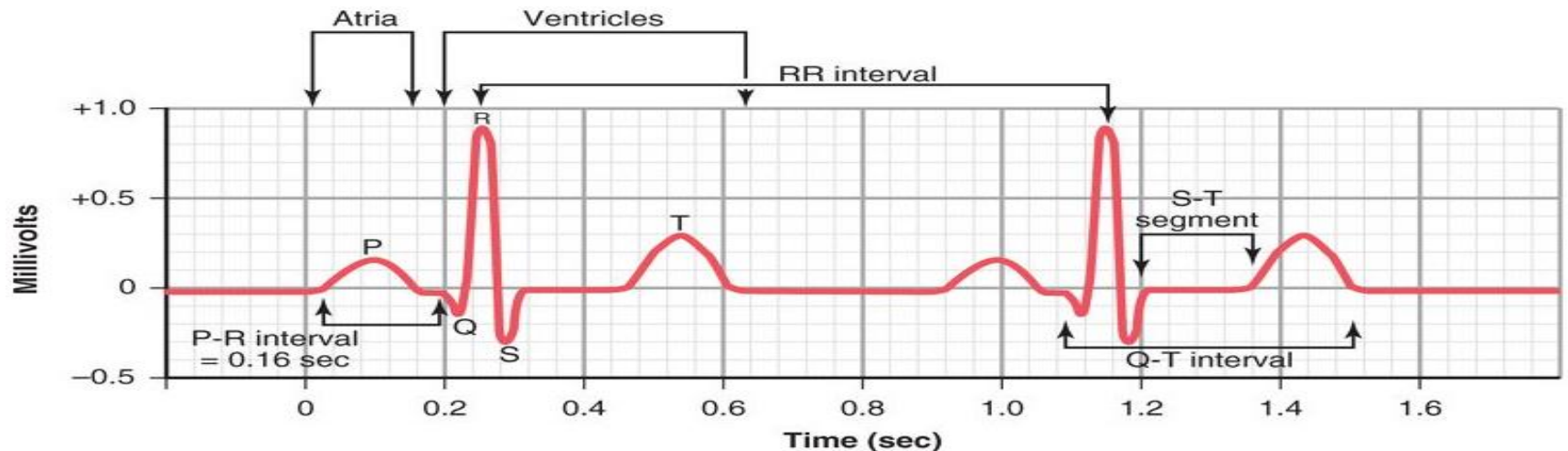
Impulse delayed at AV node



Impulse passes to heart apex; ventricular excitation begins



Ventricular excitation complete



ECG Recordings:

□ 3 Bipolar Limb Leads

I = RA vs. LA(+)

II = RA vs. LL(+)

III = LA vs. LL(+)

□ 3 Augmented Limb Leads

aVR = (LA-LL) vs. RA(+)

aVL = (RA-LL) vs. LA(+)

aVF = (RA-LA) vs. LL(+)

□ 6 Precordial (Chest) Leads: In different electrode (RA-LA-LL) vs. chest lead moved from position V_1 through position V_6 .

ECG Recordings (QRS vector points leftward, inferiorly and posteriorly)

3 Bipolar Limb Leads (Fig. 18-16 in textbook)

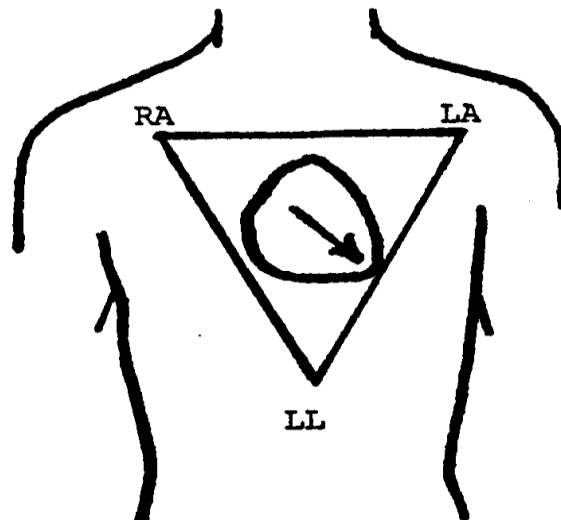
I = RA vs. LA(+)



II = RA vs. LL(+)



III = LA vs. LL(+)



3 Augmented Limb Leads

aVR = [LA-LL] vs. RA(+)



aVL = [RA-LL] vs. LA(+)

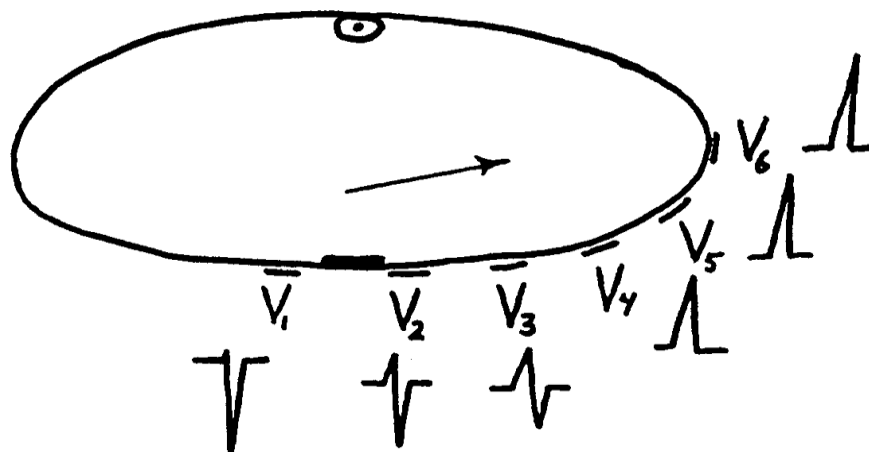


aVF = [RA-LA] vs. LL(+)



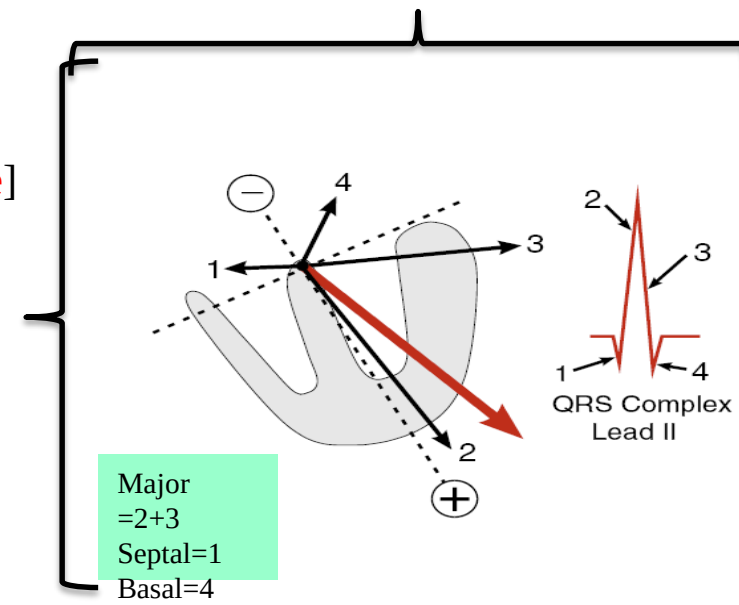
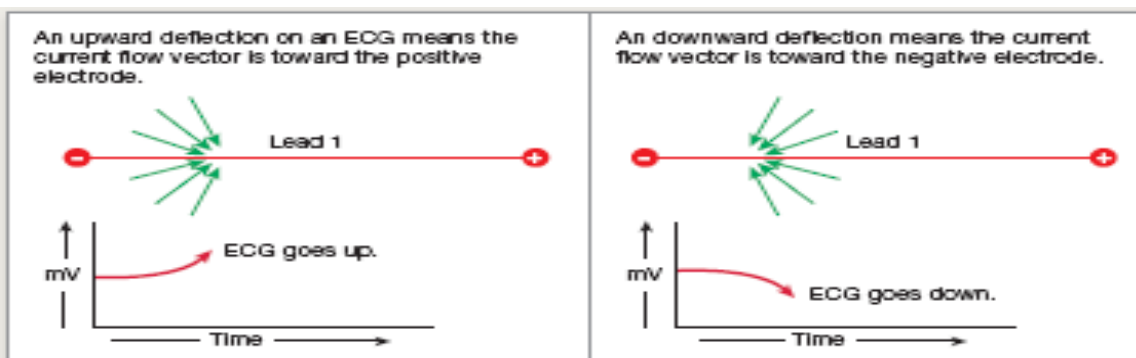
6 Precordial (Chest) Leads

Indifferent electrode [RA-LA-LL] vs. chest lead in positions V₁ through V₆



Polarity Convention [Rules of ECG machine]

- ❑ A wave of depolarization traveling towards a positive electrode results in
 - ✓ A **positive deflection** in the ECG tracing. [**P-wave**]
- ❑ A wave of repolarization traveling towards a negative electrode results in **positive deflection**. [**T-wave**]
- ❑ When **no electrical activity occurs** or the activity is too weak to measure, the waveform looks like
 - ✓ A straight line, called **isoelectric** waveform. [**PR-segment**]
- ❑ Complete depolarization or repolarization of the myocardium allows the galvanometer to deflect toward zero (Isoelectric line). [**ST-segment**]
- ❑ More tissue depolarized results,
 - ✓ High amplitude (height) of the complexes. [**R-wave**]



ECG Conventions

- ❑ Electrode attachment sites → wrist, ankle, skin on chest, and right leg used as ground
- ❑ The ECG paper speed is ordinarily 25 mm/sec.
- ❑ Each small horizontal box corresponds to 0.04 sec.
- ❑ One large box (containing 5 small boxes) represent 0.2 sec.
- ❑ Vertically, the ECG graph measures the **amplitude** of a given wave [in mV]
 - ✓ Each small square represents 0.1 mV in voltage, & 10 small boxes equals 1 mV.

❑ + to + → → Deflection +

❑ + to - → → Deflection -

❑ - to + → → Deflection -

❑ Small tissue depolarized → → Small deflection or isoelectric line

❑ 1 cardiac cycle = 1 beat = 1 PQRST = systole + diastole

ECG standard paper

❑ Consists of horizontal & vertical grid lines.

❖ The horizontal axis represents time.

✓ 300 big sq = 60 seconds

✓ 1big sq = 5 small sqs=5 mm

✓ The ECG = 25mm/sec paper speed

✓ **1 big sq = 0.2 second**

✓ 5 small sq.=0.2 second

✓ **1 small sq.= 0.04 second**

❖ The vertical axis measures amplitude in 'mm' or in mV.

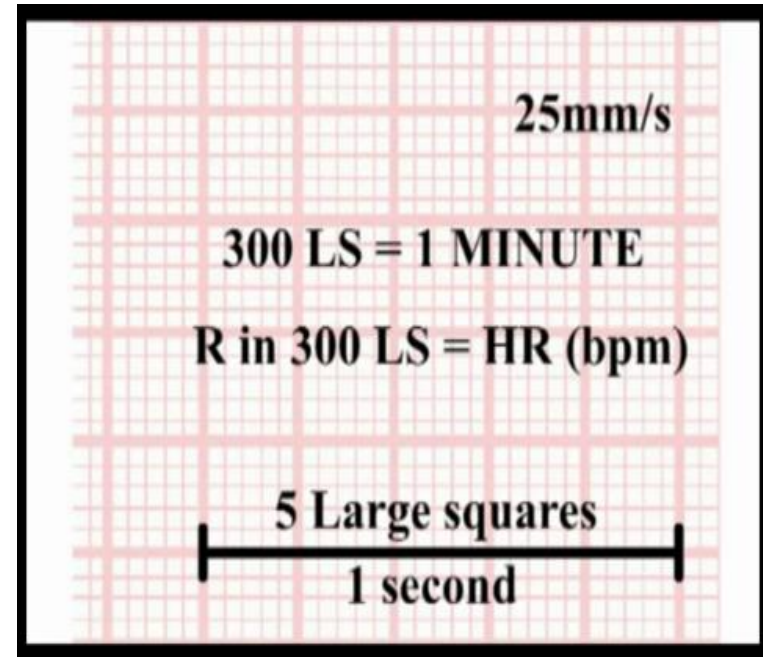
✓ Each small square represents 1mm = 0.1 mV.

✓ Each large square represents 5 mm = 0.5 mV.

❑ **1 big sq = 5mm = 0.2 sec**

= 5 small sqs=0.04 sec

= **1 small sq. = 0.04 second**



- Big square = Large square [LS]
- **R = Q' R' S**

Characteristics of normal ECG [The ECG tracing consists of W,S,I & complex]

A. Waves [W]

- ✓ P wave
- ✓ Q wave
- ✓ R wave
- ✓ S wave
- ✓ T wave

D. Complex

- ✓ QRS

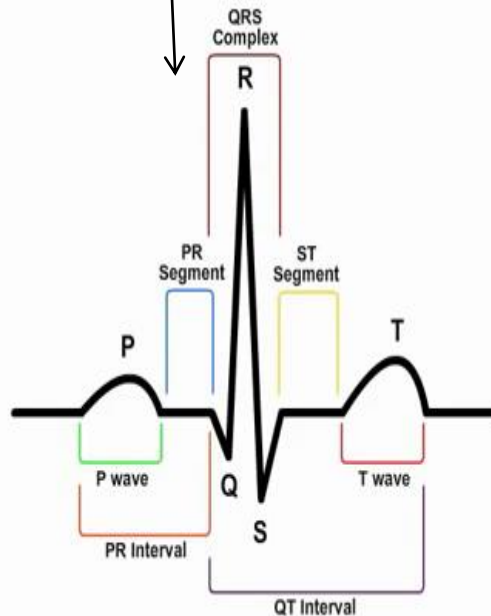
B. Segments [S]

- ✓ PR segment
- ✓ ST segment

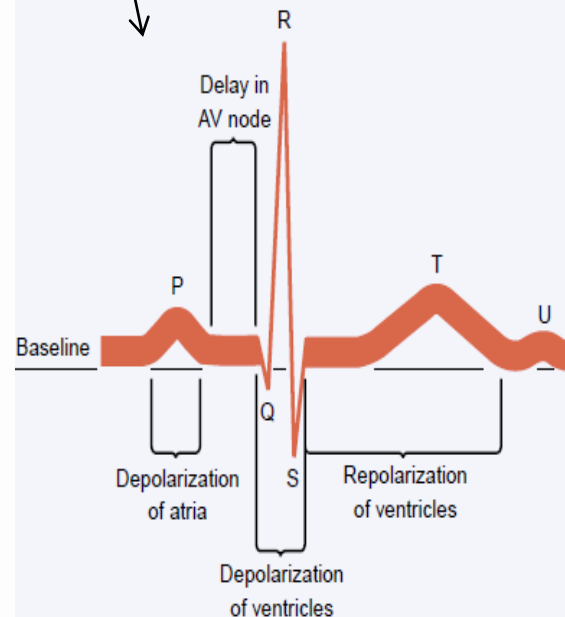
C. Intervals [I]

- ✓ PR interval
- ✓ QT interval

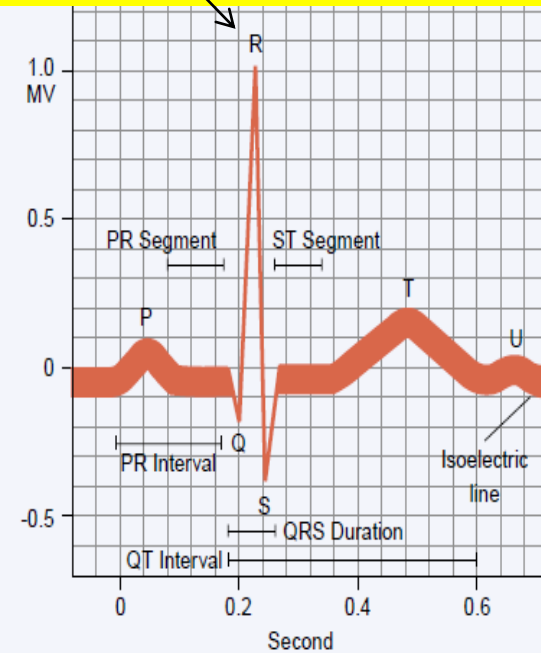
Beat-1



Beat -2



Beat-3



The ECG Leads

- **To record a routine ECG, 12 leads are used:**
 - **On the limbs:** I, II, III, AVR, AVL, and AVF
 - **On the chest:** V1, V2, V3, V4, V5, V6
- Each lead records the **same electrical impulse on the ECG** but from a different position in relation to the heart.
- Each can be read separately, or they can be read in combination.
- **The limb leads** show the current flow from one area of the body toward another.
- **Lead I** show electrical activity from Rt arm to Lt arm, **lead II** from Rt arm to Lt leg, and **lead III** from Lt arm to Lt leg.
- **The abbreviation AVR** stands for augmented vector right, an added point of reference for diagnosis. **AVL & AVF** represent the left-side and left-foot positions respectively.

ECG Leads, cont'd

- In diagnosing arrhythmias, lead II & the right-chest leads are commonly used because they most consistently show an easily visible P wave.
- Many times, P wave provides the key to success in determining the identity of an arrhythmia.
- The most important leads to remember in relation to the anatomy of the heart are:
 - V1, AVR.....Right side of the heart.
 - V2, V3, V4.....Transition between right and left.
 - V5, V6, I, AVL..... Left side of the heart.
 - II, III, AVF..... Inferior aspect of the heart.
- The area pathology shown in the ECG of can be localized by analyzing tracing from different leads. For e.g. if an infract shows up in leads II, III, and AVF only, it is located in the inferior aspect of the heart.

The Basic principles of volume conductors apply to the recording of ECG

- There are **three types of ECG recording**:

1. Bipolar limb leads: Record voltage b/n two electrodes (leads) placed on the wrists and legs.

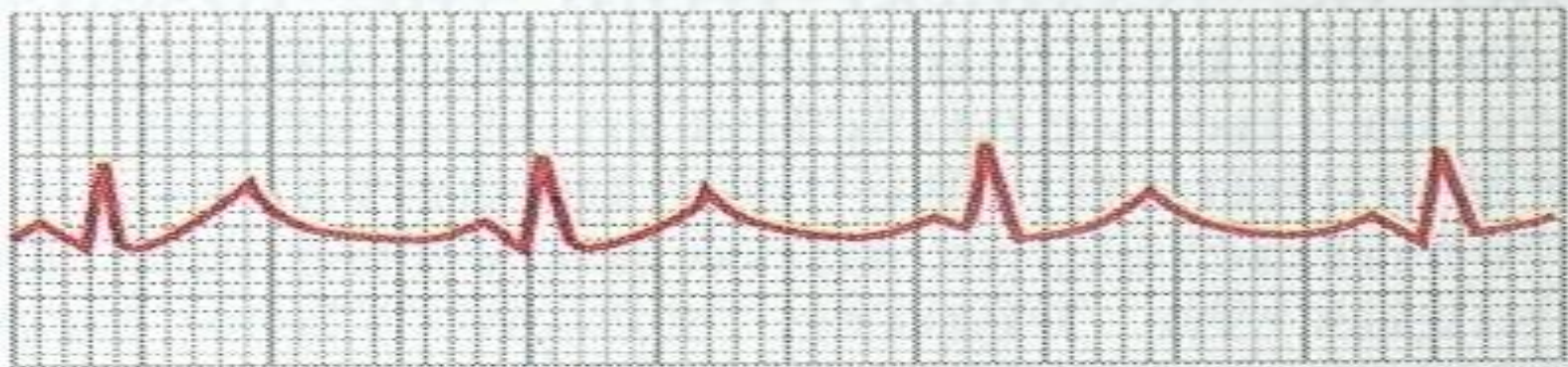
These leads include:

- **Lead I = RA-LA**, electric potential difference b/n Rt & arm Lt. arm
 - **Lead II = RA-LL**, electric potential difference b/n Rt arm & Lt leg
 - **Lead III = LA-LL**, electric potential difference b/n Lt arm & Lt leg
- Each lead shows waves of depolarization & repolarization: **P-wave, QRS complex, T-wave**, and occasionally **U-wave**.

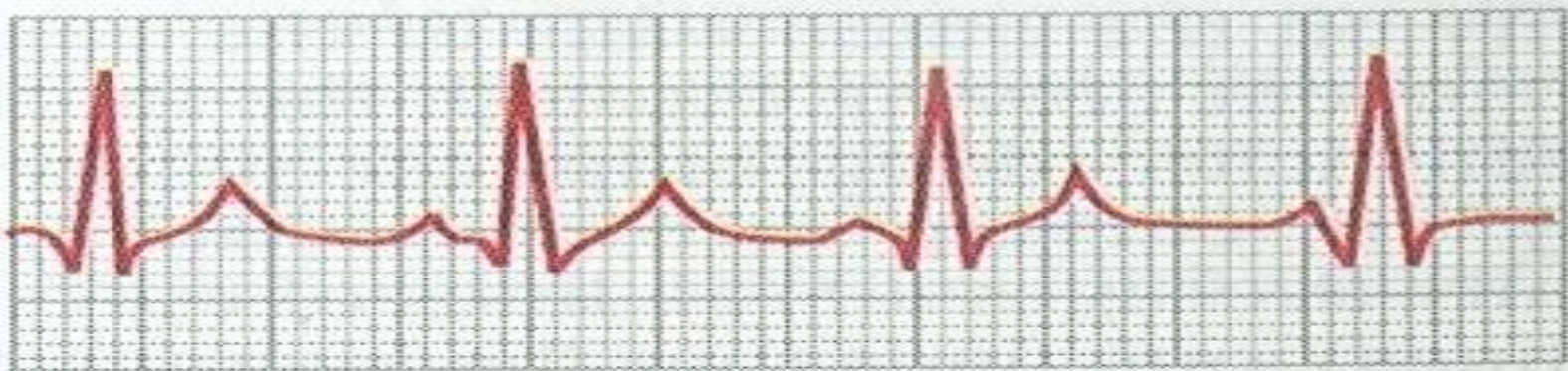
ECG Waves

- **P-wave:** - represents atrial depolarization;
- **QRS complex:** - is caused by ventricular depolarization. Following a short interval, a
- **T- wave:-** appears in the ECG and corresponds to repolarization of ventricular muscle mass.
- ECG returns to iso-electric line.
- These ECG waves are fore-runners to cardiac contraction.

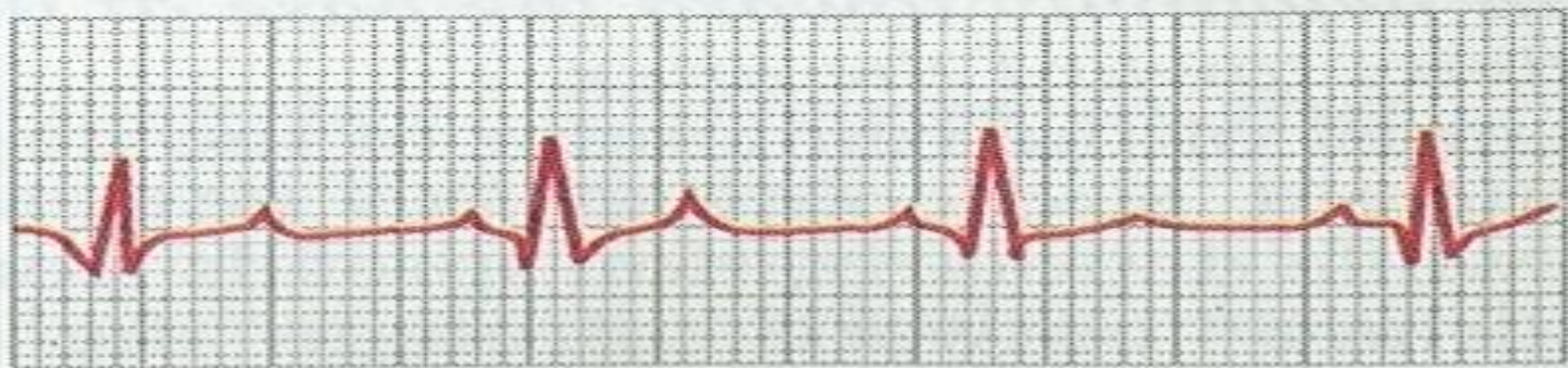
I



II



III



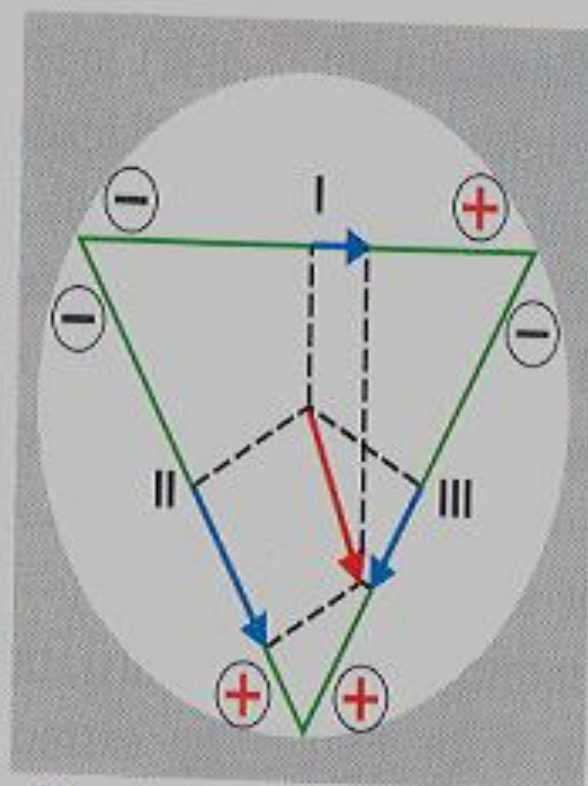
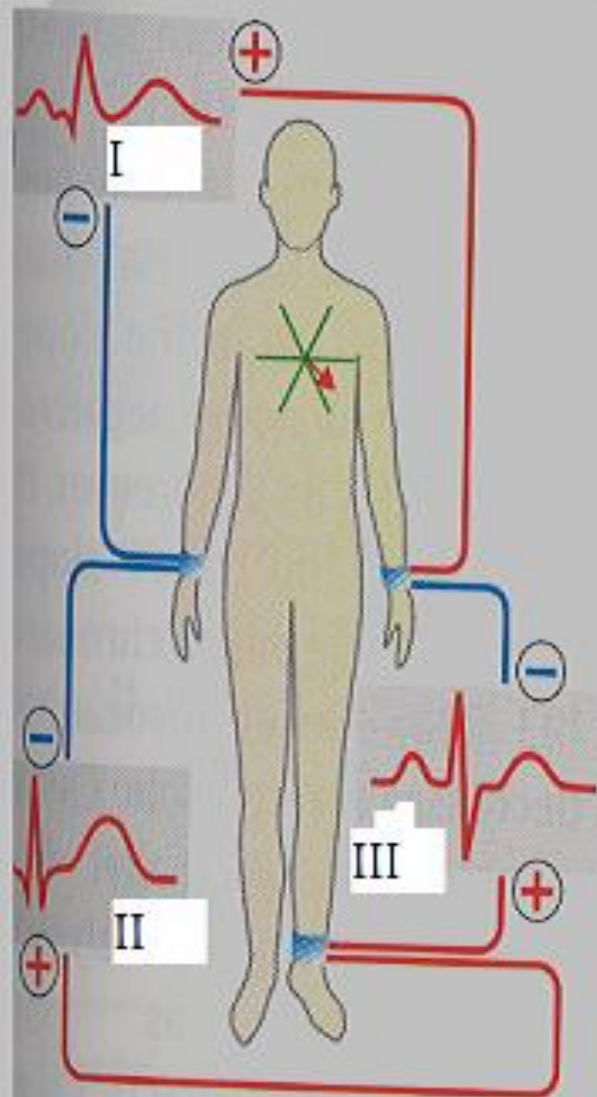
Bipolar limb leads, cont'd

- These leads are as shown in the equilateral triangle of Einthoven (fig.) drawn around the heart.
- The triangle shows that the **two arms and the left leg form apices of a triangle surrounding the heart.**
- The two arms with electrodes connect electrically with the fluid around the heart and the left leg with another electrode also connects with the fluid.

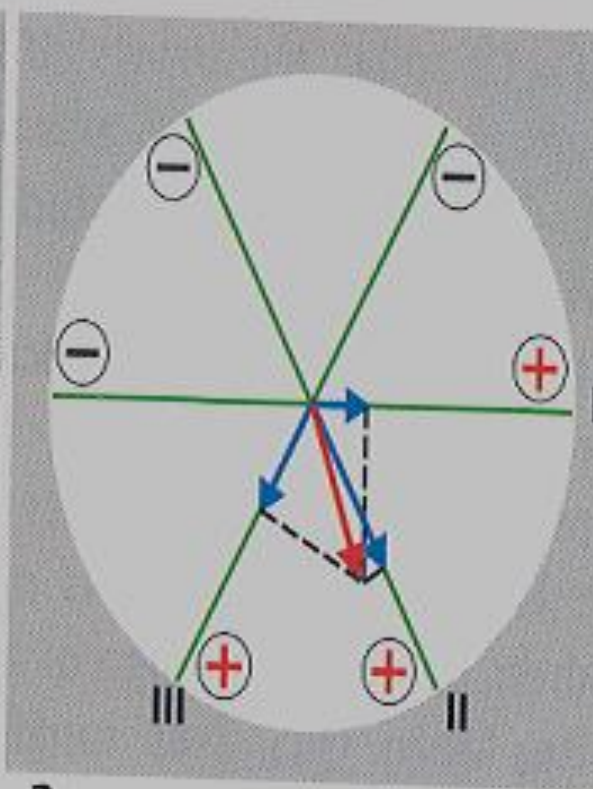
Bipolar limb leads, cont'd

- **Einthoven's law:** If electrical potentials of any two of the **three leads** are given, the 3rd one can be determined.
- The relation, $\text{Lead II} = \text{Lead I} + \text{Lead III}$, can be used to estimate electric potential of each of the leads.

Einthoven leads I, II and III (bipolar)



2



3

Triaxial system

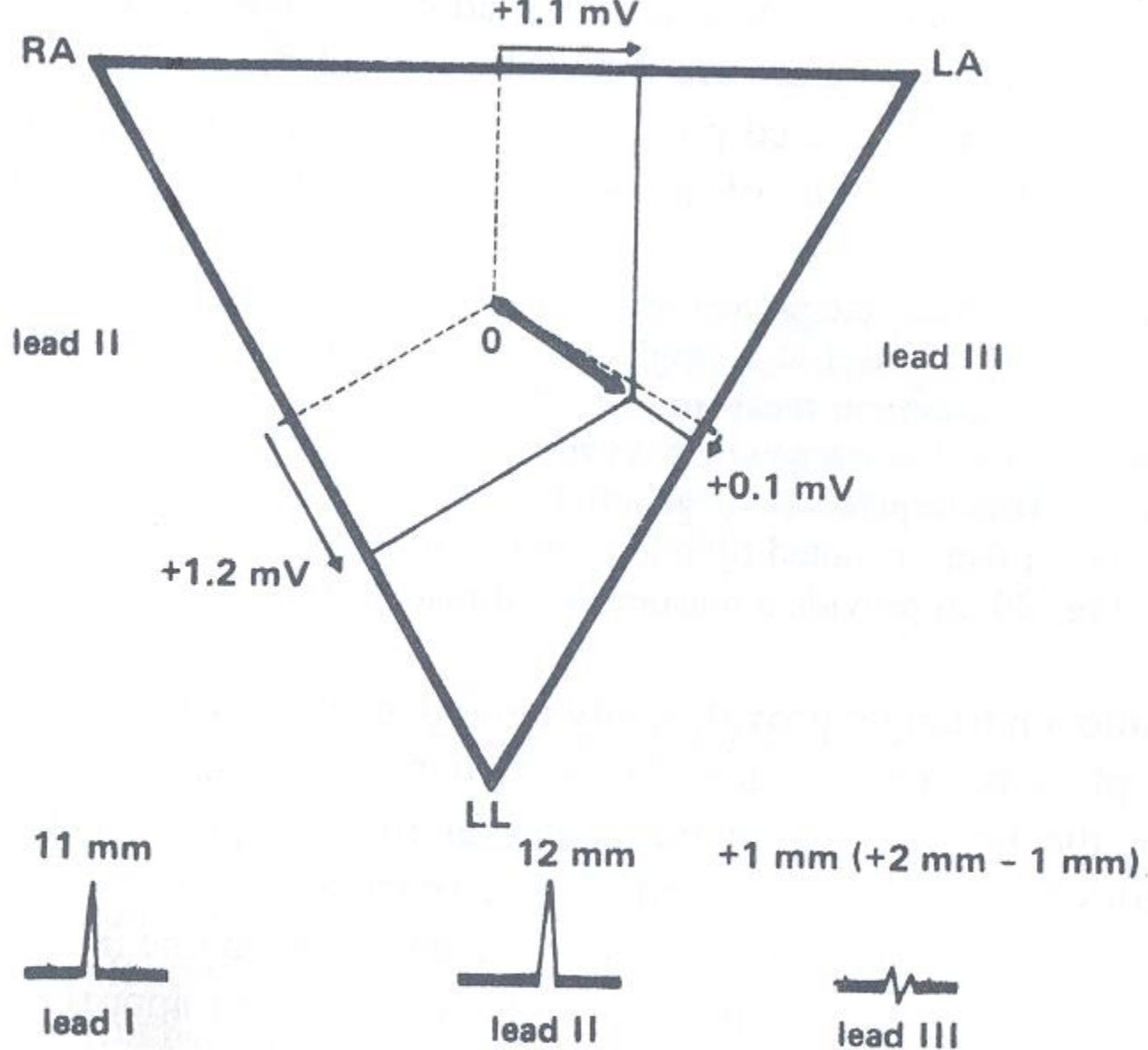
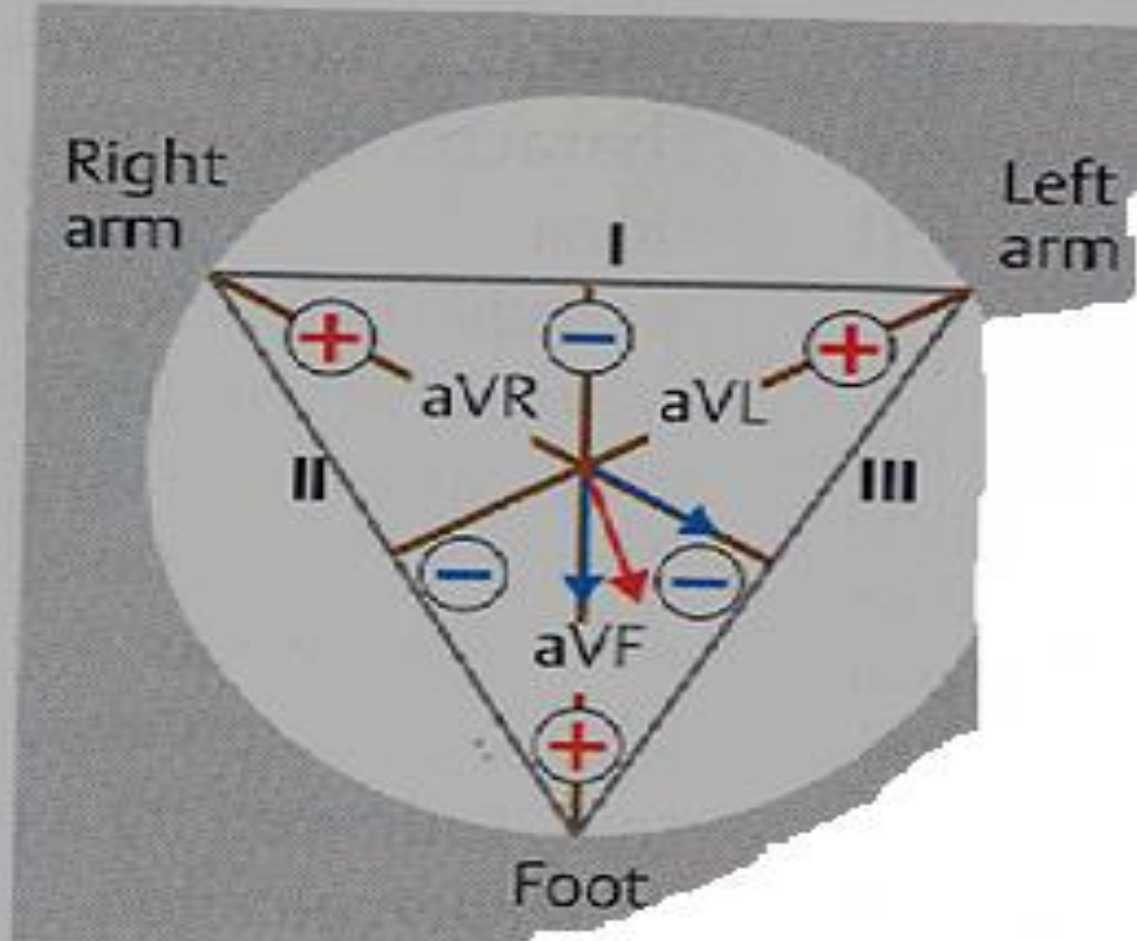
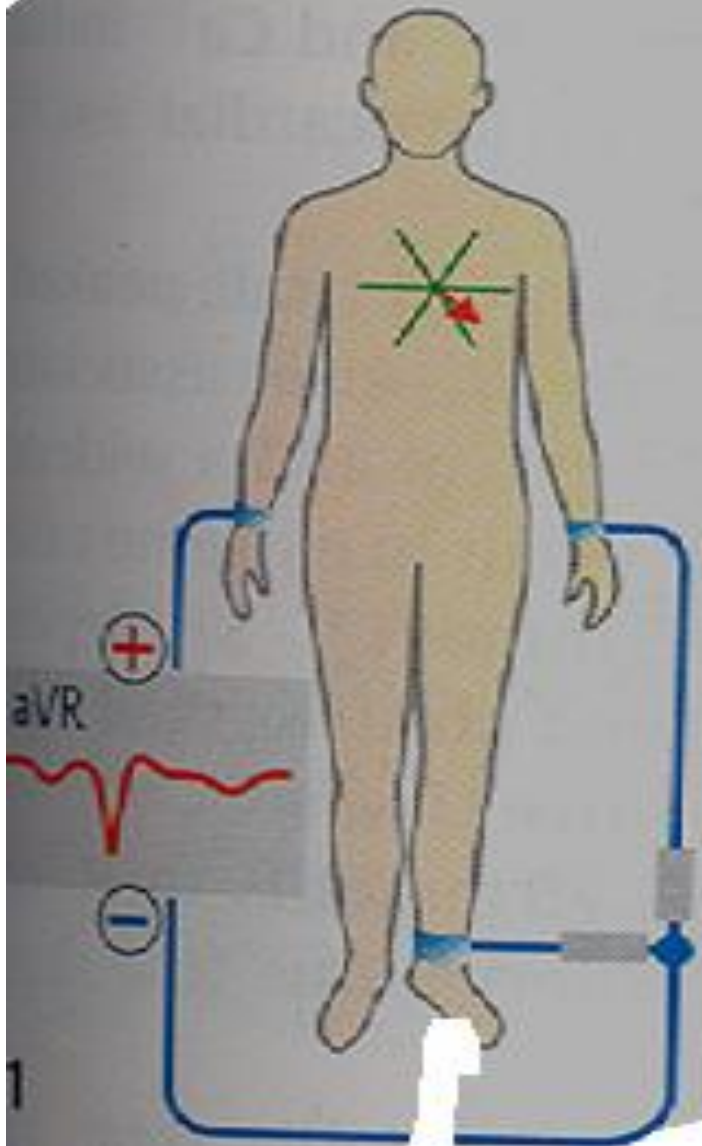


Fig.16. A normal mean QRS vector projected on the Einthoven triangle, each side of which represents one of the three bipolar limb leads (I, II, III).

2. Unipolar Limb leads

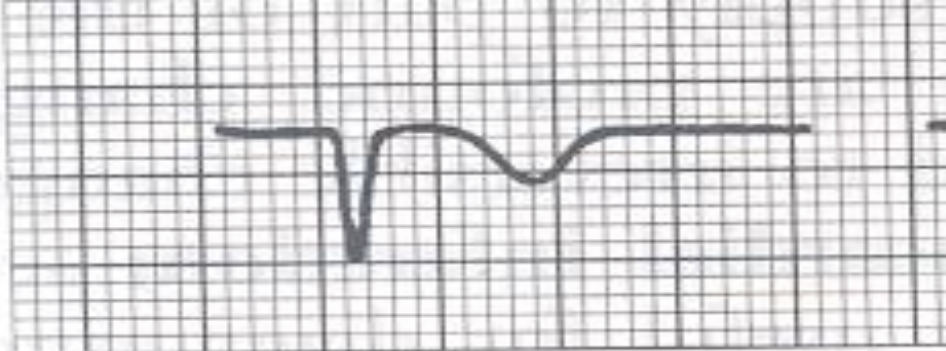
- Are augmented leads.
- Voltage is recorded b/n a single “exploratory electrode” placed on the body and an electrode that is built into the electrocardiograph and maintained at zero potential (ground).
- In this system, two of the limbs are connected through electrical resistances to the **negative (indifferent) terminal** of the electrocardiograph and the 3rd limb is connected to the **positive terminal**.
- When the **positive terminal** is placed on right arm, the lead is designated as **aVR**; when on the left arm **aVL**, and when on left leg, **aVF** (Fig. below).

Goldberger limb leads (unipolar)



2
Hexaxial system

aVR



aVL



aVF

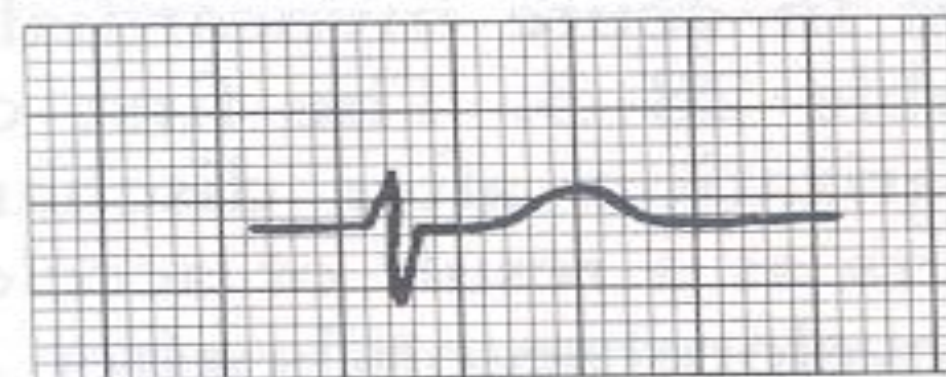


Fig..Unipolar (Goldberg) limb leads (augmented)

3. Precordial (Chest) Leads

Unipolar leads labeled V1 through V6.

- One electrode is connected to the **positive terminal** of the Electrocardiograph and the **negative electrode (indifferent electrode)** is connected through equal electrical resistances to the right arm, left arm, and left leg.
- Electrodes are placed on the chest as shown below.

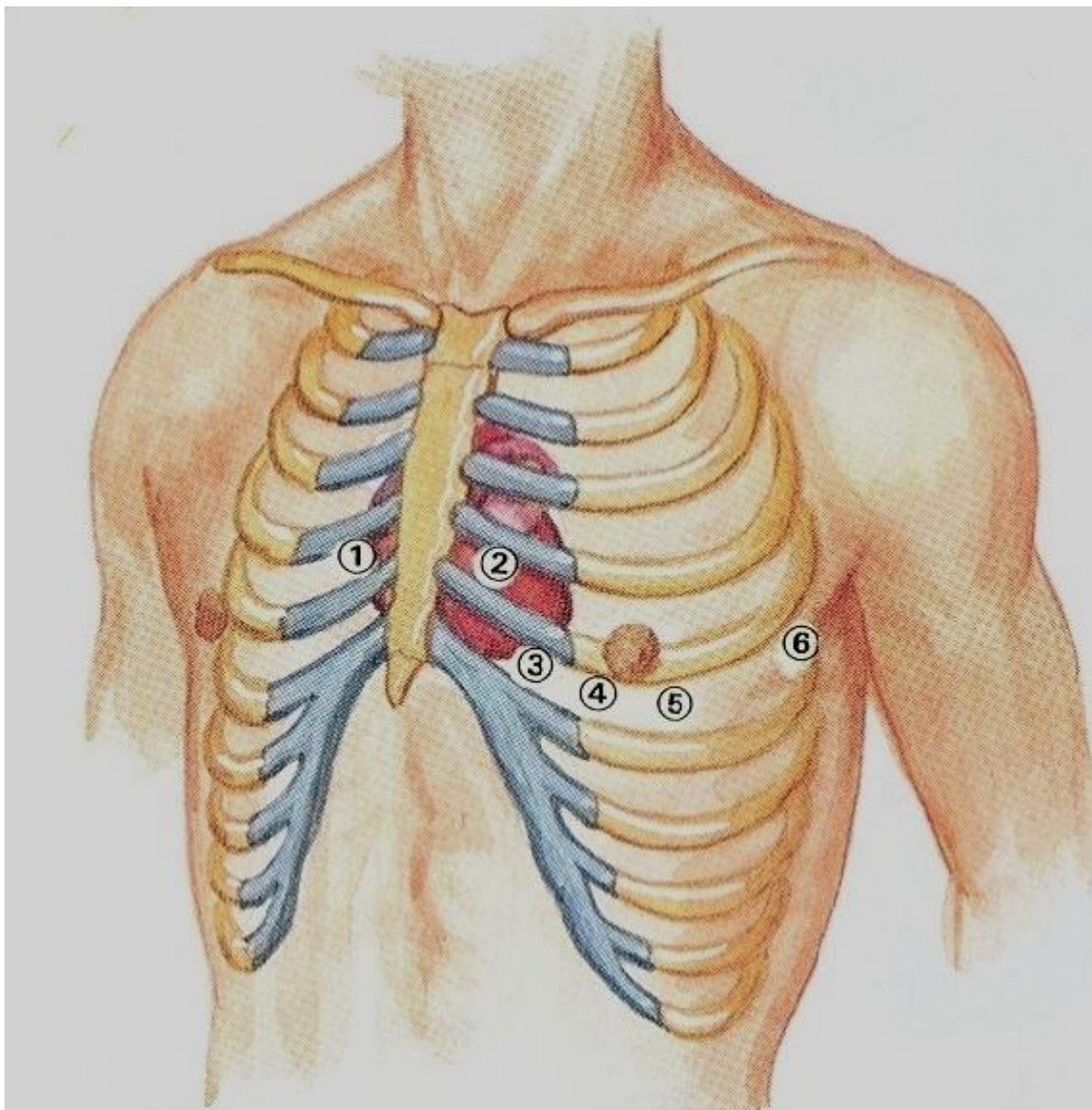


Figure. Placement of electrodes on the chest

Chest Leads, cont'd

- Heart surfaces are close to the chest wall and each chest lead records mainly the electrical potential of the cardiac muscle immediately beneath the electrode, i.e., **relatively minute abnormalities in the ventricles can cause marked changes in the ECG.**
- As shown below, **QRS in V1, V2**, are **negative** because the chest electrodes are nearer the base of the heart (**direction of electronegativity**). **V3 is in b/n.** **QRS of leads v4-v6** are **positive** because they are nearer the apex (**direction of electropositivity**).

Standard chest leads

Lead	Position of exploring electrode
V ₁	Fourth intercostal space just to the right of the sternum
V ₂	Fourth intercostal space just to the left of the sternum
V ₃	Midway between V ₂ and V ₄
V ₄	Fifth intercostal space at the left <u>midclavicular line</u>
V ₅	Left <u>anterior axillary line</u> horizontally to the left of V ₄
V ₆	<u>Midaxillary line</u> horizontally to the left of V ₄ and V ₅

Application of ECG

- By analyzing electric potential fluctuations, the physician can get some insight relating into:
 - anatomical orientation of the heart,
 - **Heart rate determination**
 - relative size of heart chambers,
 - a variety of disturbances of rhythm and conduction arrhythmia and conduction block
 - extent, location and progress of ischaemic damage (myocardial infarction)
 - Hypertrophy
 - Pericarditis and myocarditis

Some examples of ECG abnormalities

1. **Axis deviation**
2. **Voltage abnormalities**

Axis Deviation

- Occurs when the mean electrical axis is deviated either to the left ($< -30^\circ$) or to the right (> 90 to 110°).

Causes:

- Change in the **position** of the heart

Left axis deviation: expiration, supine position, obesity.

Right axis deviation: inspiration, upright position, tall and lanky stature, overload and hypertrophy of right ventricle.

- Pathologic conditions:** congenital heart disease, right ventricular hypertrophy, acquired disease of myocardium: ischaemic heart disease, degeneration of the conducting system, e.g. BBB.

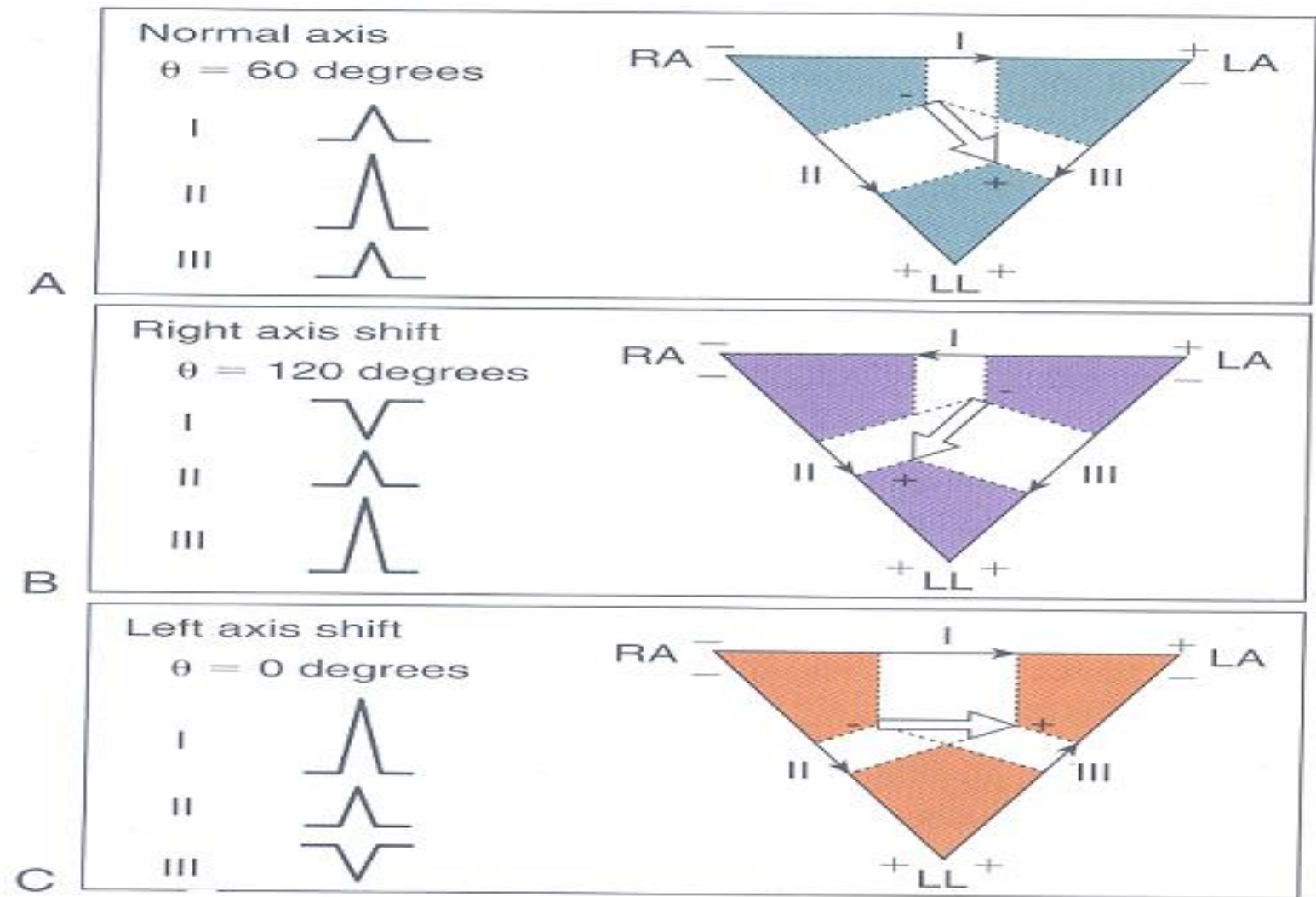


Fig.. Magnitude and direction of the QRS complexes in limb leads I, II, and III when the mean electrical axis (Q) is 60 degrees (A), 120 degrees (B), and 0 degrees (C).

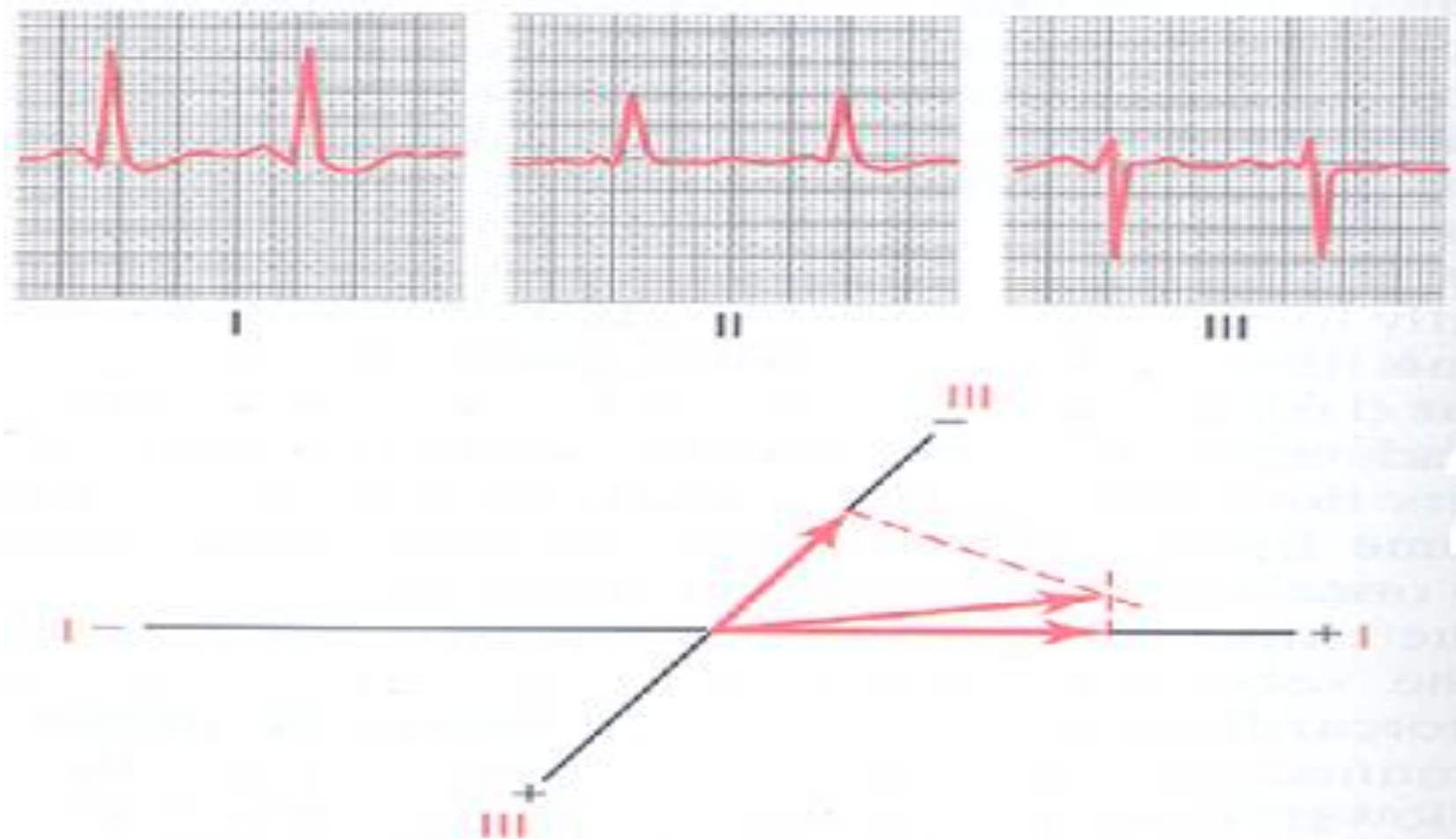


Fig..Left axis deviation in a hypertensive heart (hypertrophic left ventricle with slightly prolonged QRS complex

Voltage abnormalities

- May be determined by making use of the convention for voltage calibration (fig.. below)
- i. Increased voltage
 - If **QRS** of 3 standard leads is greater than 4 mV, then the heart is said to be hypertrophic.
 - Hyperkalaemia → **Spiked T-wave**
- ii. Decreased voltages are caused by:
 - Muscle abnormalities-like myocardial infarction
 - Conduction problem like 1st to 3rd degree heart block
 - Ant. rotation of cardiac apex
- iii. Prolonged duration of **QRS** is caused by:
 - Delayed conduction of impulses
 - Block of Purkinje system

Conduction Block

- This may occur anywhere between the SA node and the ventricles. One of the more common sites is AV node.
- Three degrees of heart block are usually distinguished.

i. First degree heart block

- PR interval prolonged, > 0.21 sec, usually due to AV nodal delay.
- May be caused by increased vagal tone and systemic diseases
- Has no effect on the pumping ability of the ventricles

Conduction Block, cont'd

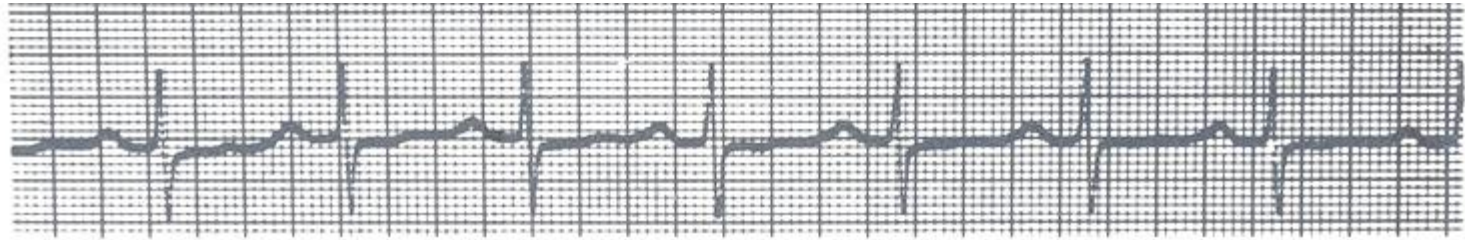
ii. Second Degree Heart Block

- **Associated with organic heart disease**
 - **Not all P waves** are followed by QRS complexes

iii. Third Degree (complete) Heart Block

1. Impulses completely interrupted b/n atria & ventricles
 2. QRS complexes dissociated from P-waves
 3. Ventricular rate much slowed (35-40 beats/min)
 4. Fainting often occurs due to cerebral ischaemia
- **Cause:** organic heart disease is often the cause.
 - **Significance:** May be associated with prolonged ventricular stand still. Cardiac arrest, if prolonged, can lead to cerebral ischemia, syncope, or death. This condition is termed **Stokes-Adams Syndrome**.

A



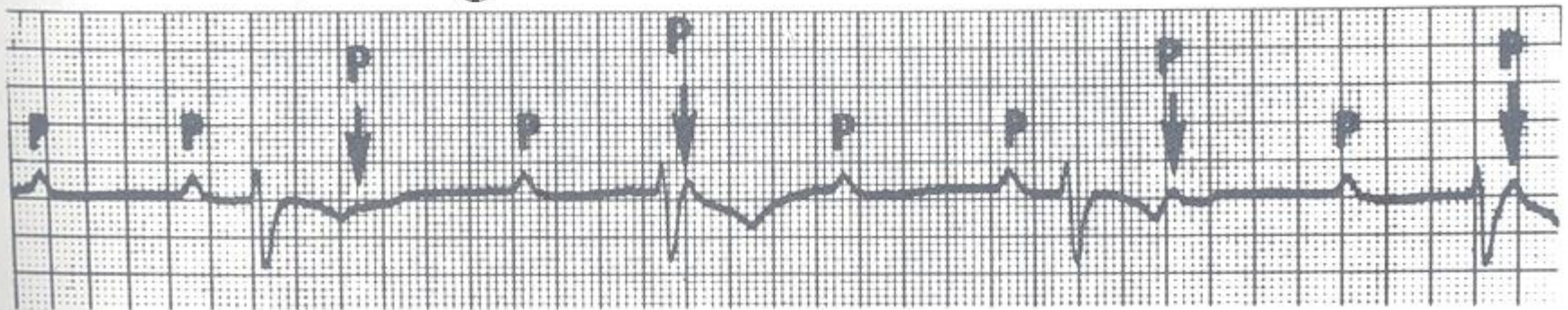
1st degree heart block

B



2nd degree heart block, 2:1

C



3rd degree heart block

Heart Rate

- ⊕ HR is the *number of cardiac cycles per minute*.
- ⊕ **Normal HR:** 60 - 100 beats/minute
 - If < 60 beats/minute = *bradycardia*, normal in the athletes
 - If > 100 beats/minute = *tachycardia*
- ⊕ **HR varies with the following factors**
 1. **Age:** - higher in new-born infants (120 beats/min)
 2. **Sex:** - higher in females (85 beats/min)
 3. **Time of the day:** - ↓ in morning, ↑ in evening.
 4. **Resting & sleep:** - ↓ the HR.
 5. **Physical training:** low in athletes (45-60 beats/min).
 6. **Body position:** - ↑ in standing & ↓ in supine positions.
- ⊕ **How to count the HR?**
 - Counting arterial **pulsation, heart sound & ECG cycles**

Regulation of Heart Rate (HR)

1. Nervous factors
2. Physical factors, e.g. temperature
3. Mechanical factors, e.g. right atrial distension
4. Chemical factors, e.g. catecholamines

Regulation of HR, cont'd

HR is determined by *rate of discharge of impulse from SA-node*.

1. Factors that directly stimulate SA-node:

- ⊕ **Body temperature** = ↑HR
- ⊕ **Right-atrial distension** → ↑Blood volume = ↑HR
 - ✓ Right-atrial distension → Stretch receptors → Medullary cardiovascular centre → Sympathetic stimulation → ↑HR
- ⊕ **Catecholamines**: AD, NAD = ↑HR

2. Factors influencing SA-nodal discharge by stimulating the medullary cardiovascular centre.

- ✓ Stimulation of higher centre (cerebral cortex, HT)
- ✓ Chemical content of the blood (↓PO₂, ↑PCO₂, ↑H⁺)
- ✓ Cardiovascular reflexes (baroreceptor, chemoreceptor)
- ✓ Stress (pain, fear, excitement)

Definition of important terms

- **Ventricular volumes:** the volume of blood in the ventricles.
- **Ventricular end diastolic volume (VEDV):** volume of blood in the ventricle at the end of **ventricular diastole** (relaxation phase).

$$\text{EDV} = 120\text{-}140 \text{ ml}$$

- **Ventricular end systolic volume (VESV):** volume of blood that remains in the ventricle at the end of **ventricular systole** (contraction phase).

$$\text{VESV} = 50\text{-}60 \text{ ml}$$

- **Stroke volume (SV):** volume of blood ejected from ventricle during **ventricular systole**.

$$\text{SV} = \text{VEDV} - \text{VESV} = 70\text{-}80 \text{ ml}$$

- **Cardiac output:** volume of blood ejected from heart per min.

$$\text{CO} = \text{SV} \times \text{HR} = 6 \text{ L/min}$$

Cardiac Output & Reserve

- @ **Cardiac Output (CO)** - is the amount of blood pumped by each ventricle in one minute.
- @ **CO** is the product of *heart rate (HR) & stroke volume (SV)*.
 - ✓ **HR** is the number of heart beats per minute.
 - ✓ **SV** is the amount of the blood pumped out by the ventricle with each beat.
- @ **Cardiac reserve** - is the difference b/n *maximal exercise & the resting CO*. **Or**
 - @ It is the **maximum percentage** that the CO can **increase above normal**.
 - CO = 5-6 liters per minute **at rest**
 - CO = 20-40 liters per minute **at maximum exercise**

Cardiac Output

⊕ **CO** is determined by SV & HR, in turn SV is determined by *three factors*:

1. **Preload/VR/EDV**
2. **Myocardial contractility**
3. **After load**

*VR = venous return

⊕ **VR** is affected by the following factors

1. **Central venous pressure**
2. **Systemic vascular resistance**
3. **Other factors**

- ▶ Sympathetic stimulation
- ▶ Blood volume
- ▶ Respiratory movements
- ▶ Skeletal muscle contraction

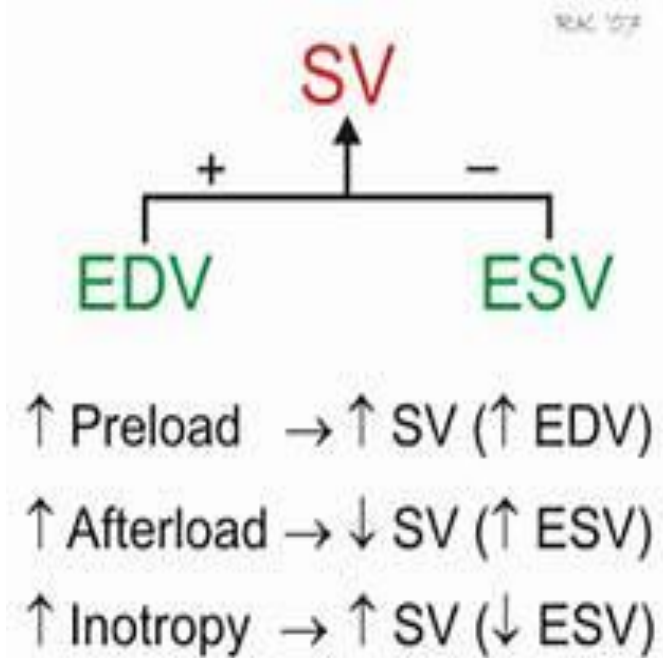
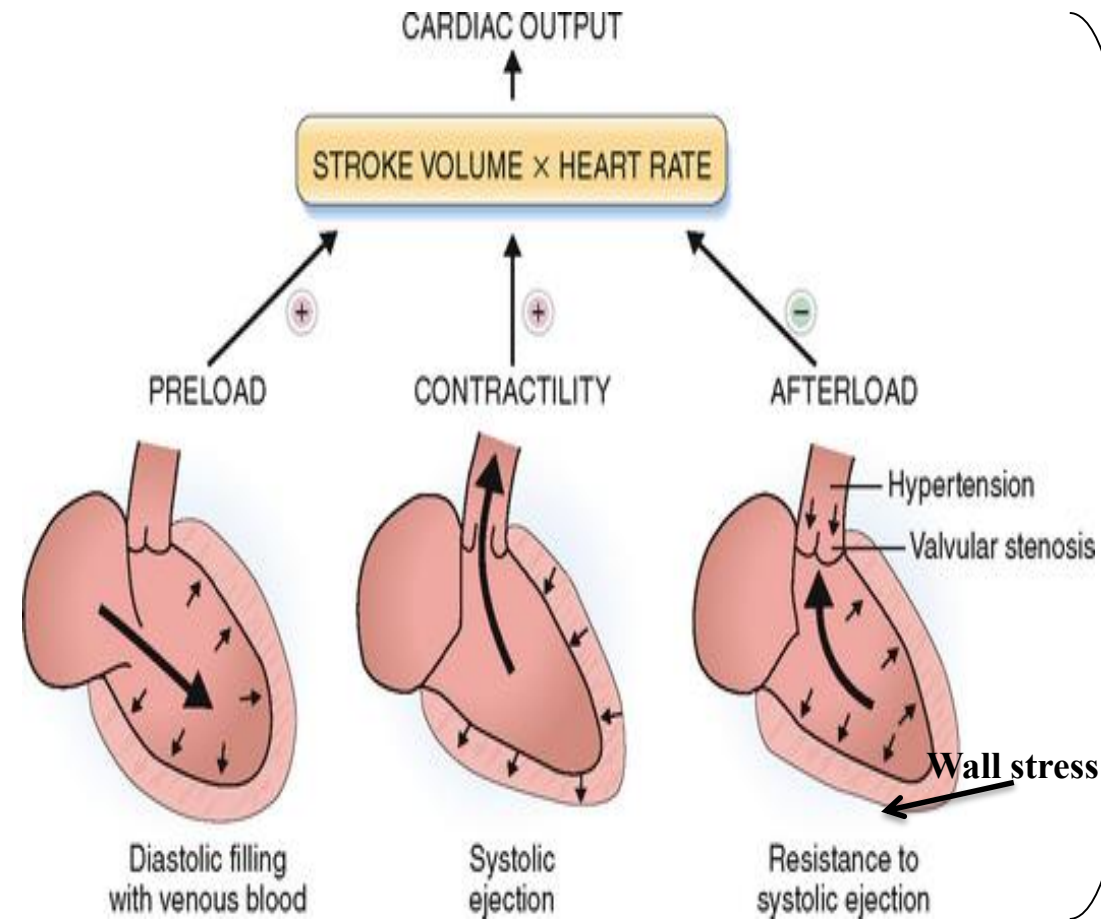
❖ During inspiration, **negative intrathoracic pressure** causes increased blood return into the right side of the heart.

Factors affecting SV

- **Preload:** – the amount ventricles are stretched by contained blood (EDV).
- **Contractility:** – cardiac cell contractile force due to factors other than EDV.
- **After load:** – pressure exerted by blood in the large arteries leaving the heart (**MAP - Mean Arterial Pressure**).

- Flow leaving either the right or left heart is **cardiac output (CO)**, whereas flow returning to the heart is **venous return (VR)**.
- At a steady state, CO & VR must be **equal**.

Factors affecting SV, cont'd



Regulation of Cardiac Output

- The CO is regulated by the two forces:
 1. **Intrinsic autoregulatory mechanism (Frank-Starling mechanism)**
 2. **Extrinsic regulatory mechanisms (ANS)**
 - **ANS:** sympathetic, parasympathetic stimulation
 - **Catecholamines:** adrenaline, noradrenaline
 - **Drugs:** cardiac glycosides
 - **Hormones:** adrenaline, glucagon & T3/T4

Intrinsic control (Frank-Starling mechanism)

- ⊕ The **Frank-Starling law** of the heart states that within the *physiological limits* the force of the cardiac contraction is proportional to its EDV (volume of the blood in the heart just before the ventricles begin to contract).
- ⊕ Increased EDV = $\uparrow$ strength of cardiac contraction, & $\uparrow$ SV.
- ⊕ This $\uparrow$ strength of cardiac contraction due to an $\uparrow$ in EDV is called the **Frank-Starling law of the heart**.
- ⊕ Increased end-diastolic volume (EDV):
 - = increased stretching of cardiac muscle
 - = increased strength of contraction
 - = increased stroke volume
 - = increased CO $\rightarrow$ increased BP

$\oplus$ EDV is the most important determinant of SV

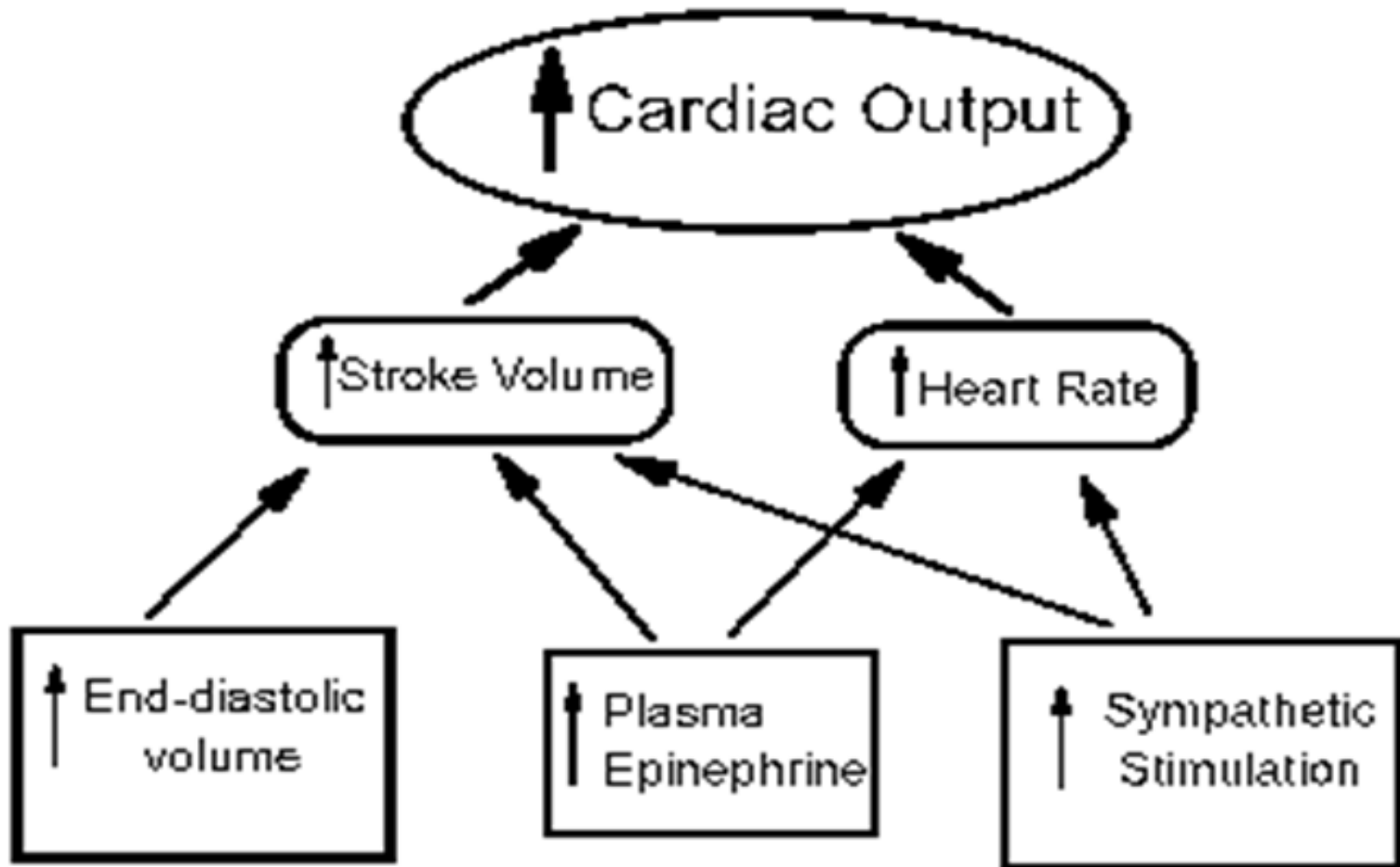
Extrinsic control

- ⊕ **Increased sympathetic stimulation**, increased the strength of contraction of the cardiac muscle.

Mechanism

- ⊕ **Sympathetic stimulation** causes release of NE which *increases permeability of muscle cell membranes to Ca^{2+} , and Ca^{2+} diffusion* causes more cross-bridges, and causes **stronger contraction**.

Extrinsic control, cont'd



Regulation of Cardiac Output

◆ **CO** (ml/min) = **HR** (72 beats/min) x **SV** (75ml/beat)

Autoregulation
(Frank-Starling “Law of the Heart”)



CARDIAC OUTPUT = STROKE VOLUME x HEART RATE

Contractility



**Sympathetic
Nervous System**



**Parasympathetic
Nervous System**



CARDIAC OUTPUT

is a function of

Heart rate

determined by

Rate of depolarization
in autorhythmic cells

is slowed by

Parasympathetic
innervation

is made faster by

Sympathetic
innervation

Epinephrine from
adrenal medulla

influence

Stroke volume

determined by

Force of contraction in
ventricular myocardium

is influenced by

Contractility

Length-tension
relationship of
muscle fibers

which varies with

Venous return

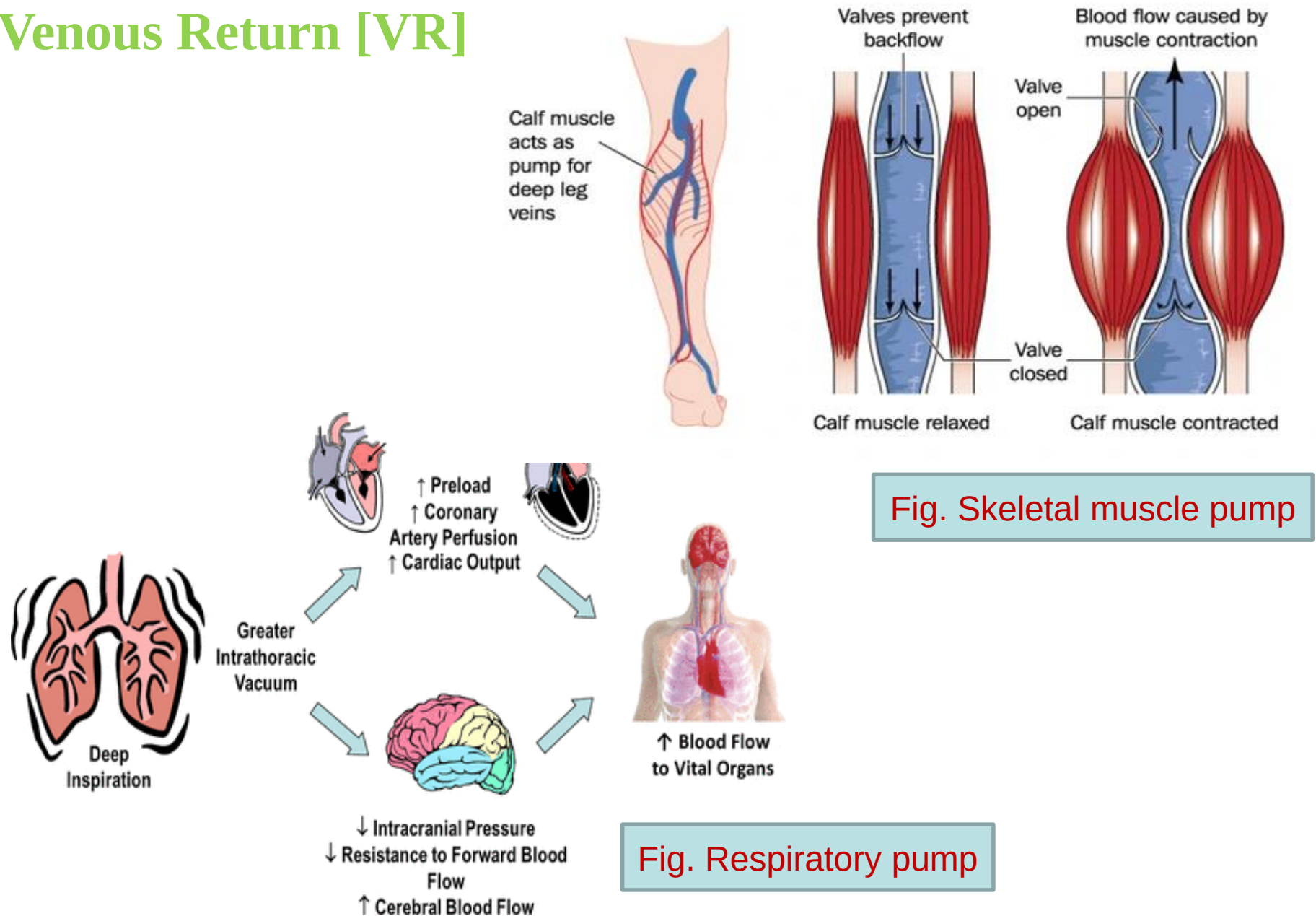
aided by

Skeletal muscle
pump

Respiratory
pump

Regulation of CO...

Venous Return [VR]



Hemodynamics

⊕ Hemodynamics is the study of the r/p b/n **flow**, **pressure**, & **resistance**, and other physical principles of blood circulation.

- **Blood vessels**: types, histology
- **Blood flow**: types & regulation
- **Blood pressure & regulation**
- **Heart Failure**

Blood Vessels: Classification

1. Elastic vessels (Windkessel vessels)

- ✓ **Example:** aorta & big arteries
- ✓ Pressure storing components (**high pressure reservoirs**)
- ✓ High ability of stretching & recoiling

2. Resistance vessels (Stop cock vessels)

- ✓ **Example:** small arteries & arterioles
- ✓ High muscular component
- ✓ Site of greatest vascular resistance in the circulation
- ✓ Regulate blood flow

3. Exchange vessels

- ✓ Made up endothelium & basement membrane
- ✓ Thin enough for exchange

4. Capacitance vessels (big to small veins)

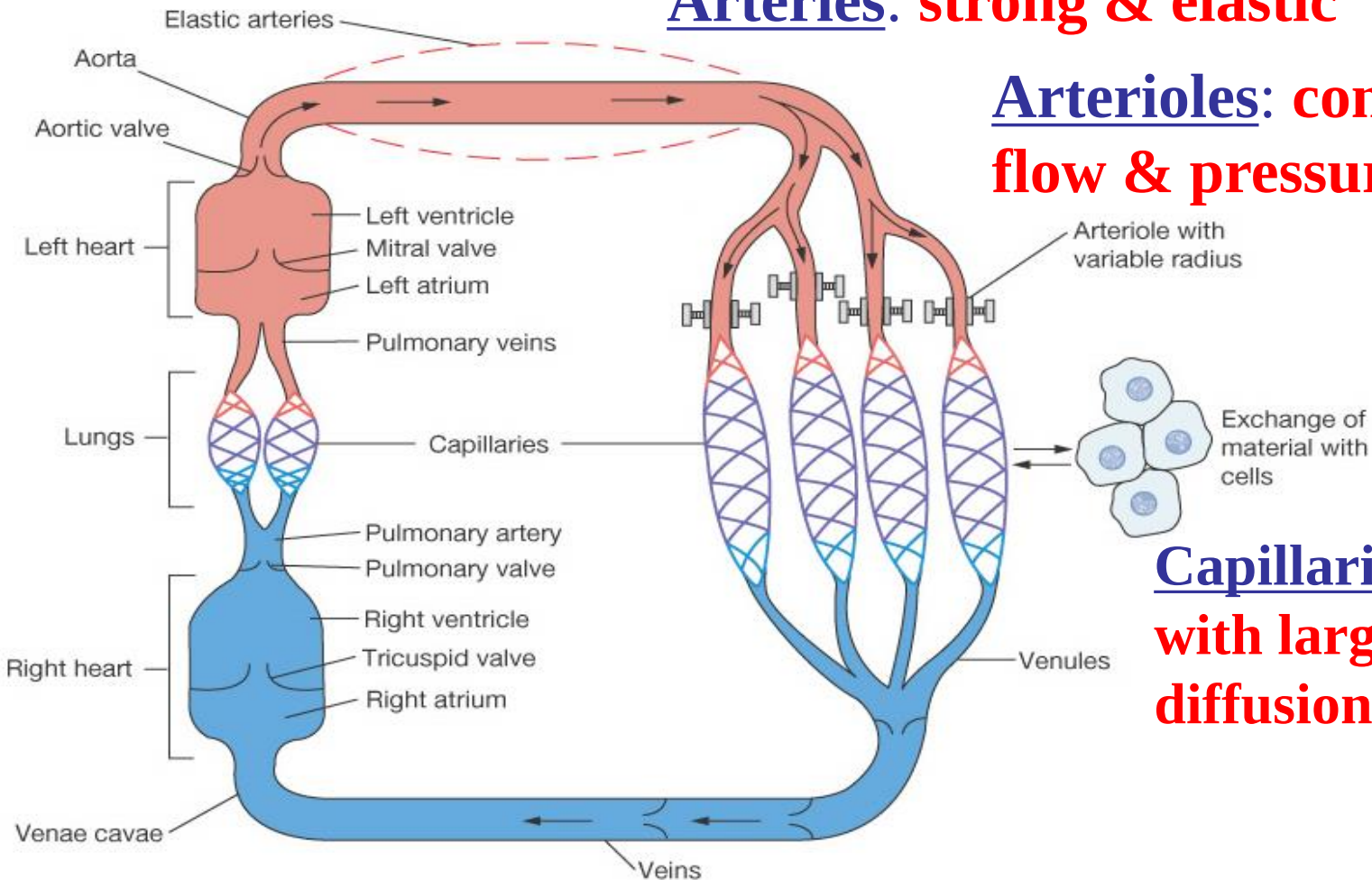
- ✓ Very high capacity of distension (**low pressure reservoirs**)
- ✓ Can accommodate large volume of blood (65% of b/d volume)

Blood vessel functions: overview

Arteries: **strong & elastic**

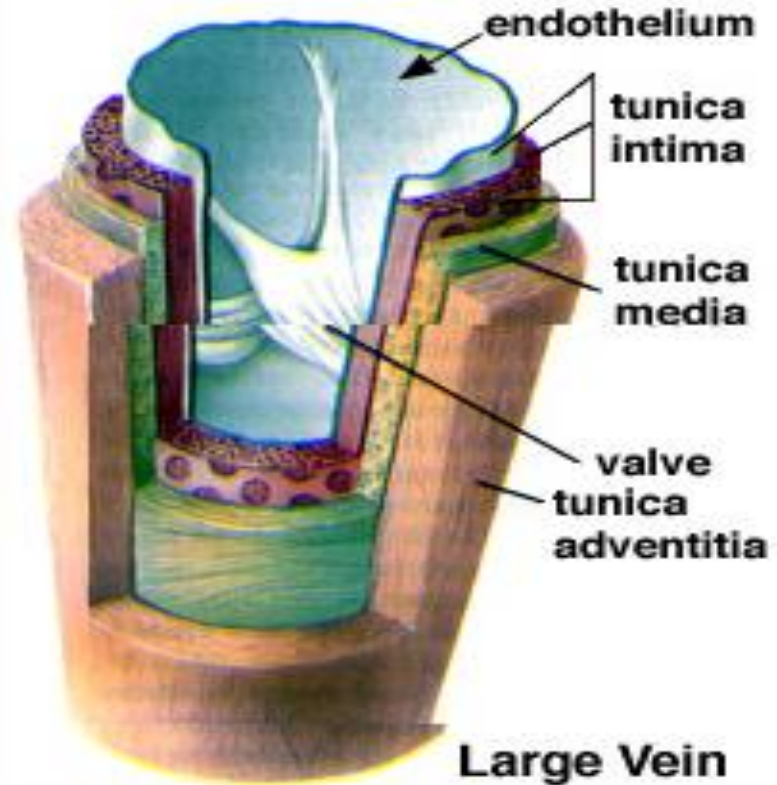
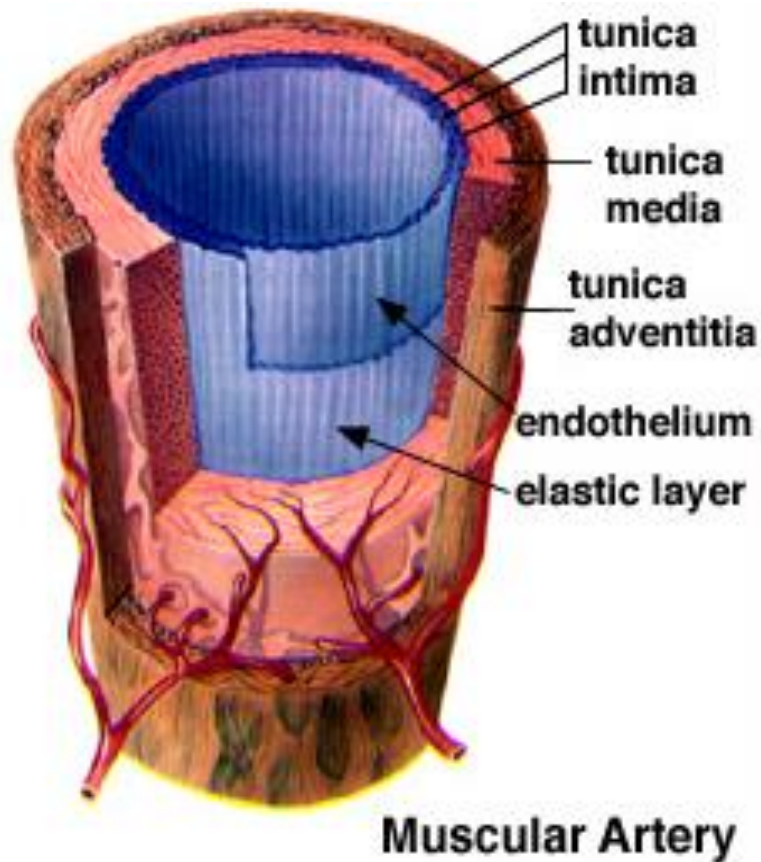
Arterioles: **control blood flow & pressure**

Capillaries: **thin & with large area for diffusional exchange**



Veins: **large compliant, low resistance. Veins with valves assures blood return.**

Histology of blood vessels



Arteries and Arterioles

- Elastic tissues
 - Rebounds
 - Evens flow
- Smooth muscles (muscular layer)
- Fibrous tissue
 - Tough
 - Resists stretch

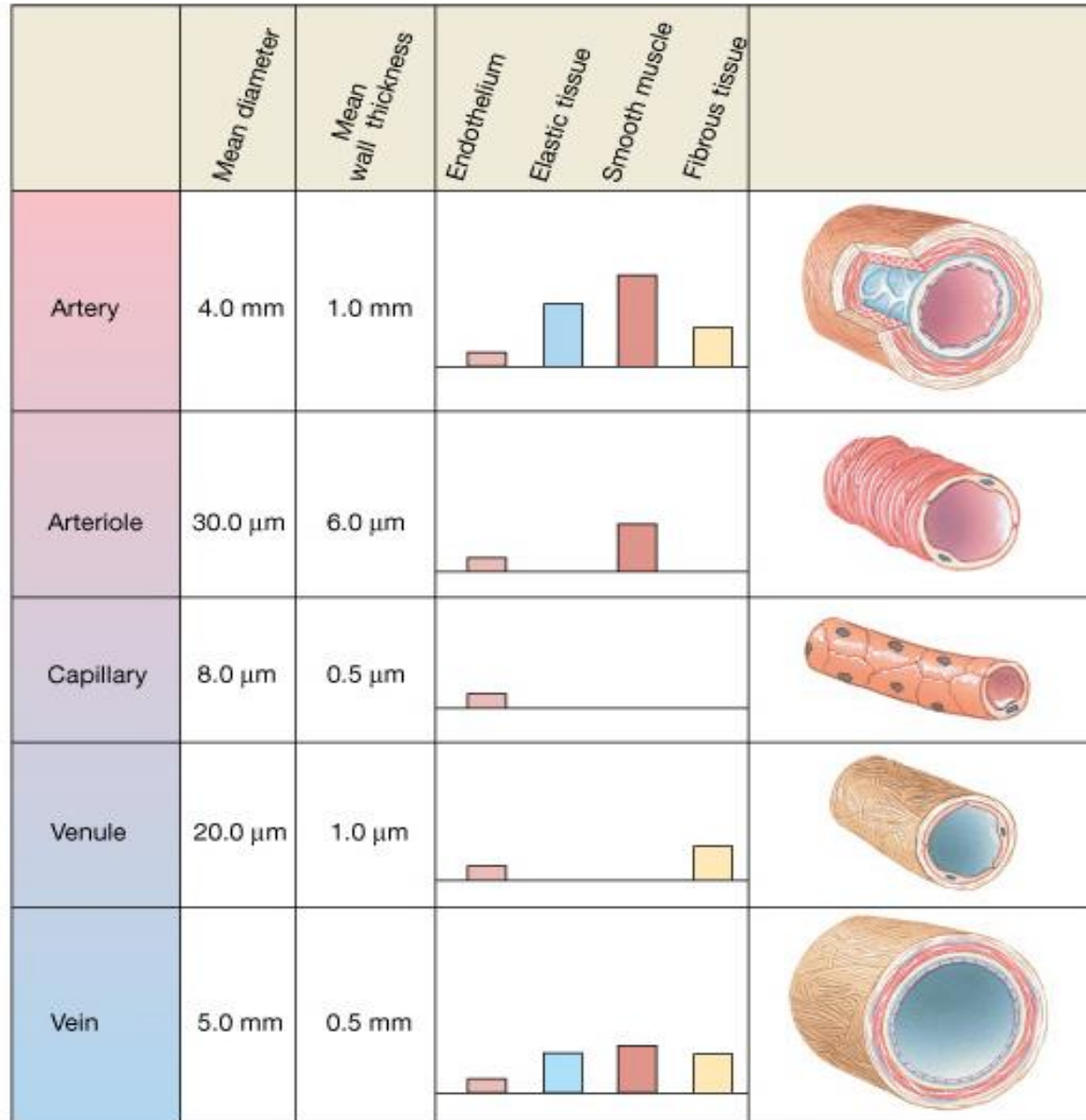
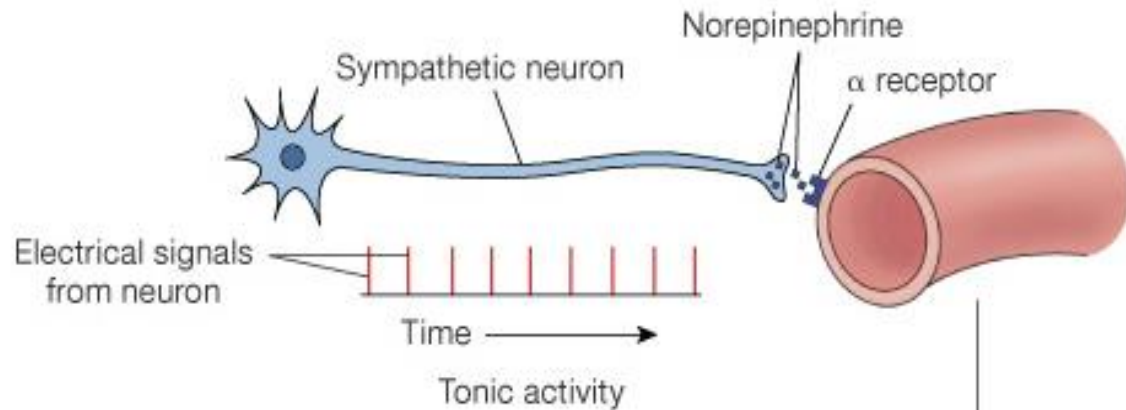


Figure : Blood vessels

Arteriole diameter is controlled by tonic release of norepinephrine.

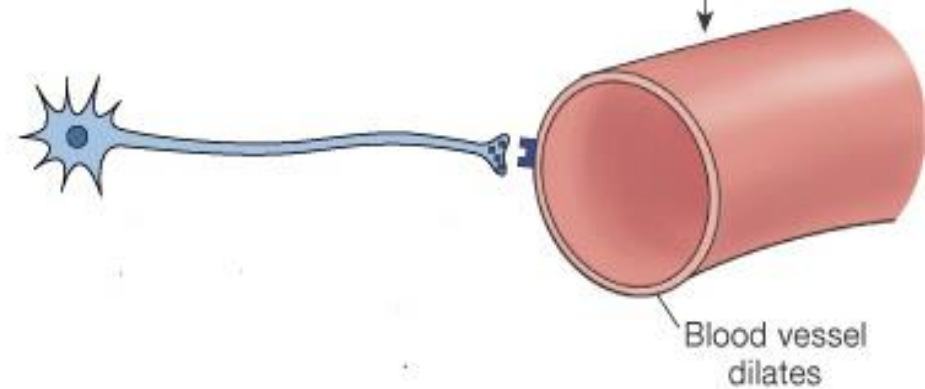
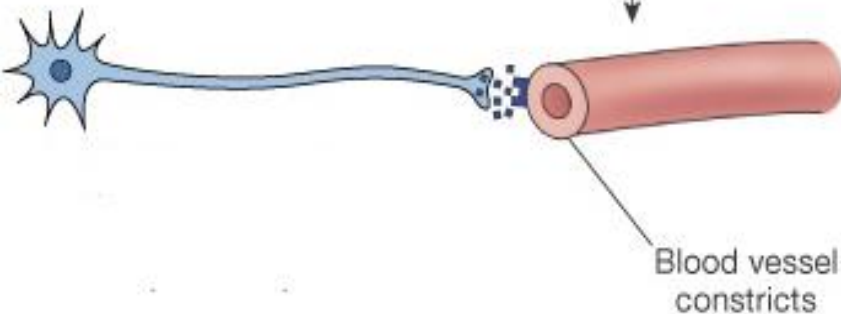


**Sympathetic nerves
Control on arteriolar
smooth muscle
contraction**

Change in signal rate

↑ Norepinephrine release
onto α receptors

↓ Norepinephrine release
onto α receptors



Capillaries

- The pressure inside is 35 to 15 mm Hg.
- 5% of the blood is in the capillaries.
- Serves for exchange of gases, nutrients, & wastes.
- Flow is slow & continuous.

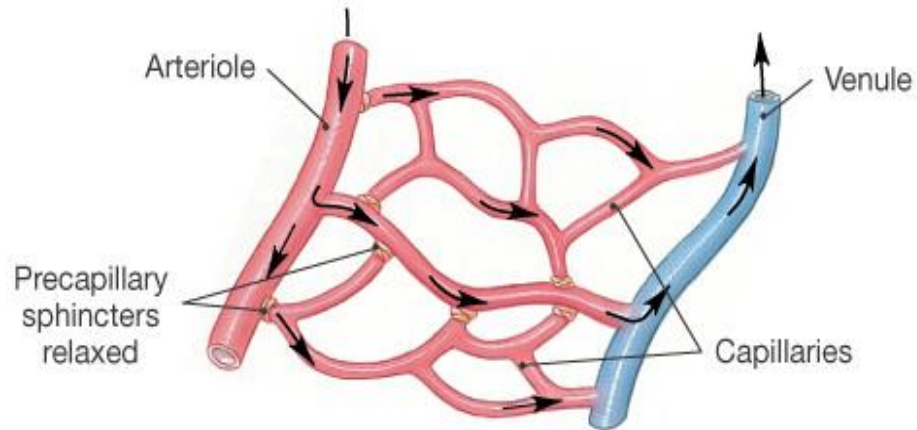
Capillary blood flow

- Has lowest velocity.
- Has largest total cross sectional area.
- Hydrostatic pressure drops slightly.

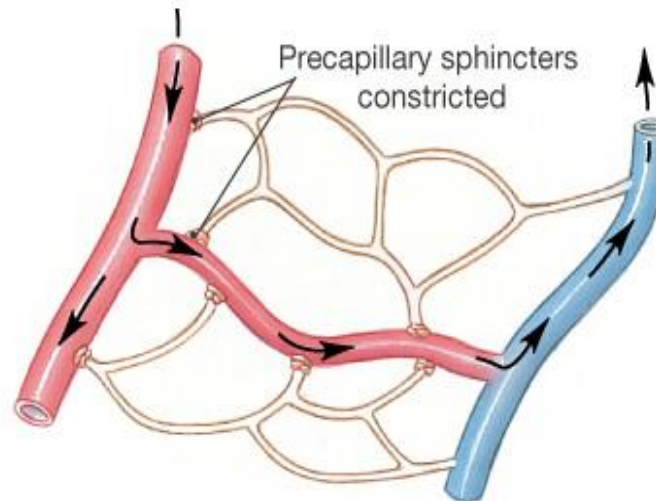
Capillaries

- Blood flow controlled by pre-capillary sphincter muscles *via* metabolic regulation
- No innervations

(a) When precapillary sphincters are relaxed, blood flows through all capillaries in the bed.



(b) If precapillary sphincters constrict, blood flow bypasses capillaries completely and flows through metarterioles.



3 structural types of capillaries

❑ Continuous, fenestrated & discontinuous

1. Continuous Capillaries → [not leaky]

- ✓ Adjacent cells that are held together with **tight junctions** [uninterrupted lining].
- ✓ E.g. Continuous capillaries of the brain:
 - ✓ Found in the blood-brain barrier [**BBB**]

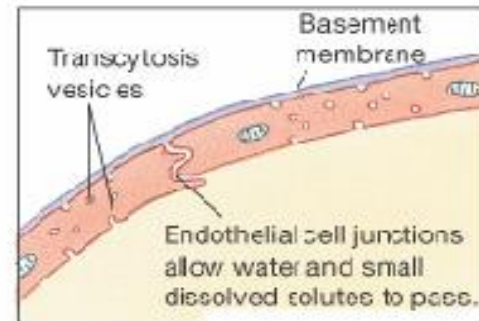
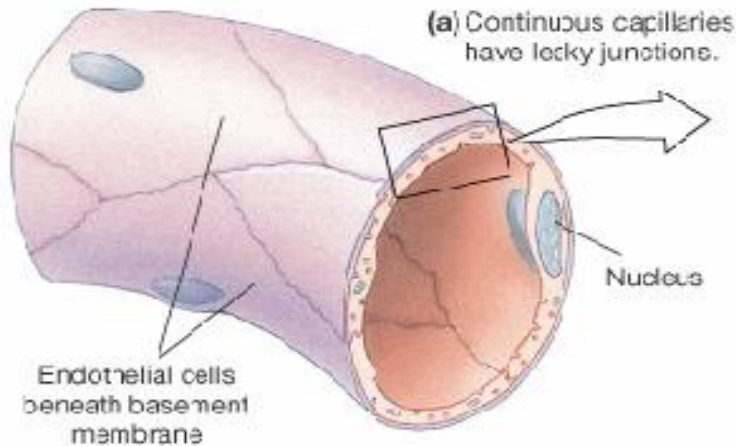
2. Fenestrated Capillaries [with pores or fenestrations] → [medium leaky]

- ✓ Found wherever **active capillary absorption or filtrate formation** occurs (e.g., in **kidneys** =glomerular capillary & in **small intestine**)

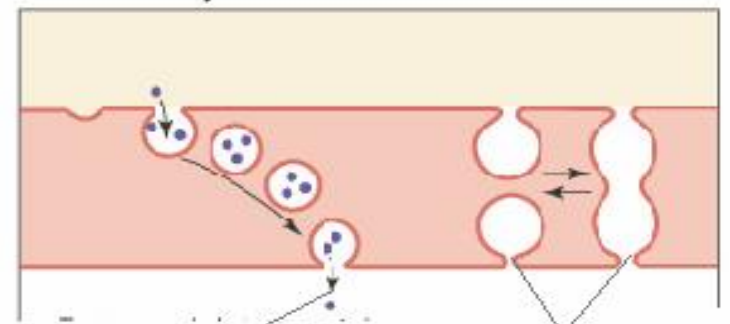
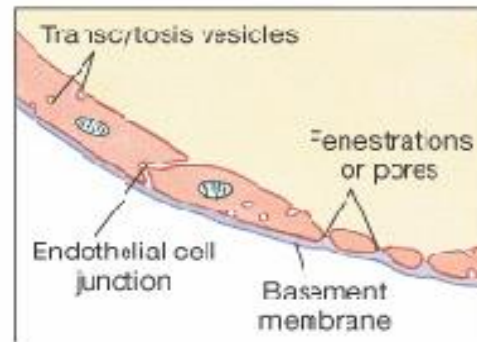
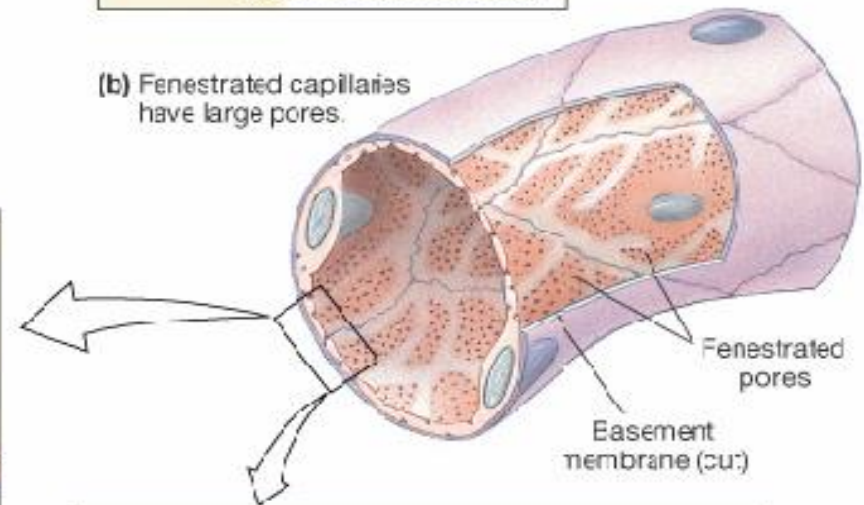
3. Sinusoids capillaries [discontinuous] → [highly leaky]

- ✓ Have **broad openings** between endothelial cells.
- ✓ Are Leaky or fenestrated capillaries with **large lumens**.
- ✓ Found in the **liver, spleen, bone marrow, lymphoid tissue....**
- ✓ Allow large molecules (proteins & blood cells) to pass

Types of Capillaries



(b) Fenestrated capillaries have large pores.

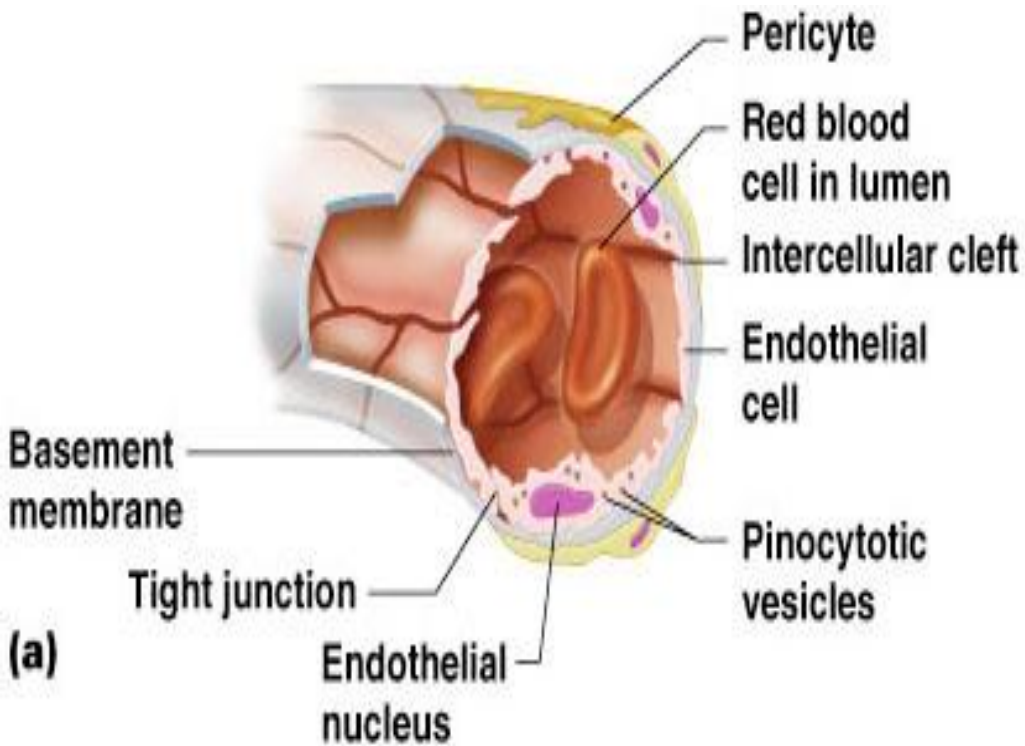


Fenestrated capillaries are found in the glomerulus of the kidney

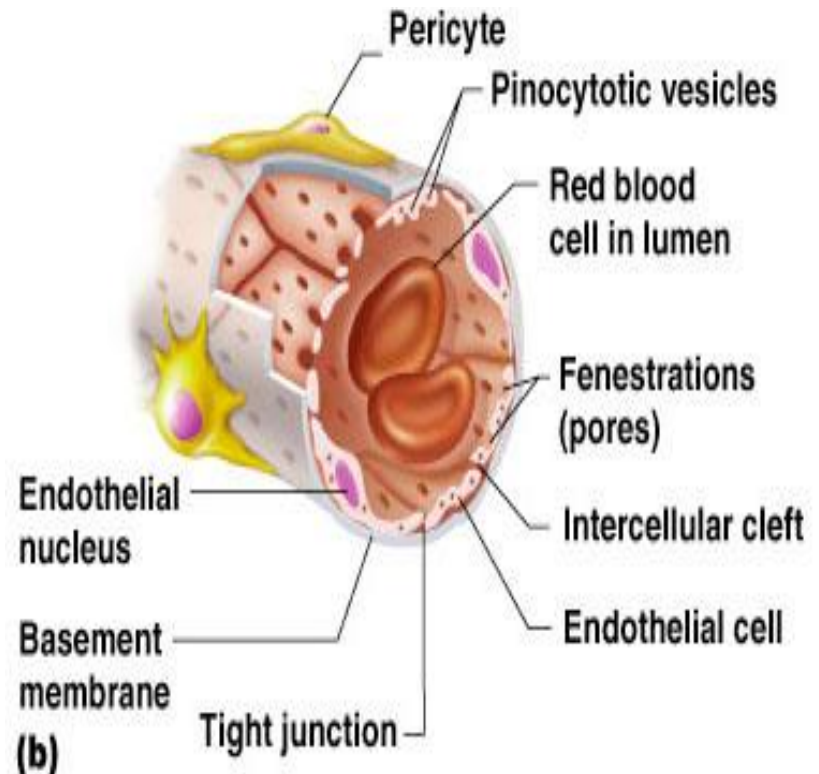
Transcytosis brings proteins across Endothelium

Some vessels form Channels

Some vessels form channels



Continuous Capillaries

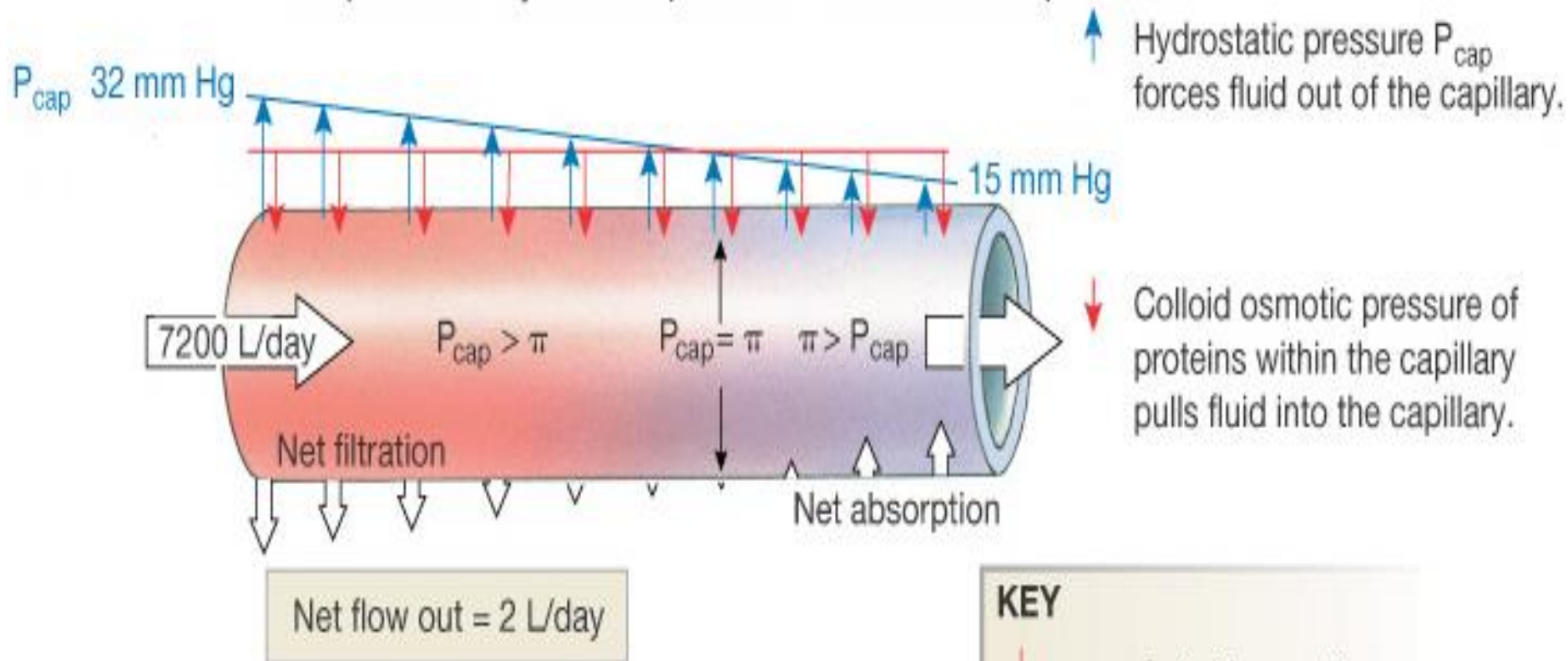


Fenestrated capillaries

Capillary Exchange:

(a) Filtration in systemic capillaries

Net pressure = hydrostatic pressure – colloid osmotic pressure



KEY

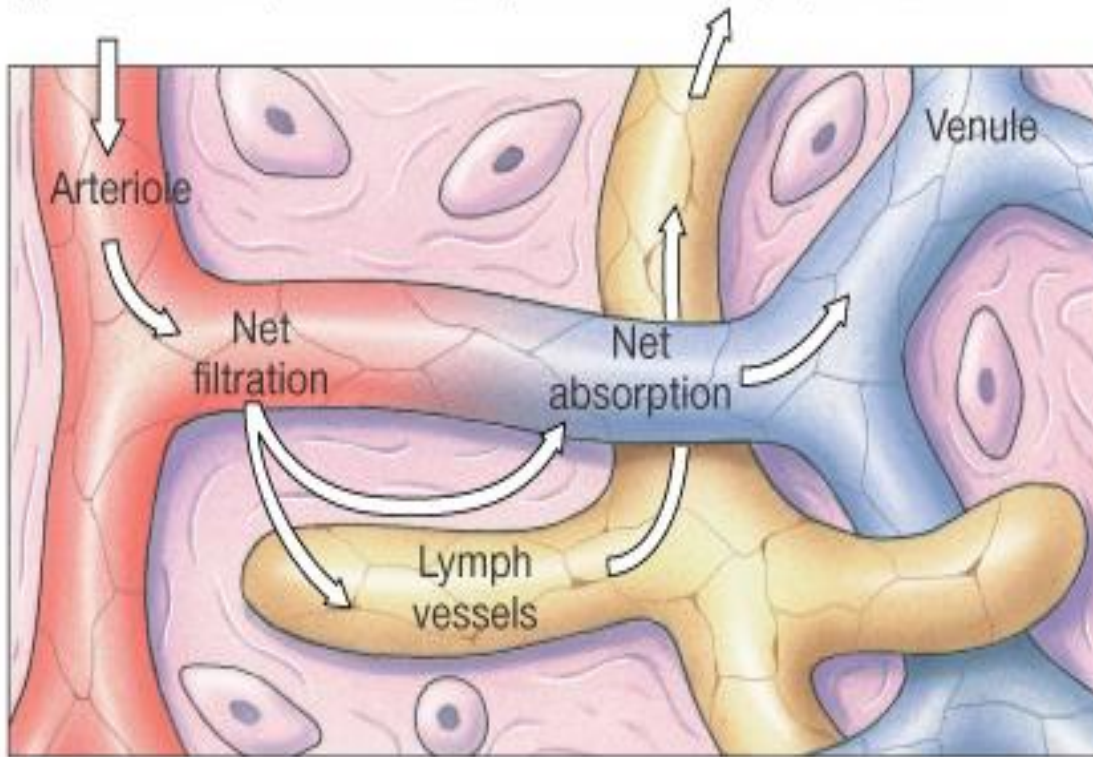
↓ π = Colloid osmotic pressure

↑ P_{cap} = Capillary hydrostatic pressure

Net outflow into ECF

- Net filtration – net absorption = net outflow
- About 2 L/day collected by lymph vessels

(b) Relationship between capillaries and lymph vessels



The excess water and solutes that filter out of the capillary are picked up by the lymph vessels and returned to the circulation.

Figure: Fluid exchange at the capillary

Capillary Blood Flow-Velocity

- Lowest velocity
- Largest total cross sectional area
- Hydrostatic pressure drops slightly

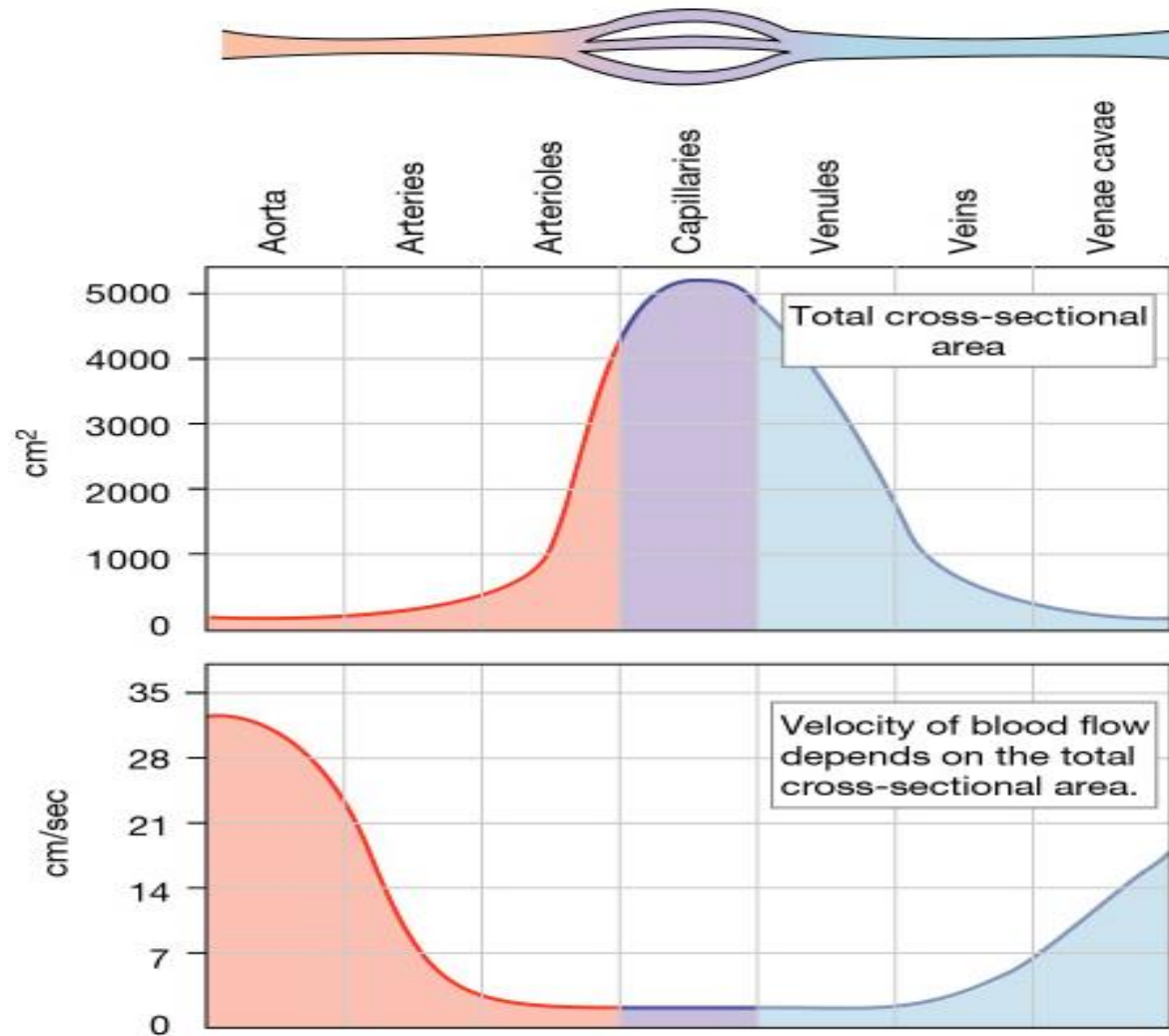
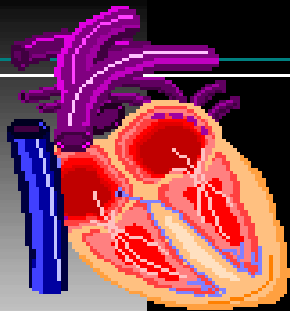


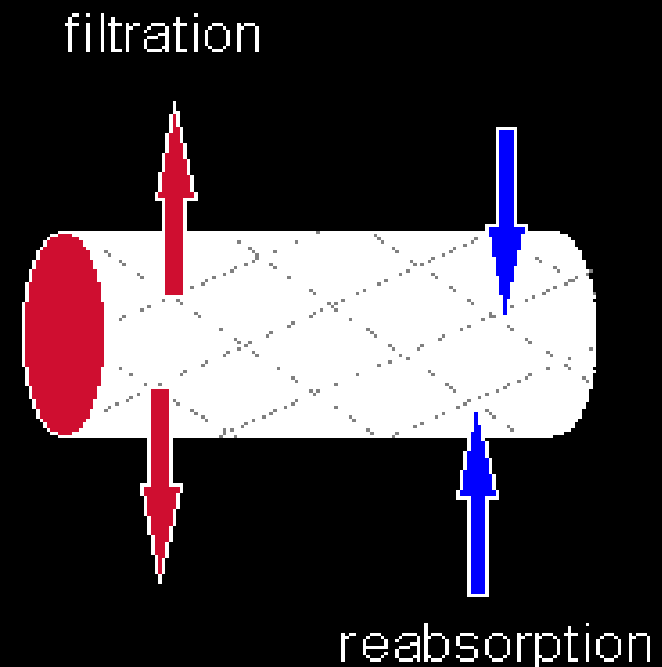
Figure. The velocity of flow depends on the total cross-sectional area

Exchange of substances in the capillaries



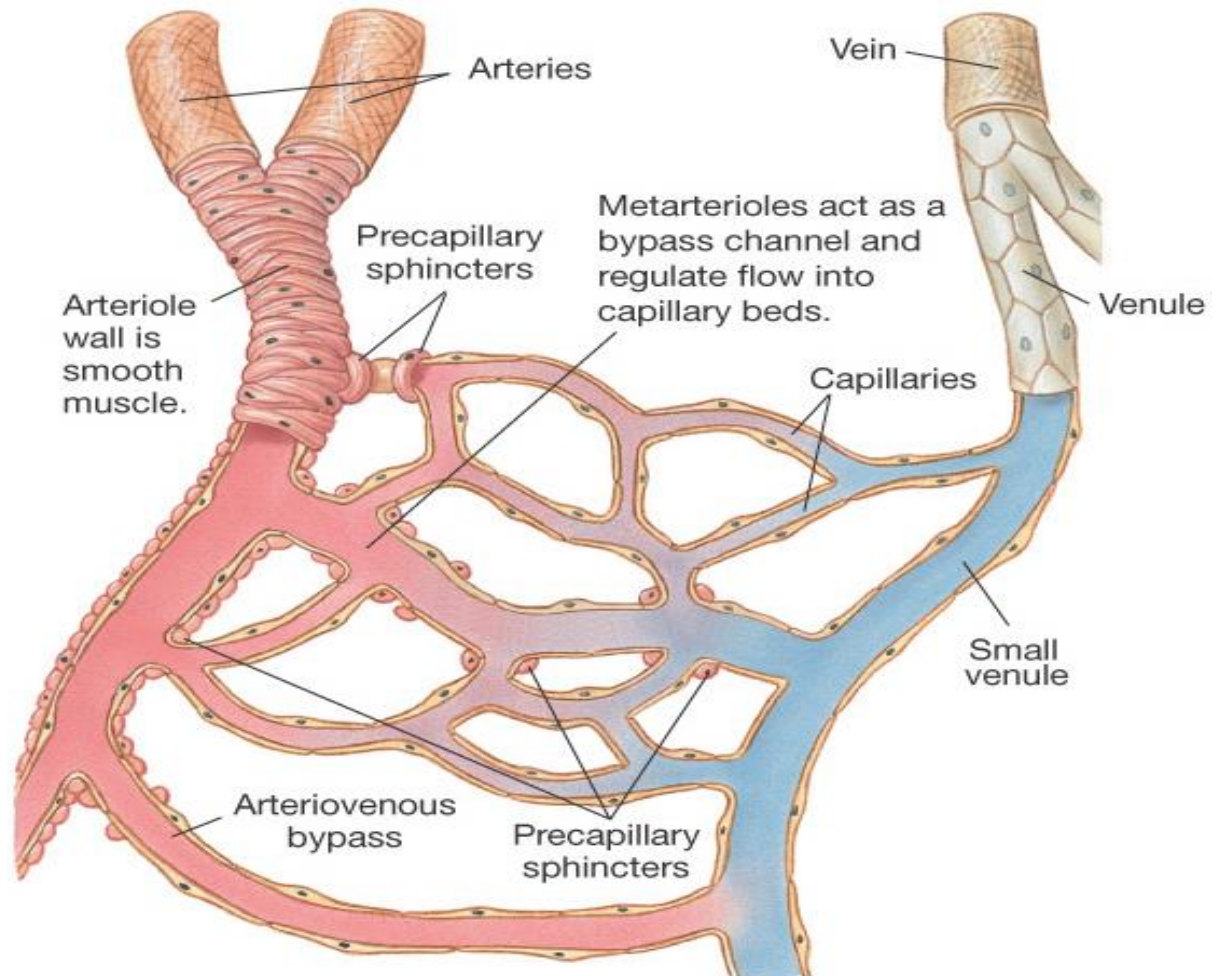
Capillaries: Bulk Flow

- ◆ Passive movement through **pores**:
 - water & solutes
- ◆ Balances the **extracellular fluid**:
 - plasma
 - interstitial fluid
- ◆ **Filtration** generally exceeds **reabsorption**

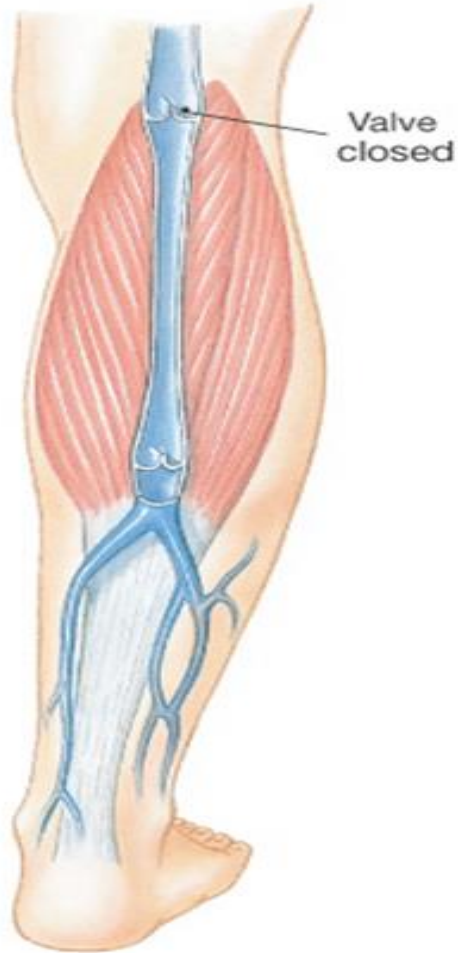


Veins & Venules (Vs. Arteries)

- ✦ Thinner walls
- ✦ Larger diameter
- ✦ Closer to skin
- ✦ Less muscle
- ✦ Less elastic
- ✦ Have **valves**



Valves in the veins prevent backflow of blood.



When the skeletal muscles compress the veins, they force blood toward the heart (the skeletal muscle pump).

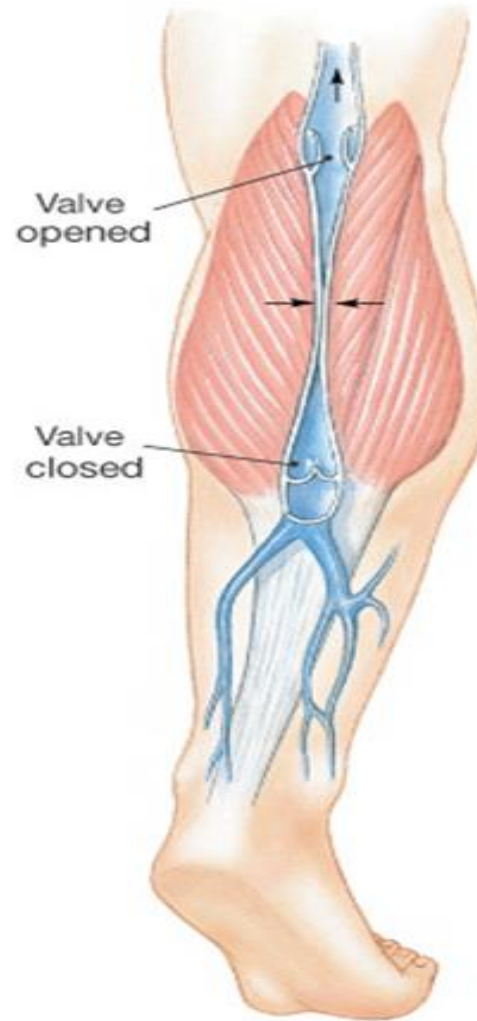
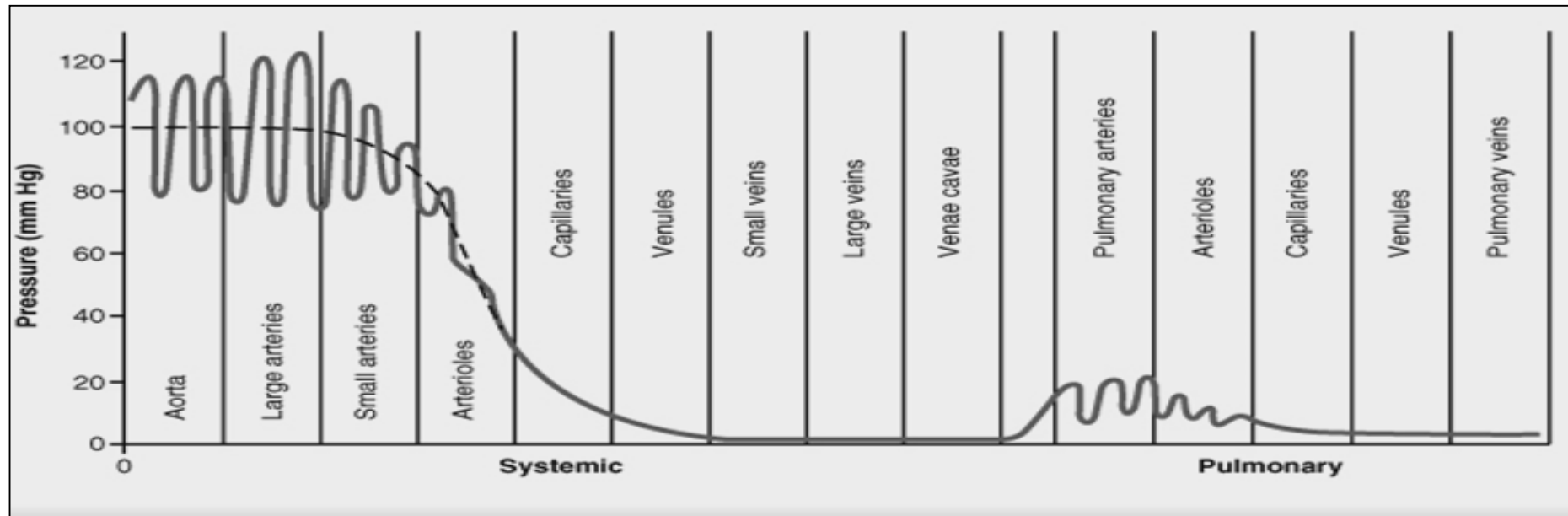


Fig. Flow-regulating valves in compliant veins

Cont'd

	Arteries	Veins	Capillaries
1.Functions	Send blood from heart =Distributer	Send blood to heart =Returner	Allow for material exchange with tissues [cells] =Supplier
2.Pressure	High	Low [less resistant to BF]	Low
3.Lumen diameter [size]	Narrow	Wide	Extremely narrow
4.Structure of the Wall	Thick	Thin	Extremely thin [one cell thick]
5.Muscle & elastic fibers	Large amounts [muscular]	Small amounts	None
6.Valves	No	Yes [prevent back flow]	No
7.Blood flow [BF]	V . Fast	Slow	V. Slow



120/80

25/10

Blood flow (BF)

- ◆ **BF** is the amount of blood that moves to a particular organ in a given time.
 - ◆ Total BF is equal to CO; 5-6 L/min.
 - ◆ ***BF is determined by 2 factors:***
 1. **Pressure difference** b/n 2 ends of the vessel, i.e., the force that pushes blood through vessels.
 2. **Resistance of flow**, expedience to flow through vessels
- Where, Q = Blood flow
 ΔP = Change in pressure
 R = Resistance
- $Q = \Delta P / R$
- ⊕ **Ohms Law**: states that BF is *directly proportional* to the ΔP but *inversely proportional* to **resistance** (R).

Blood flow

$$BF = \frac{\Delta P}{R}$$

Blood Flow

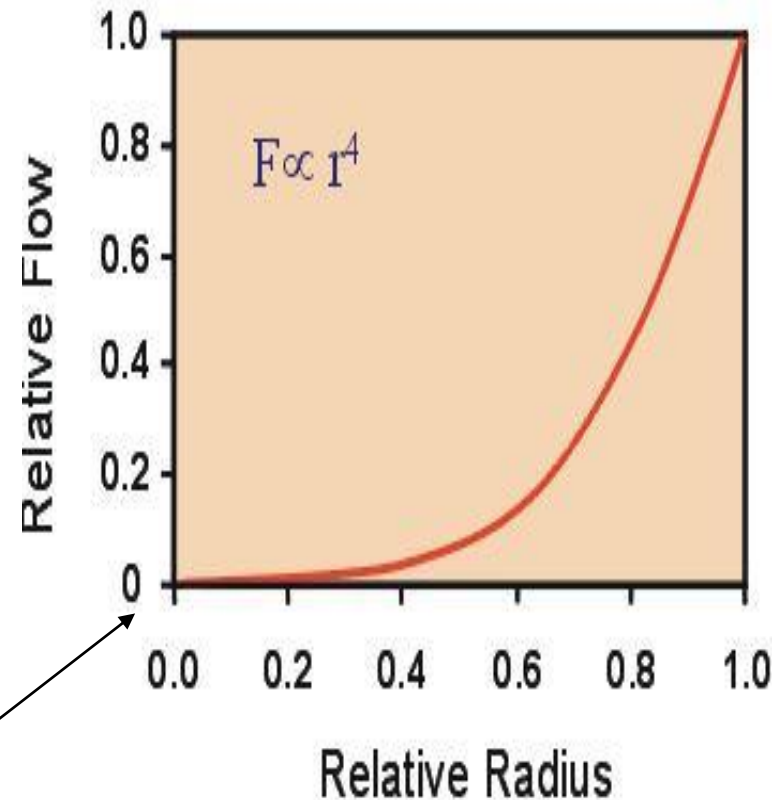
Blood flow (F) depends on:

① Pressure Gradient (ΔP) - heart

② Resistance (R) - blood vessels

- viscosity
- vessel length
- vessel diameter

Vessel diameter is the main determinant of vascular resistance.



Types of blood flow

⊕ **There are two types of blood flow** inside blood vessels:

- 1. Laminar flow**
- 2. Turbulent flow**

1. Laminar blood flow also called streamline flow

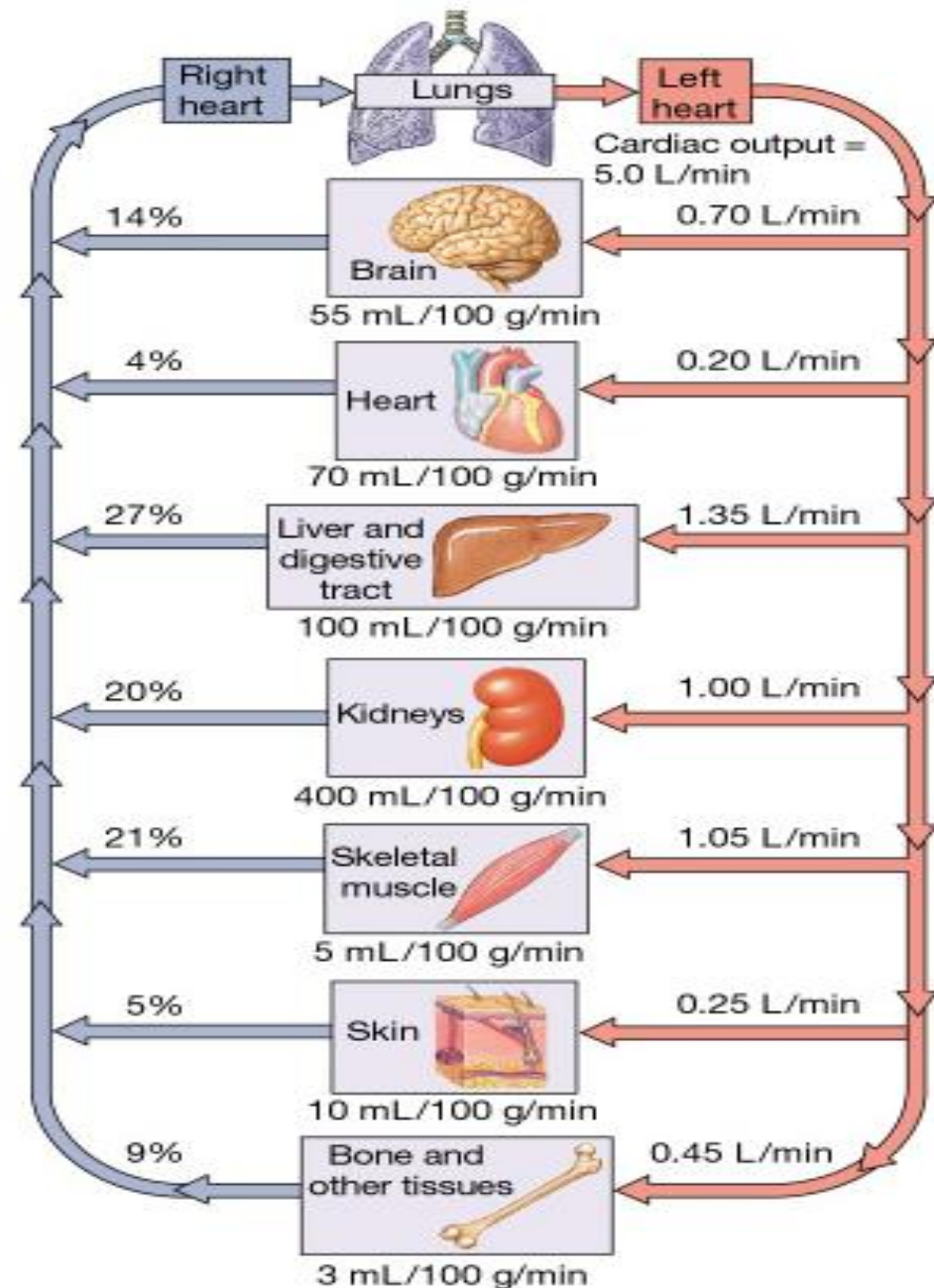
- Blood flow in a steady rate through long smooth vessels.
- Blood cells move in a straight line.
- Blood cells at the centre of vessel move faster.
- Laminar flow is **silent**.

2. Turbulent flow

- Blood flow in all directions in a vessel, and continuously mixed.
- Turbulent blood flow produces **sound**.
- Occurs at vessels with high elastic content & during high velocity.

Distribution of blood to the body organs:

- ▶ 65% in veins & venules
- ▶ 25% in arteries
- ▶ 5% in capillaries
- ▶ 5% in heart



Regulation of blood flow

- ◆ Each organ can control the rate of BF based on the degree of requirement of O₂, nutrients & removal of wastes.
- ◆ *Regulatory mechanism of regional BF* are classified into two categories: **short term & long term** regulatory mechanisms.

A. Short term regulatory mechanisms:

- Minute by minute adjustment of local BF according to metabolic demand by producing vasodilators.
 - a. Metabolic auto-regulation
 - b. Nervous regulation: SNS, PNS
 - c. Humoral regulation: vasoactive substances

B. Long term regulatory mechanism: mechanisms that adjust BF over long time. e.g. **Angiogenesis**.

Regulation of blood flow

VASOMOTION = Intermittent flow due to constriction-relaxation cycles of precapillary sphincters or arteriolar smooth muscle (5 - 10/min).

AUTOREGULATION OF VASOMOTION:

1. Oxygen Demand Theory (Nutrient Demand Theory)

- O₂ is needed to support contraction

2. Vasodilator Theory

- Vasodilator substances produced (via ↓ O₂)

e.g. **Adenosine** → Heart

CO₂ → Brain

Lactate, H⁺, K⁺ → Skeletal Muscle

3. Myogenic Activity

Arterial blood pressure (ABP)

- BP is a force exerted by the blood on the wall of blood

Patients with a very narrow PP are a cause for concern. A narrow PP could result from a reduced SV. Reduced SV can result from inadequate BV as might occur after hemorrhage or from a failing heart.

ic blood pressure.

m Hg)

. 60 – 90 mm Hg (60 mm Hg)

PP: SBP – DBP

MAP = DBP + 1/3PP (SBP - DBP)

- Hypotension vs Hypertension**

- Three factors determine ABP:** $ABP = CO \times TPR$

1. Cardiac output
2. Blood volume
3. Peripheral vascular resistance

Variations in the ABP: Causes

1. **Age:** ABP rises as age increases
Newborn: 70/50, infants: 90/60, children: 100/70
Adults: 120/80 & old age (>60 years): 150/90
2. **Sex:** After the age of 50, $F > M$
3. **Time of the day:** ↓morning, ↑evening
4. **Body wt:** ABP higher by in obese
5. **Gravity:** every 1 cm distance below the level of the heart increases pressure by 0.77 mm Hg.
6. **Exercise:** increases ABP by 40 – 50 mm Hg
7. **Emotion:** increases ABP by 10 – 30 mm Hg
8. **Deep sleep:** decreases ABP by 20 mm Hg
9. **Thermal stress:** decreases ABP

Pulse and Mean Arterial Pressures

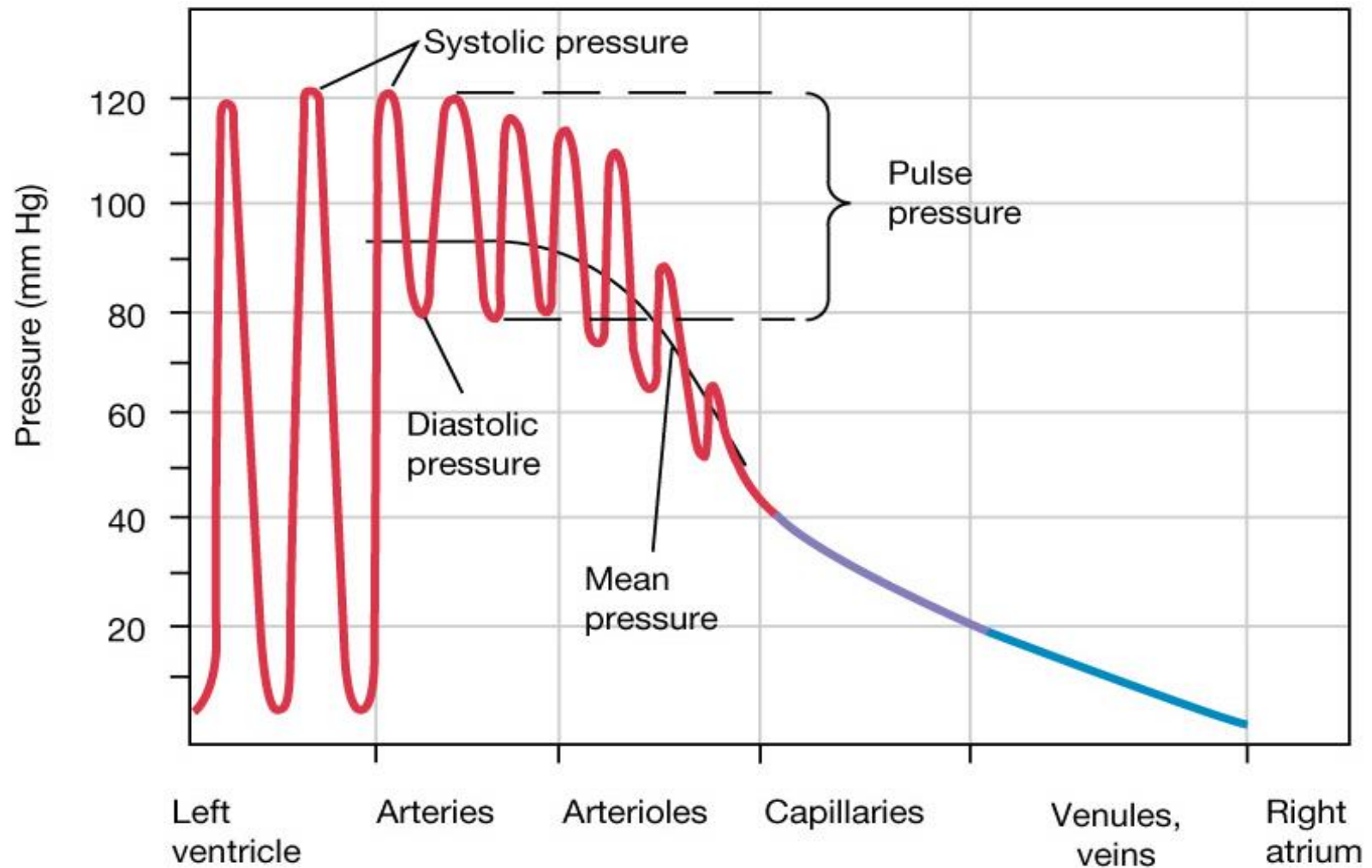
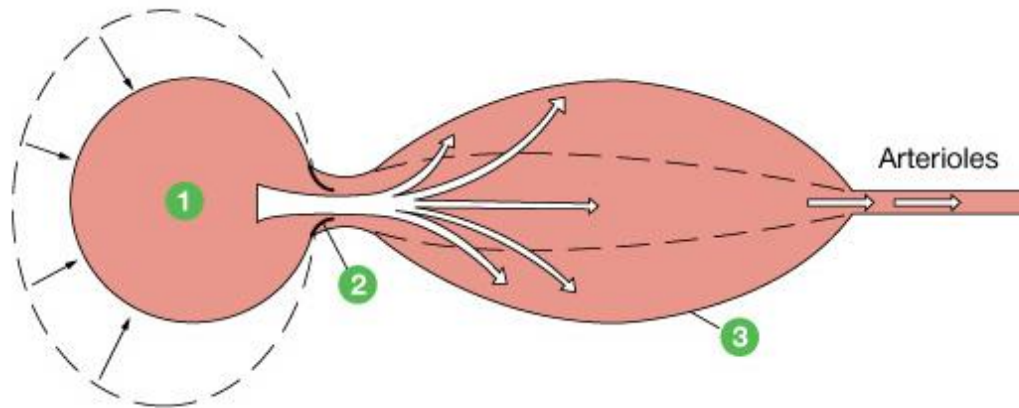


Figure: Pressure throughout the systemic circulation

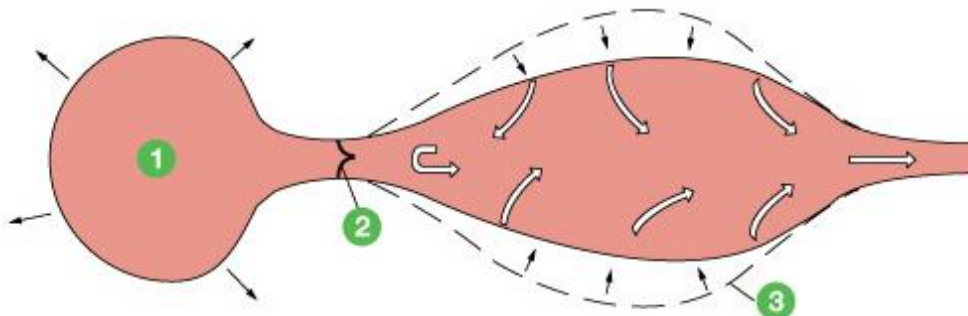
Large arteries: blood distribution and maintenance of blood pressure during **diastole**.

(a) Ventricular contraction



- 1 Ventricle contracts.
- 2 Semilunar valve opens.
- 3 Aorta and arteries expand and store pressure in elastic walls.

(b) Ventricular relaxation



- 1 Isovolumic ventricular relaxation
- 2 Semilunar valve shuts.
- 3 Elastic recoil of arteries sends blood forward into rest of circulatory system.

Methods of Measurement of the ABP

A. Direct (invasive) method

❑ It measures ABP internally.

✓ By using a sensitive IV catheter inserted into the artery.

❑ Due to its **invasiveness** & its **inability** to be applied to large number of patients;

✓ this method is not routinely practical.

B. Indirect (non-invasive) method

1. **Oscillometric method** (Automated)

✓ The cuff fitted on the upper arm, &

✓ Inflated & released by an electrically operated **pump** & **valve**.

✓ Uses an electronic pressure **sensor** with a numerical **readout** of **BP**.



Methods of Measurement...

2. Auscultatory method (Manual) →→ Hearing Korotkof sound

□ This method BP determine using **sphygmomanometer** & **stethoscope**.

1. **Sphygmomanometer**: an instrument that measures BP w/c consists of

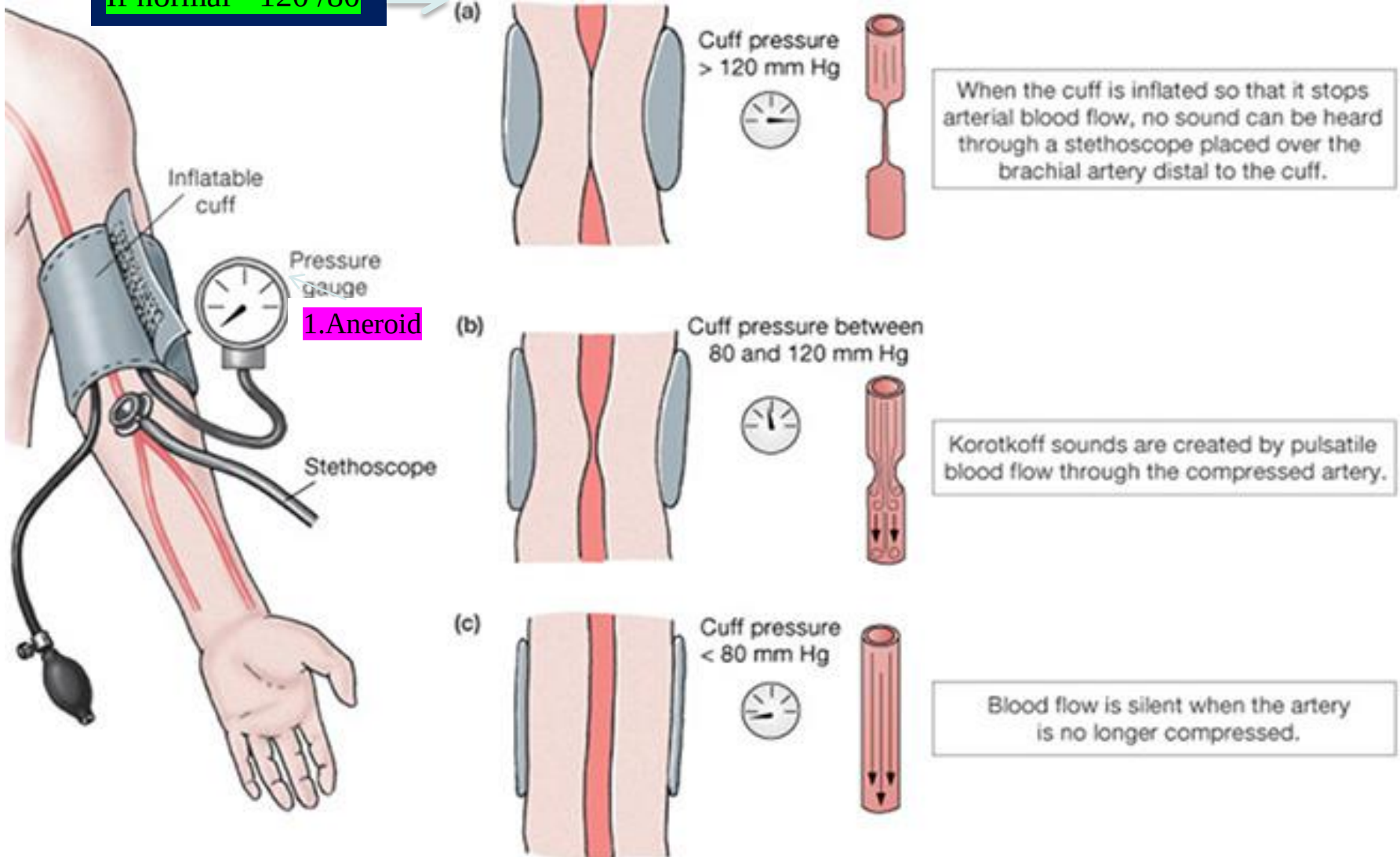
- ✓ A cuff
- ✓ Inflating bulb & valve
- ✓ Pressure gauge

2. **Stethoscope**: is an instrument used to hear sound

Measurement of “ABP” by Auscultatory method

If normal = 120 / 80

1. Aneroid



Measuring arterial blood pressure

Clinical Note

Measuring Blood Pressure

Blood pressure is measured in the human noninvasively with a device called a *sphygmomanometer*. Basically, an inflatable cuff is wrapped around the upper arm and inflated to about 170 mm Hg. A stethoscope is placed over the brachial artery just below the cuff and the air is slowly let out of the cuff. To begin with, the brachial artery is collapsed by the pressure. As the pressure in the cuff falls below the systolic blood

pressure, the vessel pops open briefly during each systole. As the flow starts and stops it is turbulent and audible. As cuff pressure falls below diastolic pressure, the vessel remains patent throughout the cardiac cycle and the sounds disappear. The cuff pressure where the sounds are first heard is taken to be the patient's systolic pressure and the pressure at which they disappear as the diastolic pressure.

Regulation of ABP

- The centers for CVS control are located in the **medulla oblongata** & lower pons, close to the centers regulating respiration.

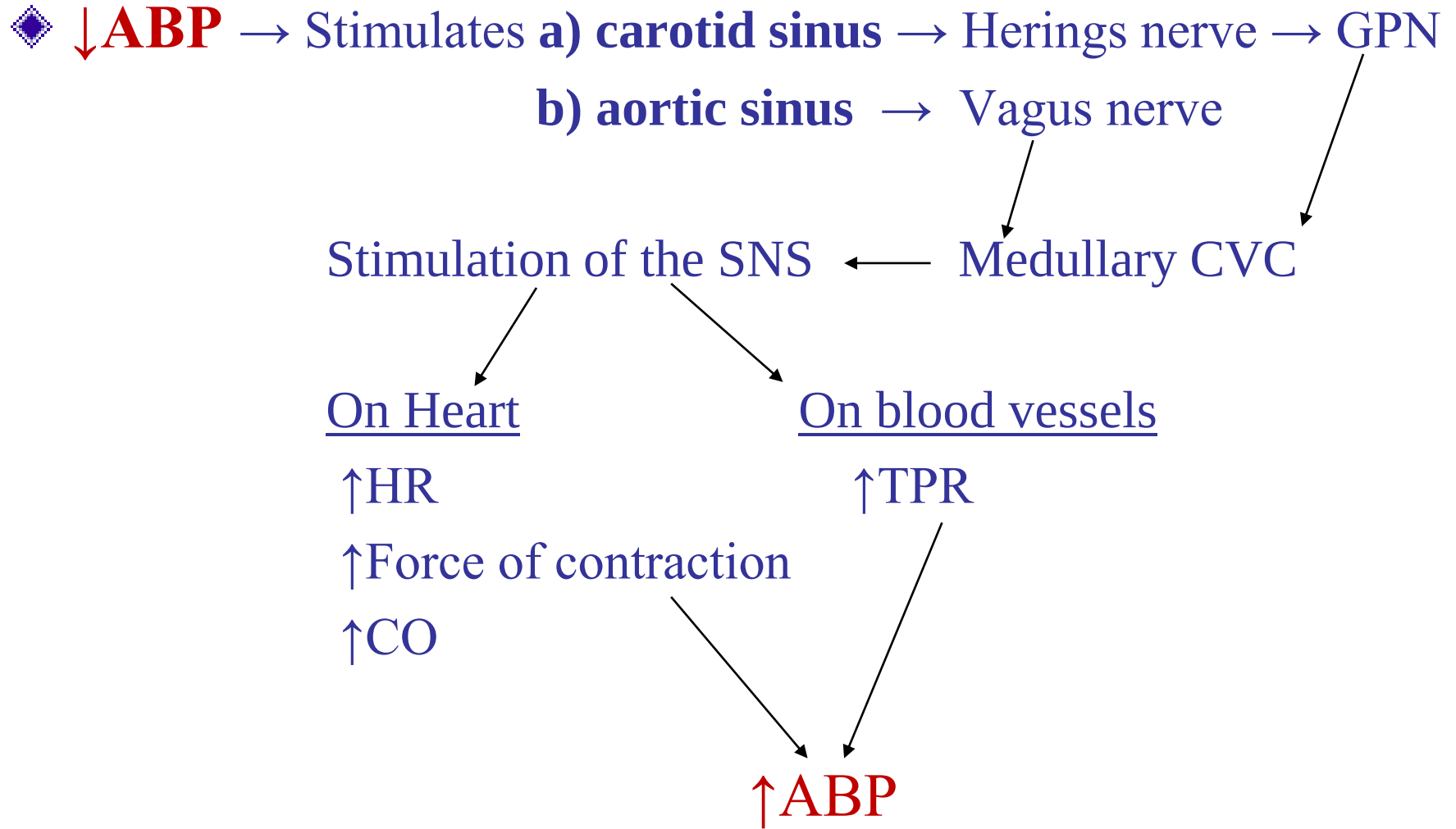
A. Short-term controlling mechanisms

- Baroreceptor reflex
- Chemoreceptor reflex
- Atrial stretch reflex
- Stress relaxation and reverse stress relaxation
- Capillary fluid shift
- Hormonal mechanisms: AD, NAD, Vasopressin (ADH)

B. Long-term controlling mechanisms

- The renal mechanism
- The renin-angiotensin-aldosterone system (RAAS)

Baroreceptor reflex



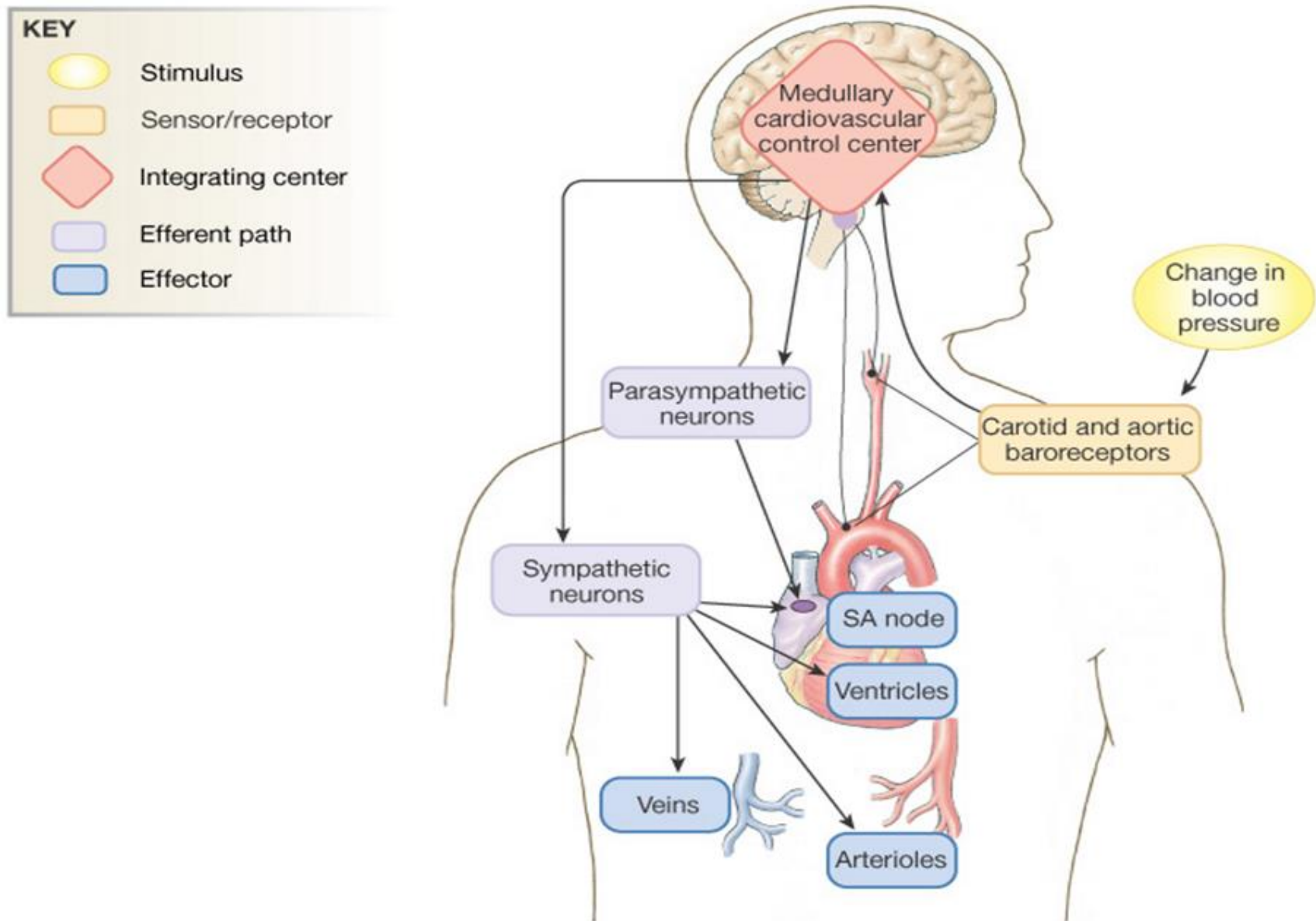


Fig. **Blood pressure regulation by baroreceptors**

Regulation of Blood Pressure

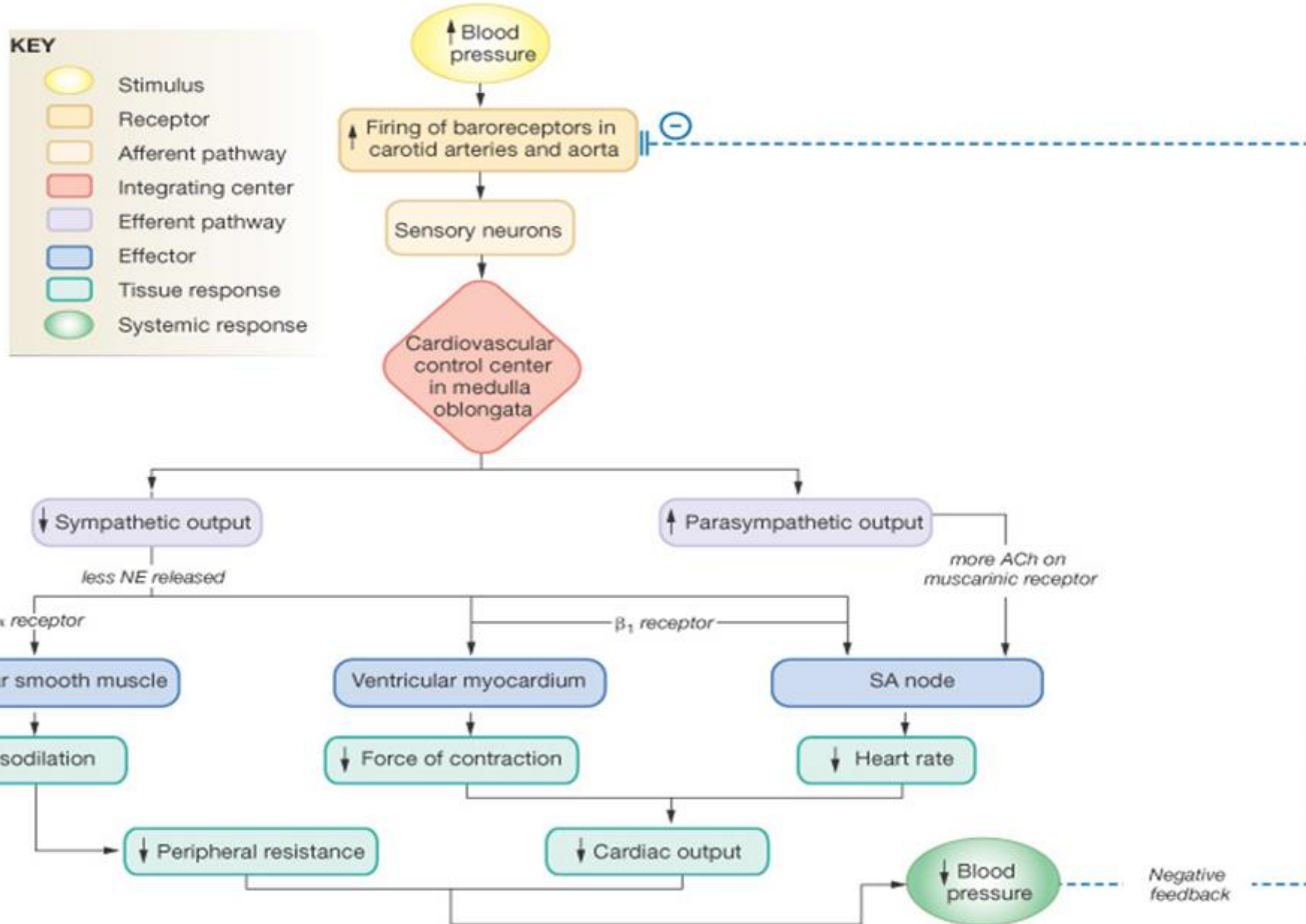
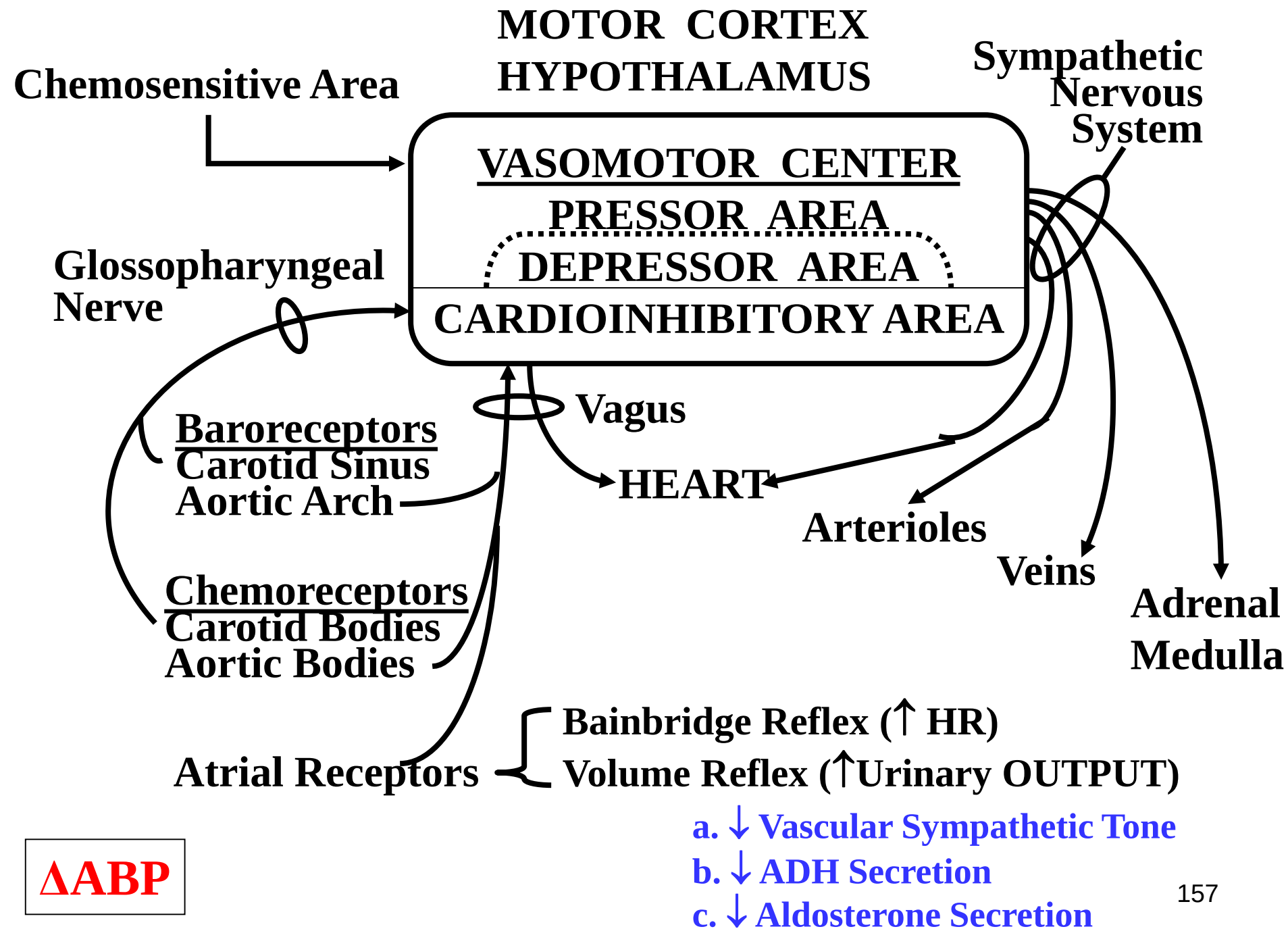


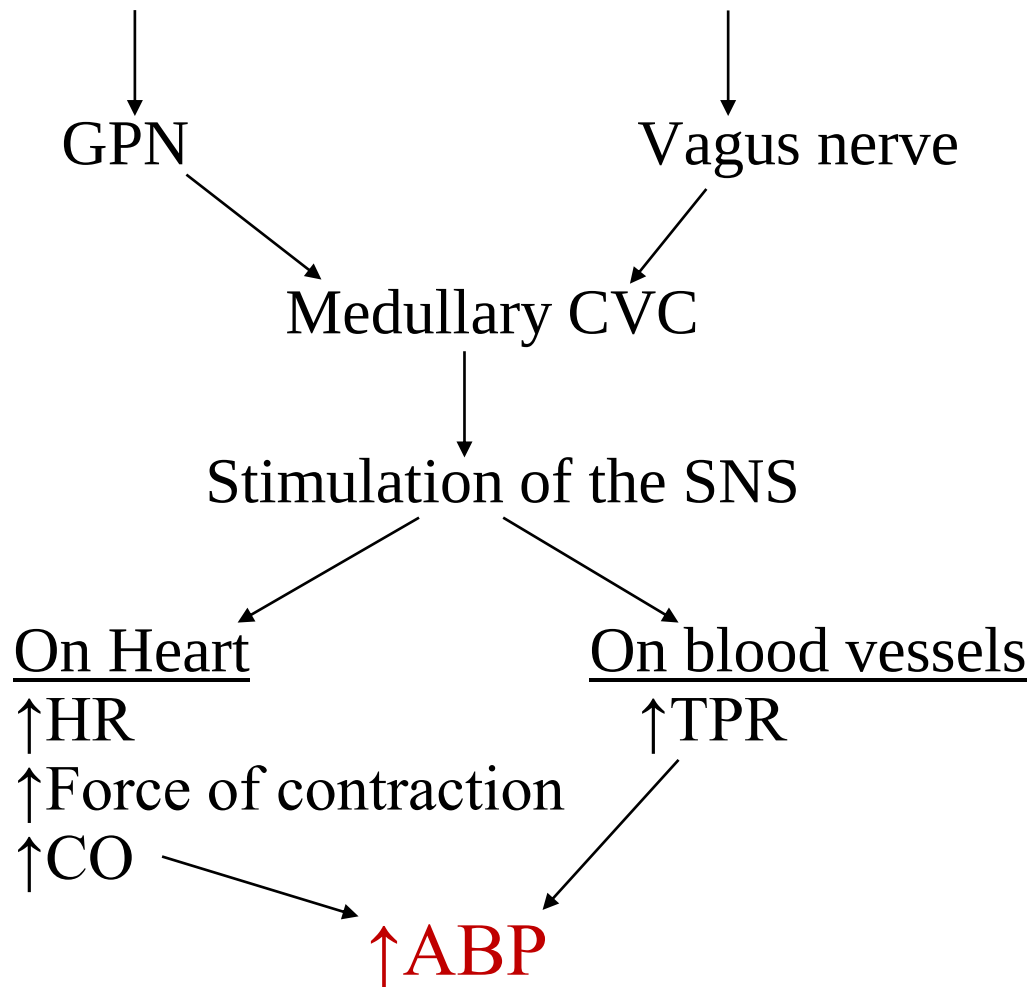
Fig. The baroreceptor reflex: the response to increased BP.



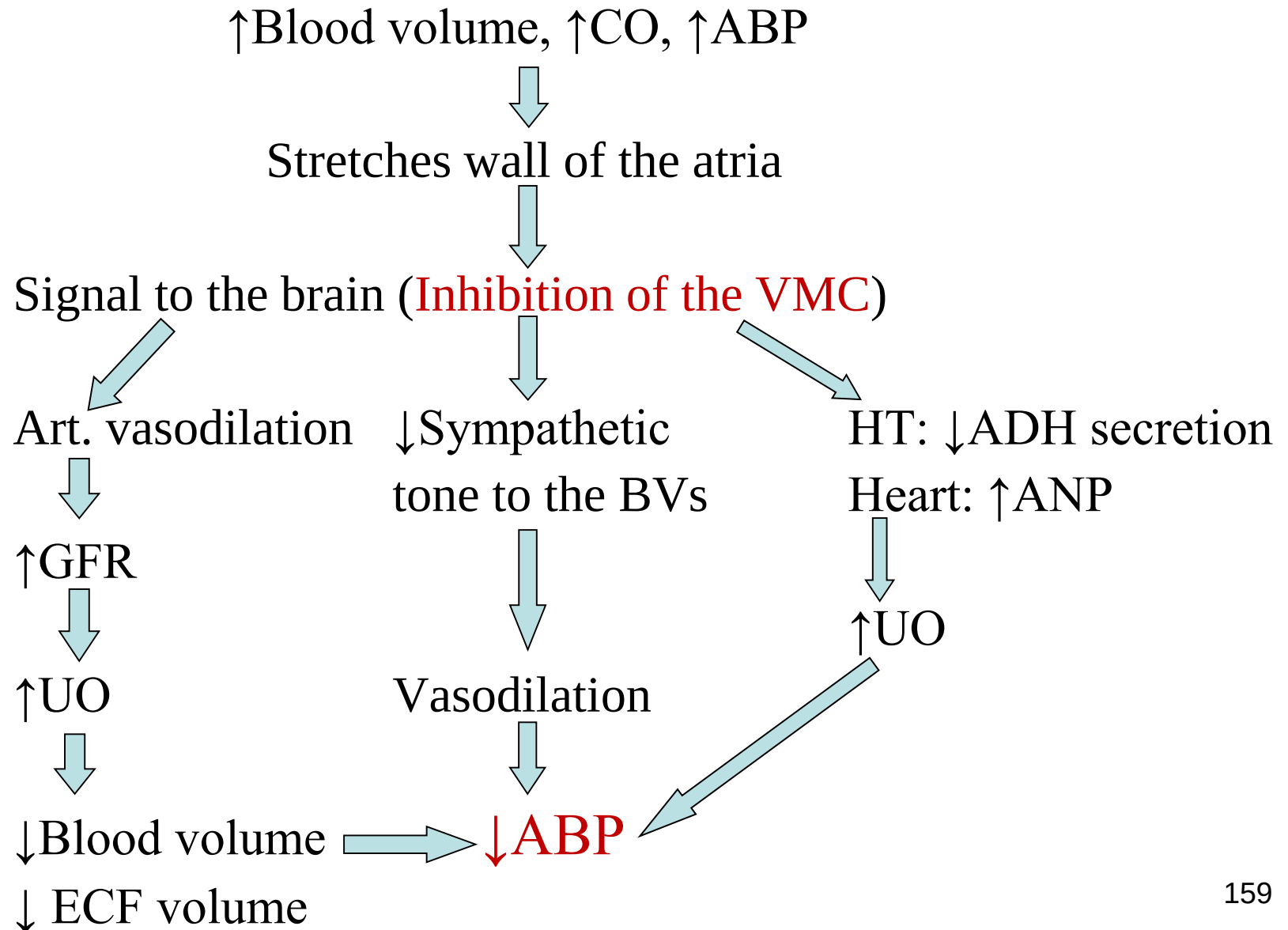
Chemoreceptor reflex

◆ $\downarrow$ ABP $\rightarrow$ Stimulation of chemoreceptors ($\downarrow$ PO₂, $\uparrow$ PCO₂, $\uparrow$ H⁺)

Carotid bodies & Aortic bodies



Atrial volume receptors



Humoral control

Catecholamines are released by adrenal medulla, which is innervated by preganglionic sympathetic fibres. The adrenal medulla secretes mostly **adrenaline** (80%)

◆ **Epinephrine & Norepinephrine**

Renin secretion is stimulated by:

- reduced renal artery pressure,
- reduced Na⁺ delivery to DT, &
- sympathetic stimulation (activation of β_1 adrenoceptors)

Aldosterone system (RAAS)

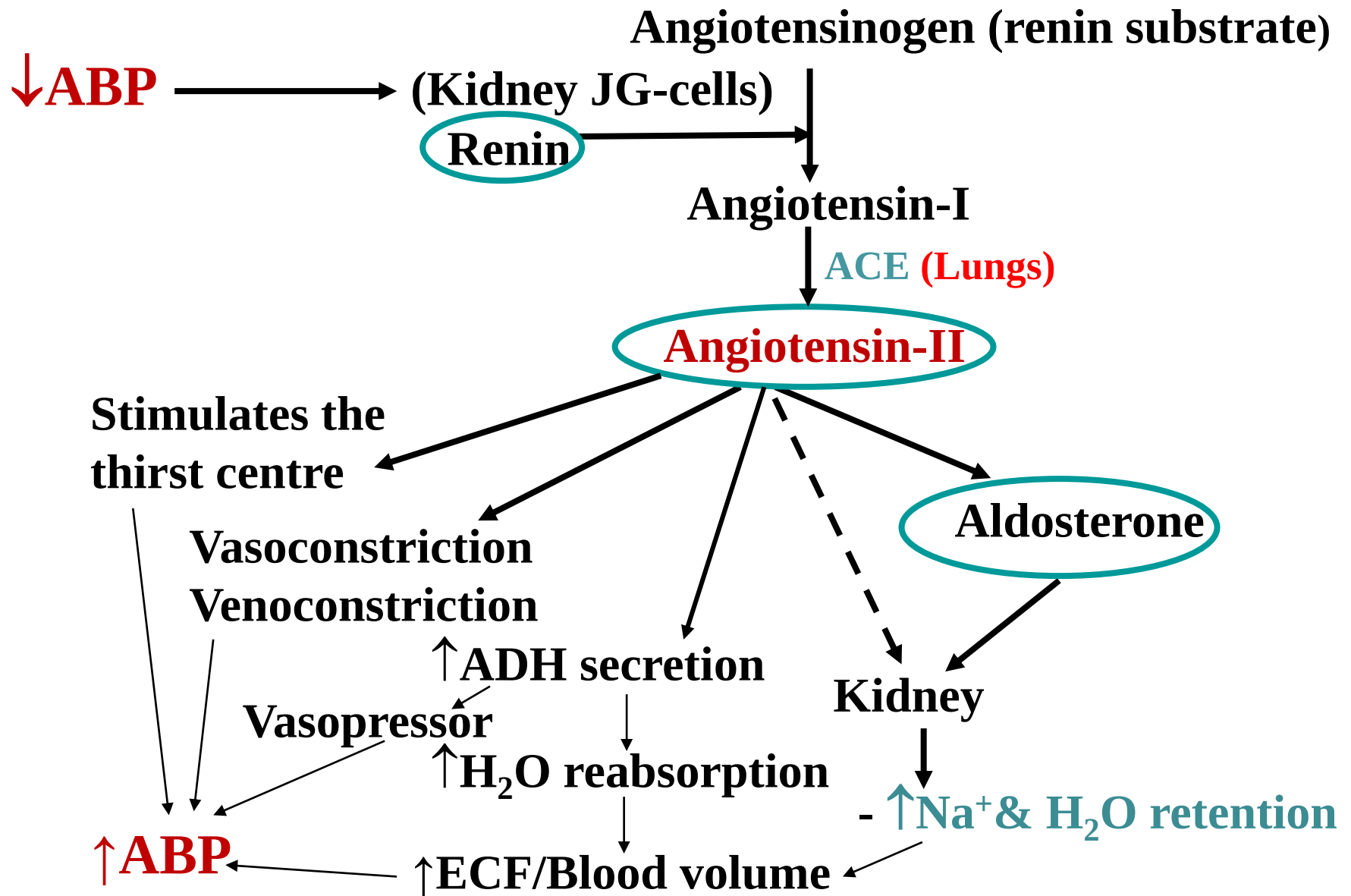
Regulator of the CVS

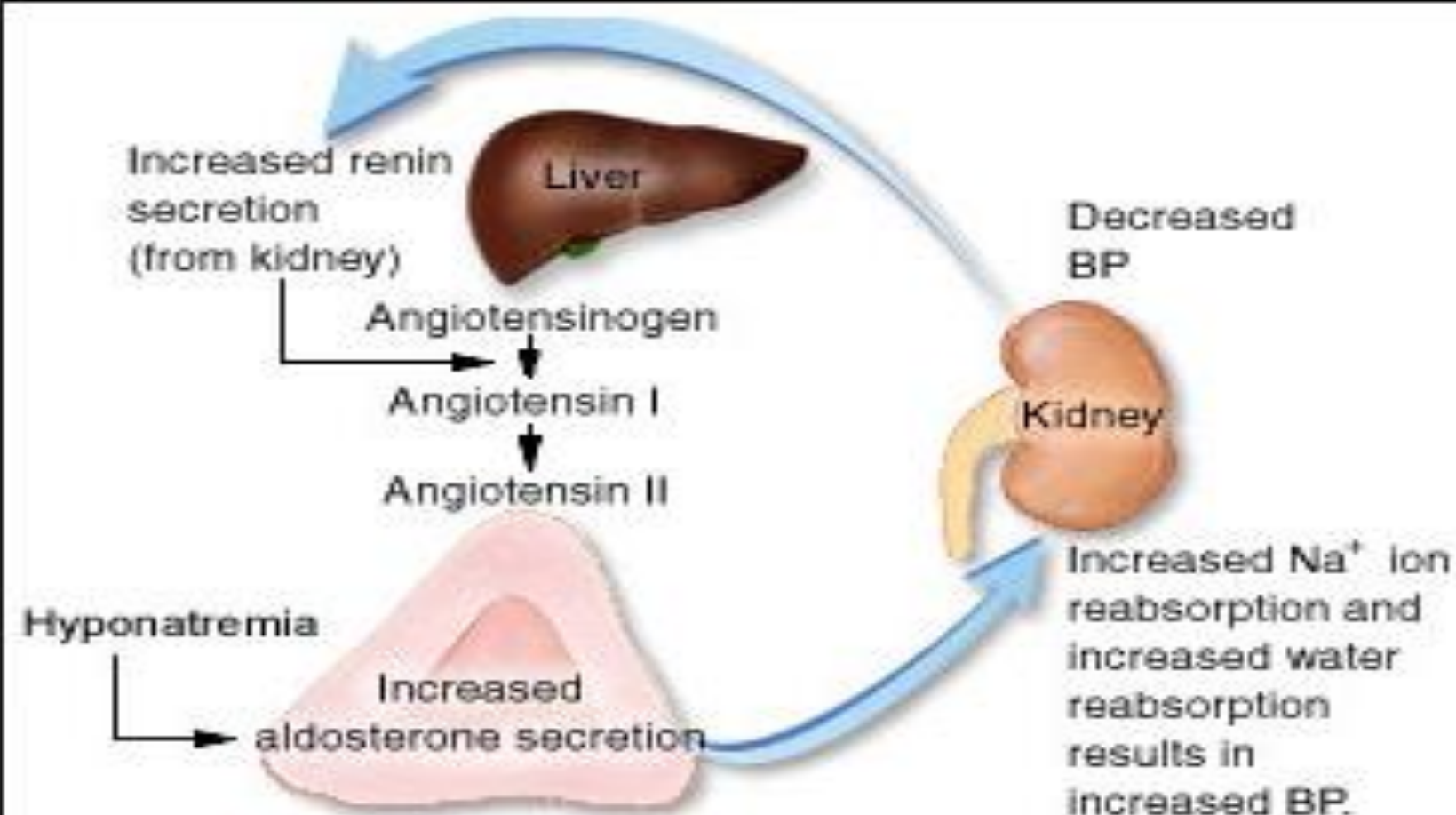
- **Renin** from the kidney (juxtaglomerular cells)
- **Angiotensinogen** - a plasma protein
- **Aldosterone** from the adrenal cortex

◆ **Vasopressin (Antidiuretic Hormone-ADH)**

- Secreted from posterior pituitary gland

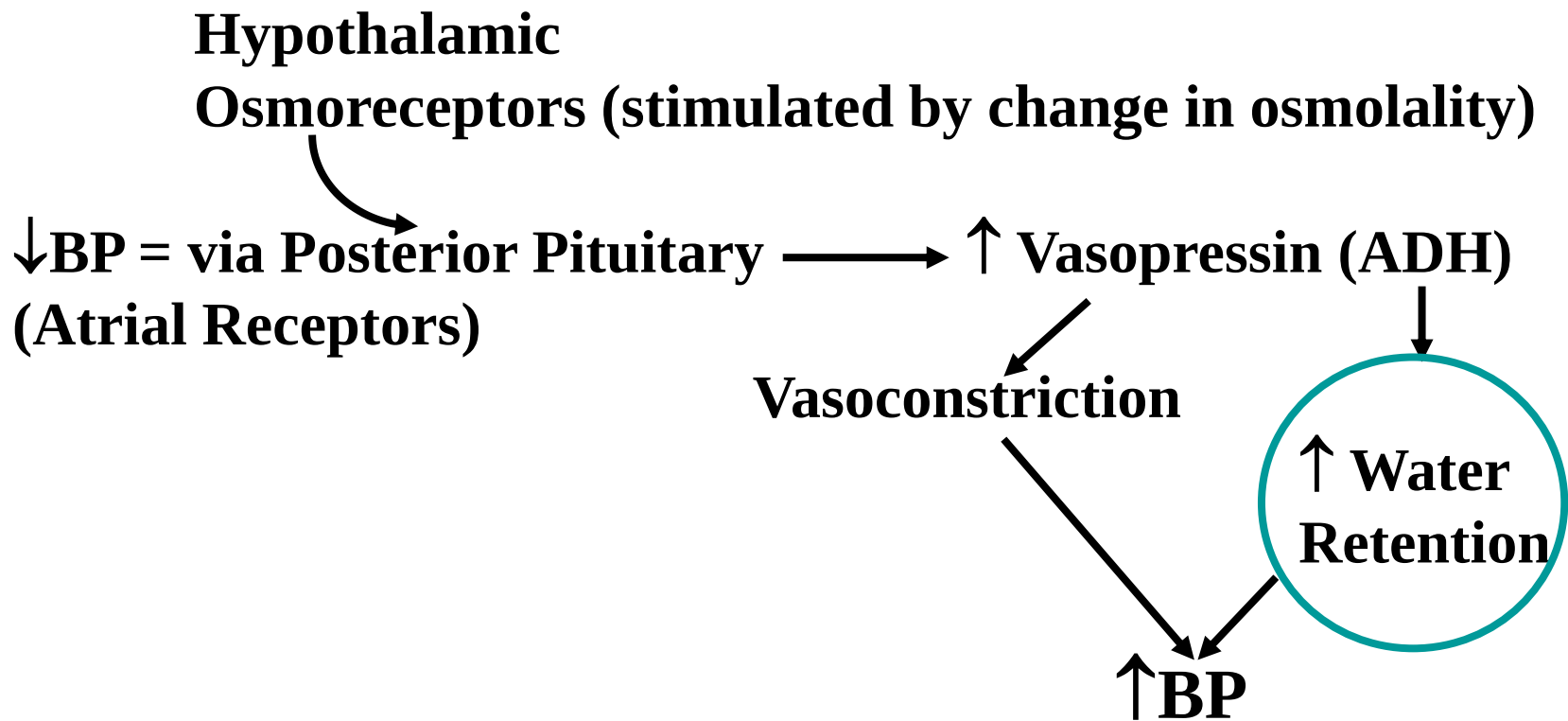
Renin-Angiotensin-Aldosterone Mechanism





Low blood pressure stimulates renin secretion from the kidney. Renin stimulates the production of angiotensin, which is converted to angiotensin II, which in turn stimulates aldosterone secretion from the adrenal cortex. Hyponatremia also stimulates aldosterone secretion. Aldosterone stimulates Na⁺ ion and water reabsorption in the kidney.

Vasopressin (Antidiuretic Hormone)



❖ *ADH mainly acts on collecting duct*

Hypertension

- ❑ **Hypertension** is a persistent elevation of the ABP, >160/95 mm Hg.
- ❑ The upper limit of ABP in different age groups is different.

Age	Upper limit of ABP (mm Hg)
10 - 20	140/90
20 - 50	150/95
Over 50	160/100

❖ **Hypertension** or **high blood pressure** is a medical condition wherein the blood pressure is chronically elevated. **Persistent hypertension** is one of the risk factors for strokes, heart attacks, heart failure and arterial aneurysm, and is a leading cause of chronic renal failure.

Types of Hypertension

A. Primary (essential) hypertension (90% of cases)

- ↑↑ABP with **no clear cause**.

B. Secondary (symptomatic) hypertension (10%)

Causes of secondary hypertension

1. Renal hypertension: renal ischemia, excessive renin

2. Endocrine disturbance

- Pheochromocytoma (↑↑AD)
- Cushing's disease (↑↑Cortisol)
- Hyperaldosteronism
- Oral contraceptives

3. Cardiovascular diseases

- **Aortic coarctation** : congenital constriction of the aorta.
- **Atherosclerosis**

4. Drug induced hypertension

Effects of Hypertension

- **Physiological effects include:**
 - Increases workload on heart
 - Enlarged left ventricle
 - Increased oxygen demand for enlarged muscle
 - May lead to **coronary ischemia**
 - Abnormal stress on vessel walls
 - Stroke, aneurysms, promotes atherosclerosis, visual disturbances
- **No obvious symptoms** for most of cases.

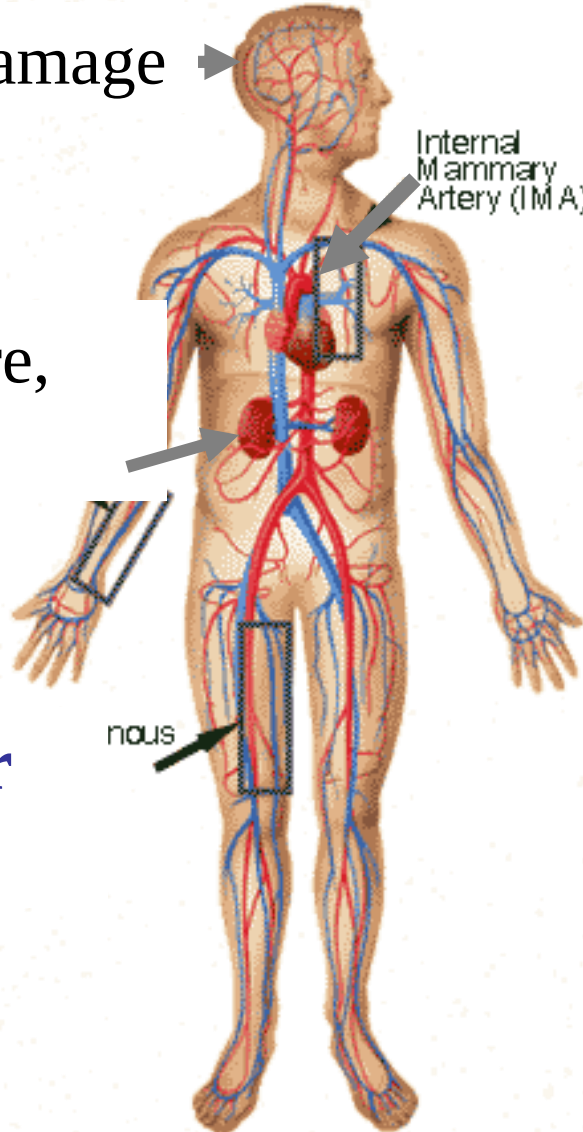
Consequences of Hypertension

CNS: stroke, retina damage ➔

Heart: CHF, angina, MI, sudden death

Kidneys: renal failure, proteinuria

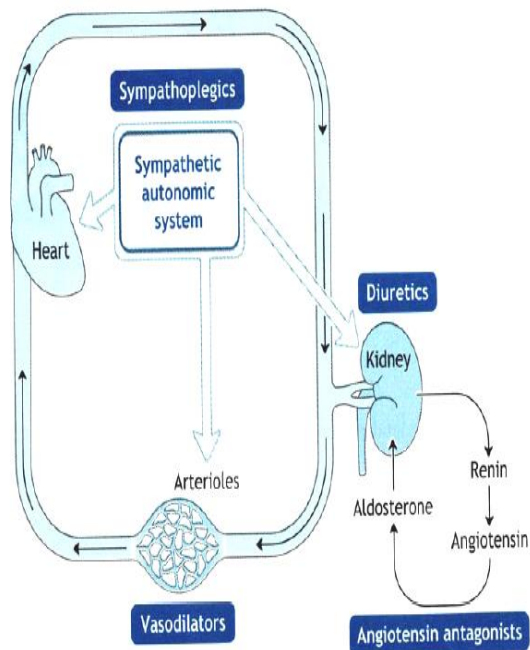
Peripheral Vascular disease



Treatment

- ❑ Lifestyle modification
- ❑ Drugs

How Do We Treat?: Drug Therapy



Major Drug Groups:

Diuretics

Thiazides → **Hydrochlorothiazide**
Loop Diuretics → **Furosemide**

Sympathoplegics

Centrally Acting Agents
Ganglion Blockers
Postganglionic Sympathetic Neuron Blockers
Adrenoceptor Blockers (α and β)

Vasodilators

Nitrovasodilators (**nitroglycerin** → **NO**)
 K^+ channel openers/agonists
 Ca^{2+} Channel Blockers → **Verapamil**

Angiotensin Antagonists

ACE inhibitors
Competitive block of AT1-receptor

Pharmacological Application: Antihypertensive Drugs

Drug classification	Generic agents	CO	TPR	PV
Diuretics		↔	↓	↓
Thiazides	Hydrochlorothiazide			
Loop diuretics	Furosemide			
K ⁺ -sparing diuretics	Amiloride			
Sympatholytics				
Central acting	Methyldopa	↓	↓	↔
β-Antagonist	Propranolol	↓	↓	↔
α-Antagonist	Prazosin	↔	↓	↔
Vasodilators		↑	↓	↑
Arterial	Hydralazine			
Arterial and venous	Nitroprusside			
Ca⁺⁺-channel blockers				
	Verapamil	↓	↓	↔
	Nifedipine	↔	↓	↔
	Diltiazem	↓	↓	↔
ACE inhibitors	Captopril	↔	↓	↔
AngII-antagonists	Losartin	↔	↓	↔

**PV- plasma volume*

Stroke

- ❑ Is a cerebrovascular accident.
- ❑ Is sudden interference of blood circulation in the brain, **starving** one or more parts of the brain with oxygen, nutrients & others.
 - ✓ This prevents the brain from getting oxygen & nutrients from the blood.
- ❑ A sudden loss of brain function caused by a blockage or rupture of a blood vessel to the brain
 - ✓ A blood vessel in the brain bursts or becomes blocked.
- ❑ **Characterized by:**
 - ✓ Loss of muscular control, diminution or loss of sensation or consciousness,
 - ✓ Dizziness, slurred speech → May result in permanent brain damage or death.

Circulatory Shock

- ◆ It is a generalized **insufficiency of blood flow** to the peripheral organs that leads to multiple organ failure.

Types/Causes

1. **Cardiogenic shock**

- Myocardial infarction
- Valvular dysfunction
- Cardiac arrhythmia
- Cardiac failure

2. **Hypovolemic/hemorrhagic/ shock**

- $\downarrow$ Blood volume $\rightarrow$ $\downarrow$ ECF vol $\rightarrow$ $\downarrow$ VR $\rightarrow$ $\downarrow$ CO $\rightarrow$ $\downarrow$ ABP $\rightarrow$ $\downarrow$ BF to vital organs.

3. **Neurogenic shock:** $\downarrow$ Vasomotor tone $\rightarrow$ $\uparrow$ Vasodilation $\rightarrow$ $\downarrow$ ABP $\rightarrow$ $\downarrow$ BF.

4. **Septic shock (distributive shock):** systemic infection

5. **Anaphylactic shock:** allergic conditions

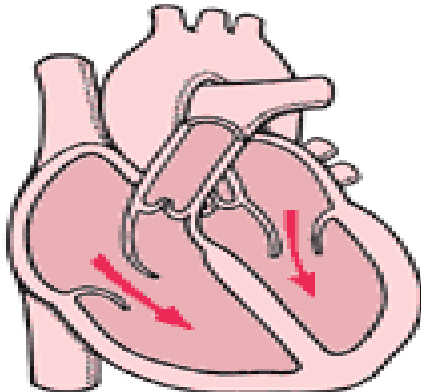
Heart Failure

- ⊕ CO (blood and O₂ supply) inadequate to meet demand).
- ⊕ **Symptoms** depend on which chamber is affected.
 - Left ventricle cannot maintain CO due to damage to muscle (from MI) or ↑ABP (hypertension).
 - Cannot keep pace with right ventricle.
 - Blood backs up into pulmonary circuit.
 - Right ventricle works harder.
 - Increased pulmonary pressures.
 - Fluid filters from capillaries into lungs → **Pulmonary edema**
 - Compromises **gas exchange**.
 - Hypoxia & shortness of breath
 - Venous congestion
 - Ankle edema
 - ↓BP

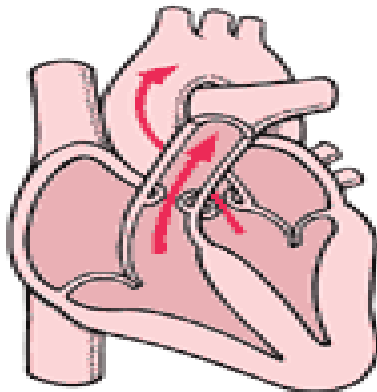
Heart Failure

Normal

Diastole
(filling)

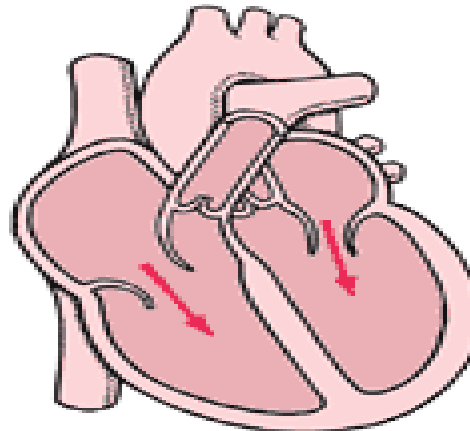


The ventricles fill normally with blood.

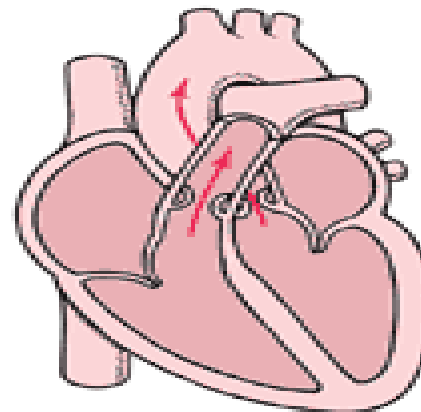


The ventricles pump out about 60% of the blood.

Systolic Dysfunction

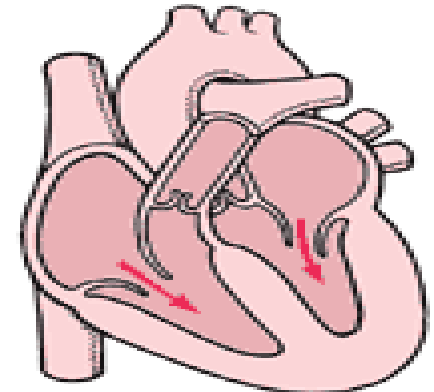


The enlarged ventricles fill with blood.

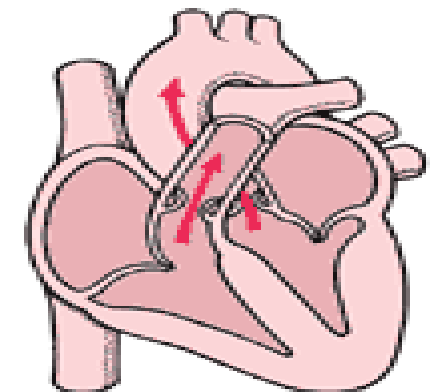


The ventricles pump out less than 40 to 50% of the blood.

Diastolic Dysfunction



The stiff ventricles fill with less blood than normal.



The ventricles pump out about 60% of the blood, but the amount may be lower than normal.

Clinical Evaluation of Heart Function

- **Ejection fraction:** proportion of end diastolic volume ejected from left ventricle
 - $\{(\text{End diastolic volume} - \text{End systolic volume}) / \text{end diastolic volume}\} * 100 = \text{ejection fraction}.$
 - $(\text{SV} / \text{EDV}) * 100$
- **Normally about 70%**
- **Ejection fraction is decreased** with heart failure.

Treatment of Heart Failure

- ◆ Bed rest
 - ↓Load on the heart
- ◆ Reduced salt intake
- ◆ Diuretics
- ◆ Digoxin (digitalis)
 - ↑Force of cardiac contraction
- ◆ Drugs to decrease peripheral resistance - *Vasodilators like ACE inhibitors & nitroglycerin*
 - ↓Load on the heart
- ◆ Cardiac transplant

Disturbances in Heart Rate & Rhythm

- Electrical origin
- Occurs because of ischemia
 - Premature contractions due to ectopic focus
 - Atrial
 - Ventricular
- Ventricular arrhythmias may be fatal
 - Tachycardia
 - Fibrillation and cardiac arrest

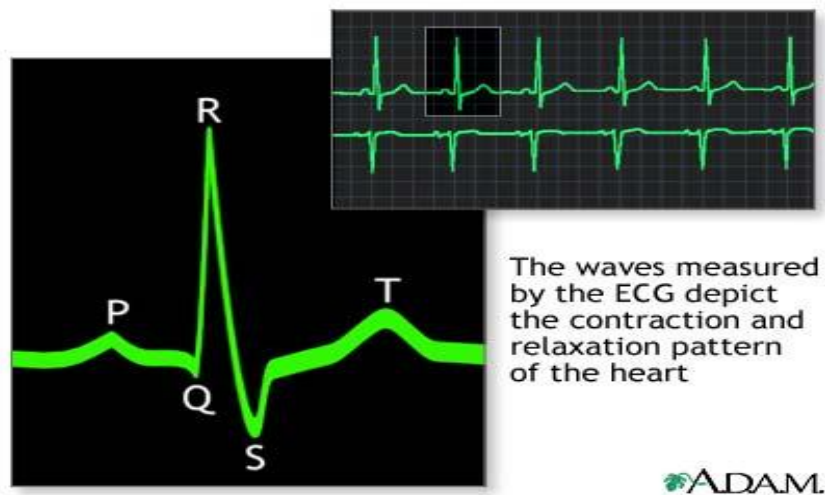
Cardiac Arrhythmia

- **Arrhythmia:** abnormal *rate* or *rhythm* of the heart beat.

Types of arrhythmia

1. **Extra systole (premature beat)**
 - Abnormal contraction produced during diastole.
 - It could be atrial, nodal or ventricular extra systole.
2. **Paroxysmal atrial tachycardia (PAT)**
 - Atrial beat = 200 beats/min
 - Ventricular beat = 170 -180 beats/min
3. **Paroxysmal ventricular tachycardia (PVT)**
 - Ventricular beat = 200 beats/min
4. **Atrial flutter:** fast & regular atrial beats (200-400 b/min)
5. **Atrial fibrillation:** fast & irregular atrial beats (400-600 b/min)
6. **Ventricular fibrillation:** fast & irregular ventricular beats (400-600 b/min)

Illustrations of Normal & Abnormal Heart Rhythms



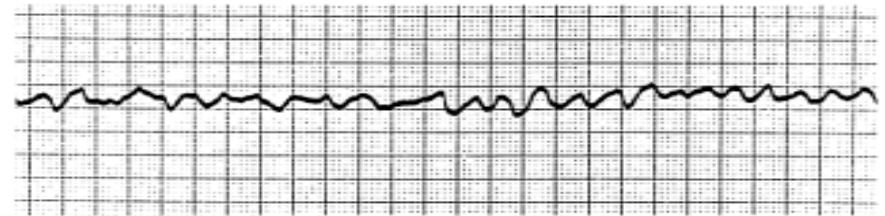
Normal sinus rhythm



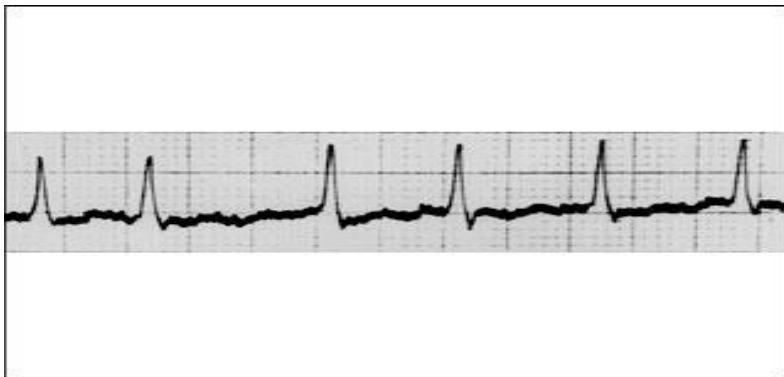
Premature ventricular contraction

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Ventricular Fibrillation



**No effective pumping of
blood from heart**



Atrial fibrillation

Treatment of cardiac arrhythmia

1. **Reduction of over activity of the ANS on the heart.**
 - **Cholinergics:-** for sympathetic over activity
 - **Adrenergics:-** for parasympathetic over activity
 - **Beta blockers:-** for ventricular arrhythmia
2. **Drugs that can prolong refractory period of the AV-node.**
 - **Cardiac glycosides:-** digoxin, ouabain